Lesson 1 The Musical Magic Mirror

"Mirror, Mirror, on the wall, who' s the fairest of them all?" I am sure you' ve been very familiar with this line since it comes from the classic childhood fairy tale *Snow White and the Seven Dwarfs*.

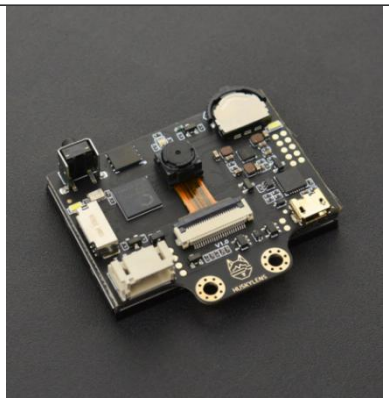
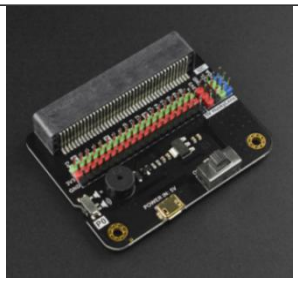


Now, with HUSKYLENS, you can make a real magic mirror like this by yourself. Getting the mirror to talk is a bit difficult, but letting a buzzer make a sound is easy. So let's make a musical magic mirror that can express its attitude by playing different music!

Learning Objectives

1. Understand the working principle and application scenarios of face recognition
2. Learn to use the face recognition function of HUSKYLENS
3. Learn to use HUSKYLENS to make a musical magic mirror

Preparation

	
HUSKYLENS ×1	IO Extender for micro:bit ×1
	
micro:bit v2 ×1	cards with a human face print on them

Tips: In order to facilitate teaching, we specially prepared cards with a human face printed for HUSKYLENS to learn and recognize. Of course, HUSKYLENS can also learn to recognize real human faces.

Learning Content

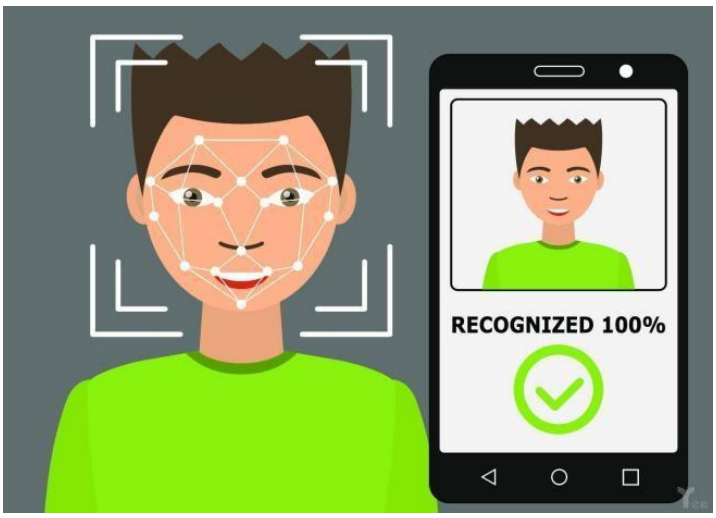
The working principle and applications of face recognition

Like other biological characteristics of the human body (fingerprint, iris, etc.), the face is inherent, unique, and not easily copied. It is the first kind of image that has been studied, and also the widely used one in the field of computer vision. This project is making use of the **face recognition** of HUSKYLENS.

1. What is face recognition

Face recognition is a biological recognition technology for identity recognition based on information about facial features. It uses a camera to collect images and videos containing human

faces and automatically detects and tracks the faces, then performs a series of analytical techniques of the detected faces.



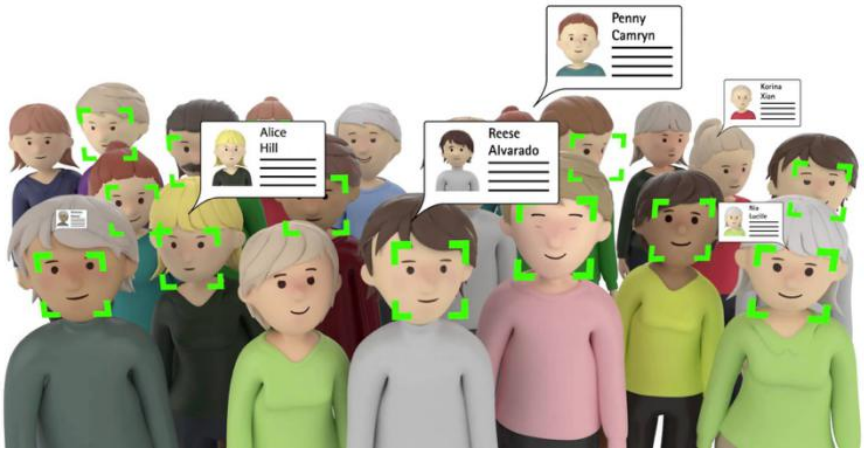
2. The working principle of face recognition

There are 4 key steps in the process of face recognition:

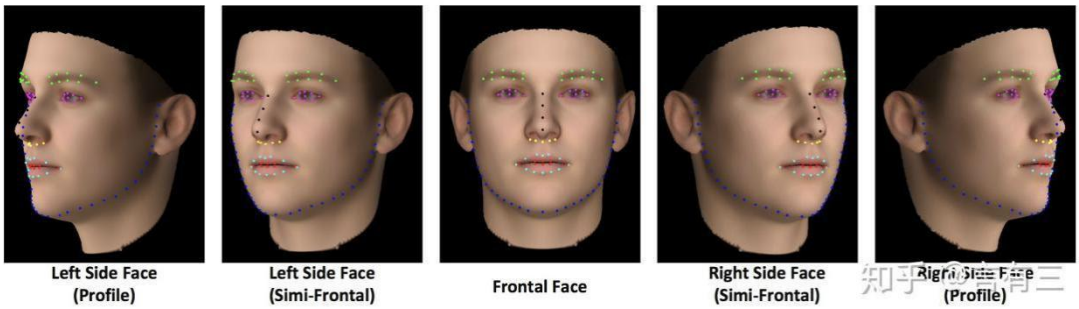


Here is a brief description of these 4 steps.

Face detection: Find the position of the face in the picture, usually marked with a box.



Face alignment: Identify faces from different angles by locating feature points on the face.



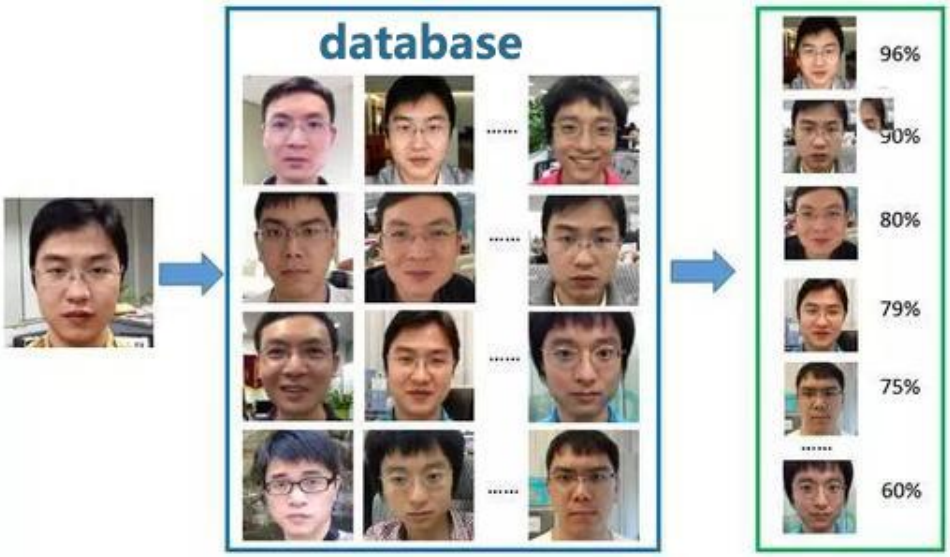
Face coding: It can be simply understood as extracting face information and converting it into information that can be understood by a computer.





[-0.23, -0.54, ..., 0.27]

Face matching: Match the face information to an existing database, which to obtain a similarity score and give the matching result.



Face recognition is also considered to be one of the most difficult research topics in the field of biometric and even in the field of artificial intelligence. **The difficulty of face recognition** is mainly caused by the characteristics of the face as a biological feature.

Similarity: There is just a little difference between different individuals. The facial structure of all faces is similar, and even the appearance and structure of human facial organs are very similar. Such a feature is advantageous for using human faces for positioning, but it is disadvantageous for using human faces to distinguish human individuals.

Variability: The shape of the human face is very unstable, and people can produce many expressions through facial changes. Observing the face from different angles, the visual image of the face will also be very different. In addition, face recognition is also affected by various factors such as lighting conditions (such as day and night, indoor and outdoor, etc.), face coverings (such as masks, sunglasses, hair, beards, etc.), age and other factors.

3. Main application scenarios of face recognition technology

Access control system: In areas protected by security, the identity of entrants can be identified through face recognition, such as prisons, detention centers, communities, schools, etc.

Camera surveillance system: For example, surveillance of people in public places such as banks, airports, stadiums, shopping malls, supermarkets, etc., to achieve the purpose of identity recognition, or installing surveillance systems at airports to prevent terrorists from boarding the plane.

Network application: Use face recognition to assist the use of credit cards for online payment, to prevent non-credit card users from impersonating the owner and to prevent counterfeit claims in social security payment, etc.

Face recognition is currently widely used in all walks of life, such as student attendance systems, cameras, unlocking mobile phones, and all-in-one verification machines.



In addition to the above several application scenarios, where else can you list the use of face recognition technology in your life?

Upgrade firmware

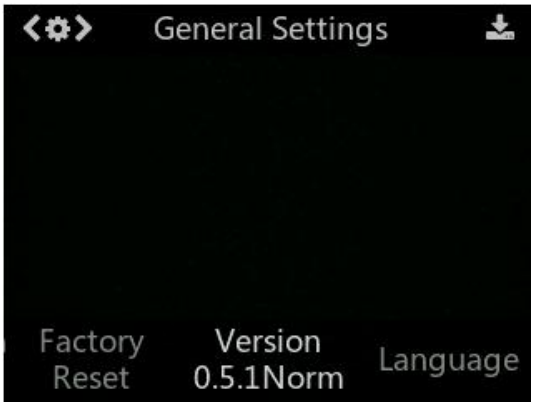
Before using HUSKYLENS, it is recommended that you upgrade the firmware to the latest version to get the latest features and the most stable experience. If your HUSKYLENS is already the latest version of the firmware, there is no need to update it.

This tutorial requires a firmware version of at least V0.5.1, this version integrates all functions (including object classification).

1. Check the firmware version number

1. Toggle the function button to the right until the last option "general setting", short press the function button to enter the secondary menu;

2. Toggle the function button to the right until you reach the "version" option, you can see the version number. As shown in the figure below, the version number is 0.5.1Norm.



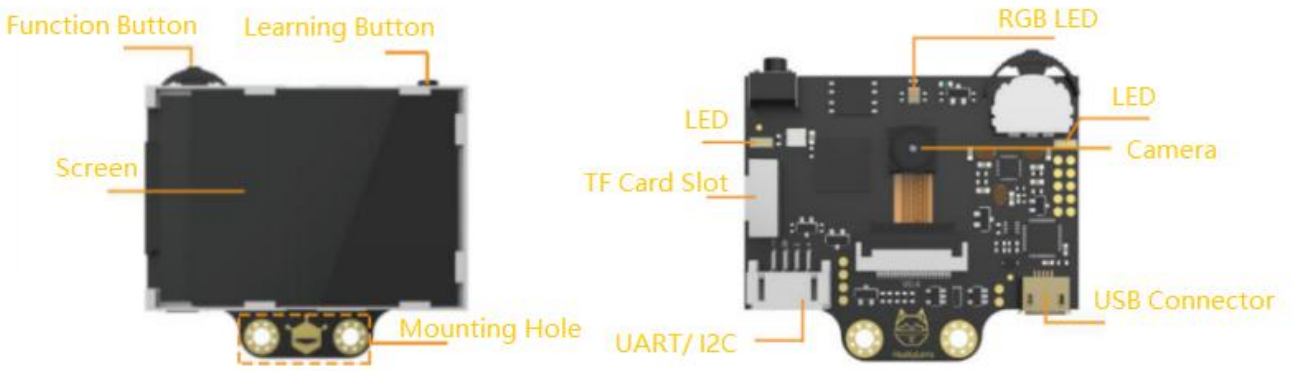
2. Firmware upgrade

If your firmware version is lower than V0.5.1 or you are a user of HUSKYLENS for the first time, please refer to the WIKI website for firmware burning and language setting.

HUSKYLENS Face Recognition Function Demonstration

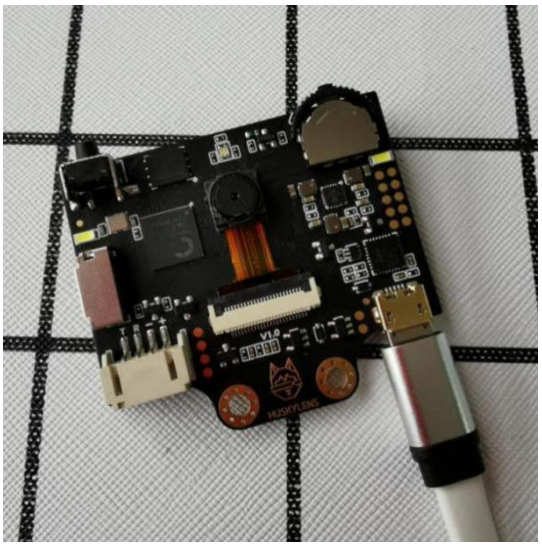
The reason why HUSKYLENS can distinguish faces is mainly because of its built-in machine learning function. It is like a database collector, which can manually input the specified face information and mark this information.

How does it work? Take out your HUSKYLENS first, and let's do it together!



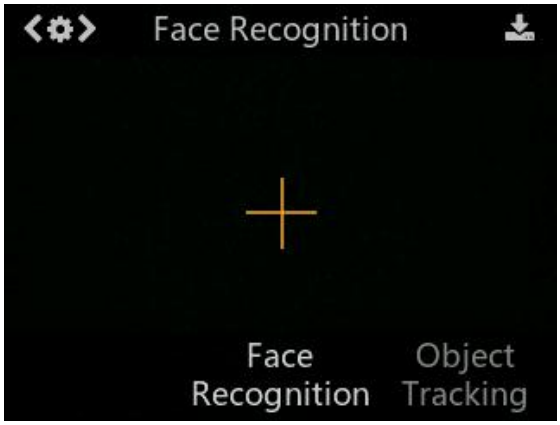
1. Power on

HUSKYLENS comes with an independent USB power port. Power on the board via a USB cable.



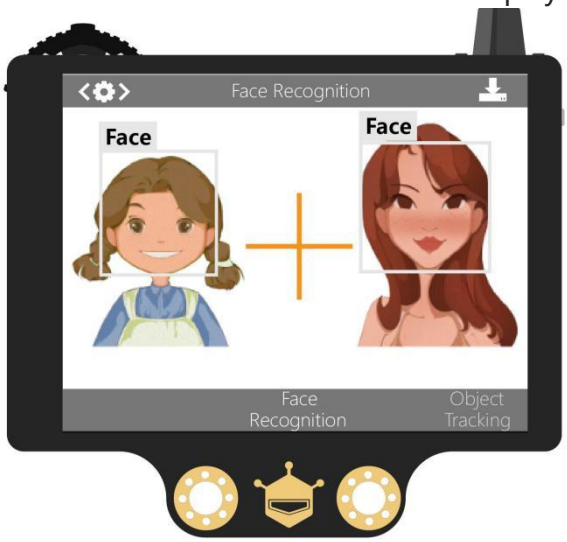
2. Select the "Face Recognition" function

Toggle the "Function Button" to the left until "Face Recognition" is displayed at the top of the screen.



3. Face learning

Point HUSKYLENS at the human face area, the screen will automatically select all the detected faces with a white frame, and the words "human face" will be displayed respectively.



Align the "+" in the center of the HUSKYLENS screen with the face that needs to be learned, short press the "learning button" to complete the learning. If the same face is recognized, a blue frame will appear on the screen and display "Face: ID1". This shows that HuskyLens has learned the face and can recognize it now.





* Keep pressing the "learning button", you can record faces from multiple angles.

* If there is no "+" in the center of the screen, it means that HUSKYLENS has been learned under this function (learned state). At this time, short press the "learning button", the screen will prompt "press again to forget!". Short press the "learning button" again before the end of the countdown to delete what learned last time.

Project Practice

After learning the basic operation of the HUSKYLENS camera, let's finish the production of the musical magic mirror together~

The first function to be realized is that when the camera recognizes faces, it can distinguish between learned and unlearned faces in the terminal. The second is to add music to achieve different music playback, at least two. At last, you can also find a small mirror to build the exterior. This will be done in two tasks.

Task 1: Distinguish human faces

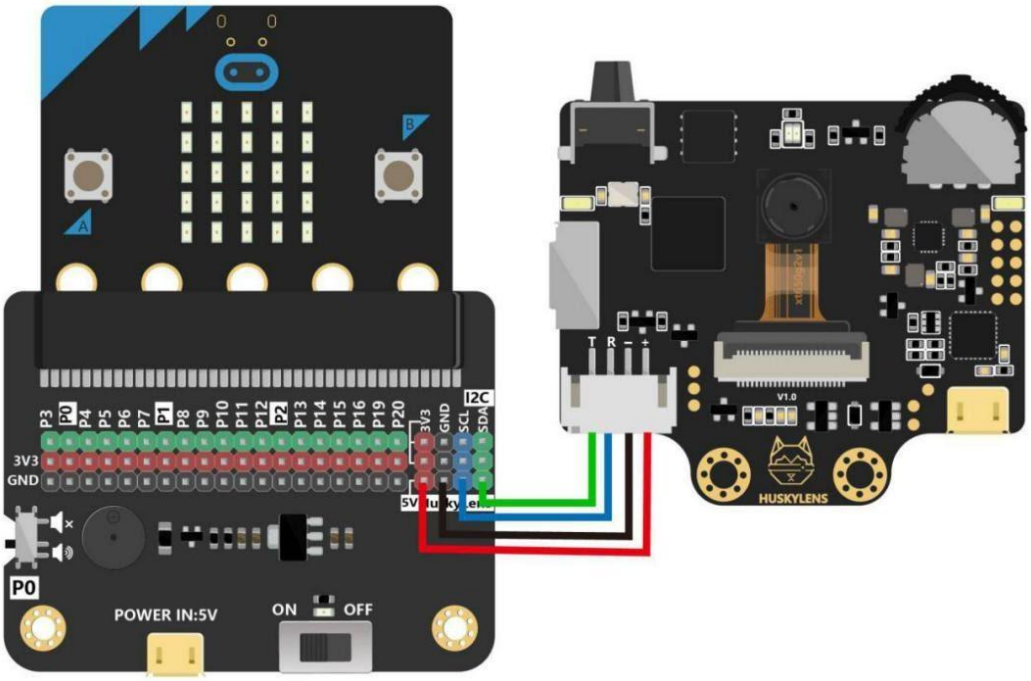
First, learn how to use the HUSKYLENS camera to recognize and distinguish human faces, also be able to make judgments. When the learned faces are recognized, there should be some feedback.

Task 2: Add music

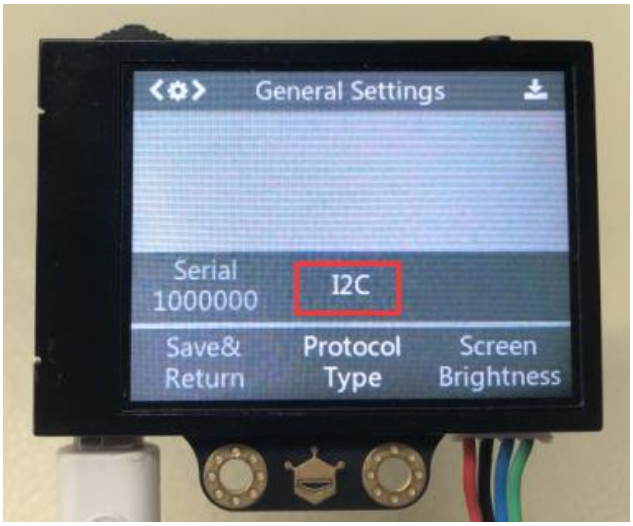
After learning how to distinguish faces and perform the feedback function, you can add more functions on this basis, such as being able to play different music after identifying different faces.

Task 1: Distinguish human faces

Hardware Connection



Note: HUSKYLENS sensor uses the I2C interface, you need to pay attention to the line sequence when wiring, and do not connect it wrongly or reversely. At the same time, you need to set the protocol type to I2C on HUSKYLENS.

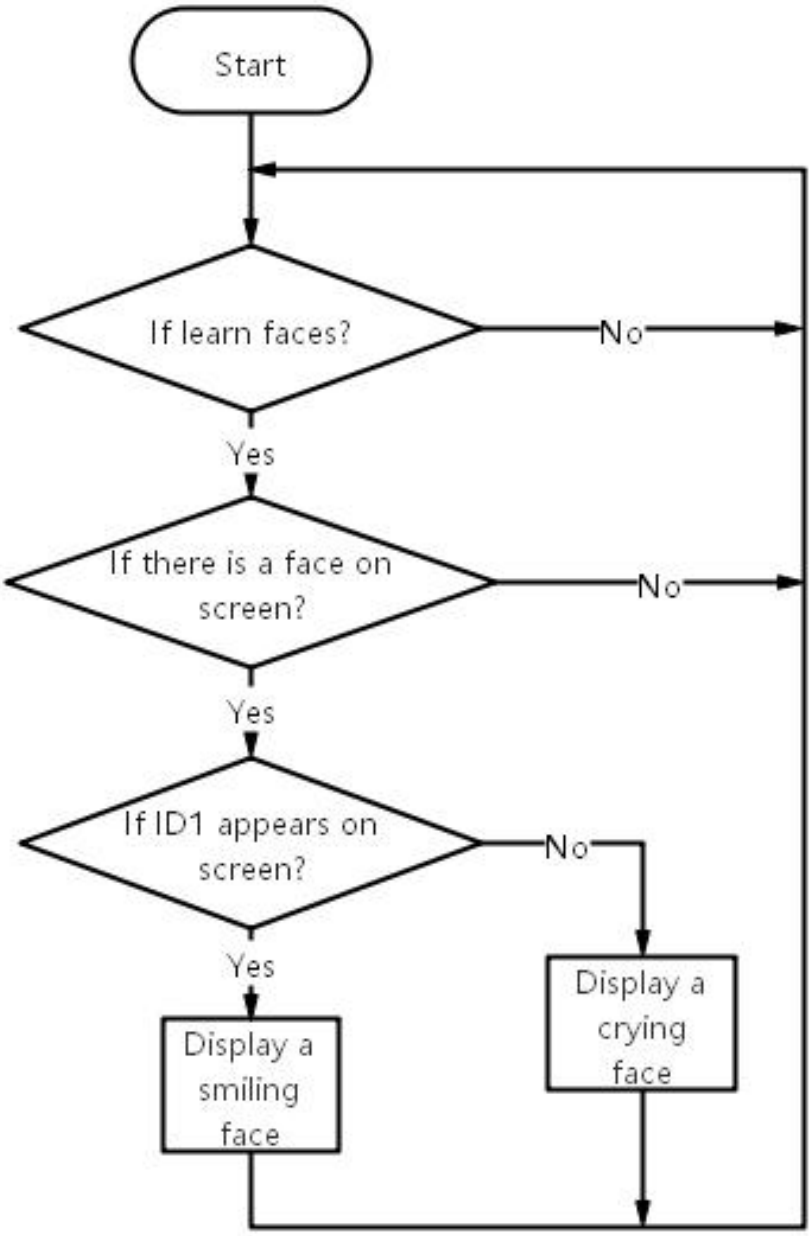


Program Design

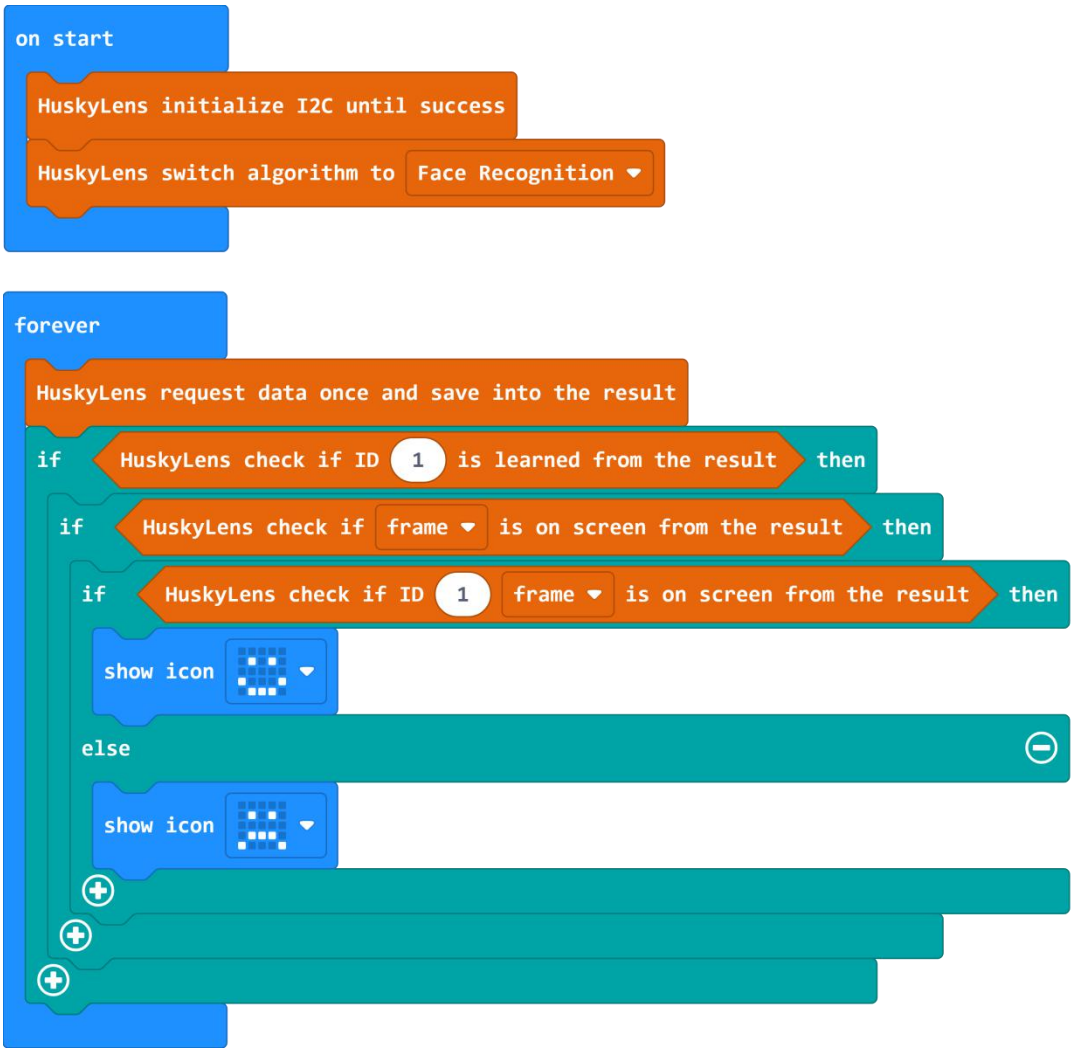
Function Description:

The default camera for distinguishing faces here has learned the specified face information (for the specific learning process, please refer to the HUSKYLENS face recognition function demonstration in this chapter). When the camera sees the face again, it will judge whether it is the learned face. By displaying different patterns on the main control screen, the display process is more intuitive. When a smiling face is displayed on the main control panel, it means that the learned face has been recognized; when a crying face is displayed, it means that the face has not been learned.

Program Flowchart:



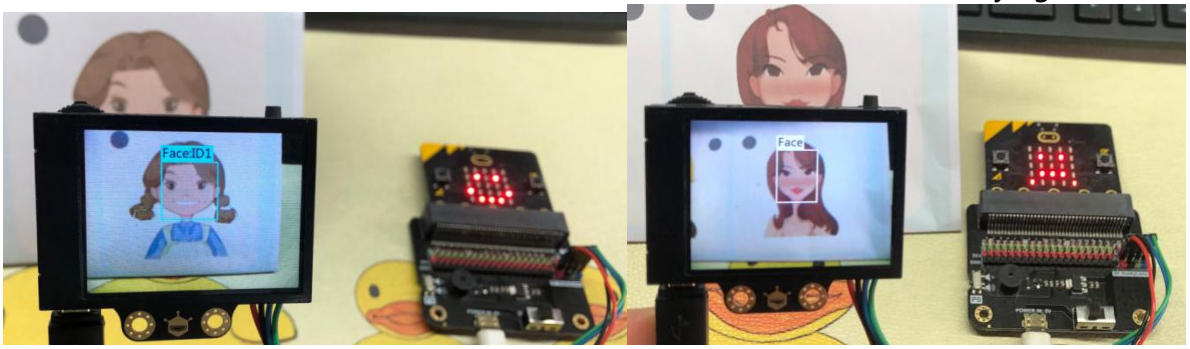
Program Example



For users who are not familiar with the use of micro:bit, please refer to the introductory tutorial of micro:bit.

Operating Effect

Run the program, when HUSKYLENS detects a face that has been learned, it will display a smiling face, and if it detects a face that has not been learned, it will switch to a crying face.



Note: When running the program, in addition to powering the IO expansion board, HUSKYLENS also needs a separate power supply.

Task 2: Add music

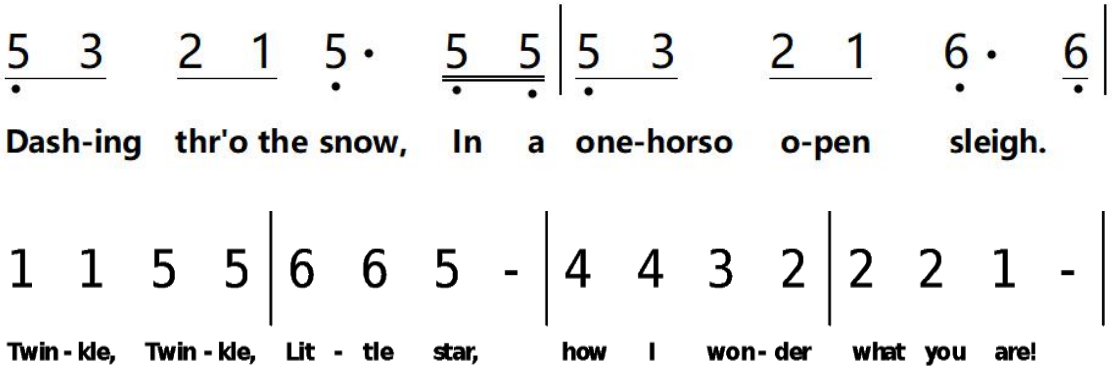
The program design

Function Description:

Here, we can add a piece of music under the smiling face and crying face. For example, "Dashing thr'o the snow, In a one horso open sleigh." in the music "Jingle Bells" and "Twinkle Twinkle little star, how I wonder what you are!" in "Little Star".

How to add music? The buzzer that comes with the IO expansion board is used here, turned on the switch that controls the buzzer while in use.

How to find the music score? Search for the notation of "Jingle Bells" and "Little Stars" on the Internet, and cut out the paragraphs you need.



How to write a program? In MakeCode, it has instructions for playing notes, which are divided into three scales of low, mid and high, and 7 different beats.

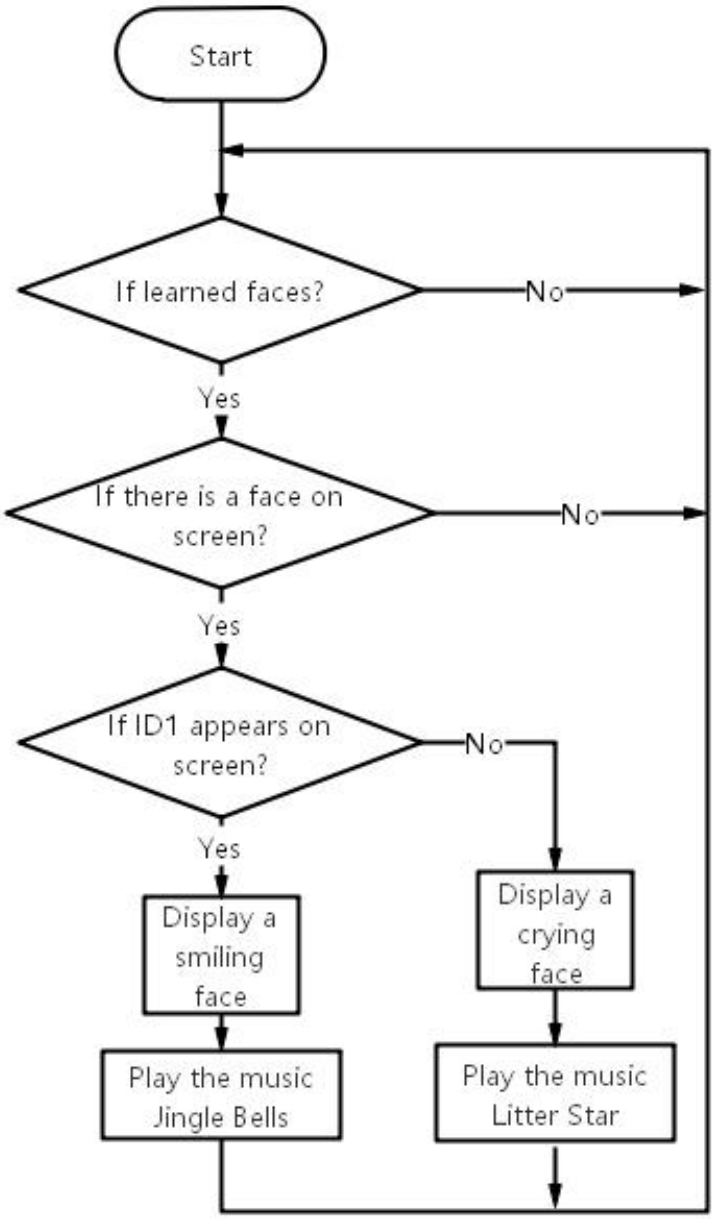




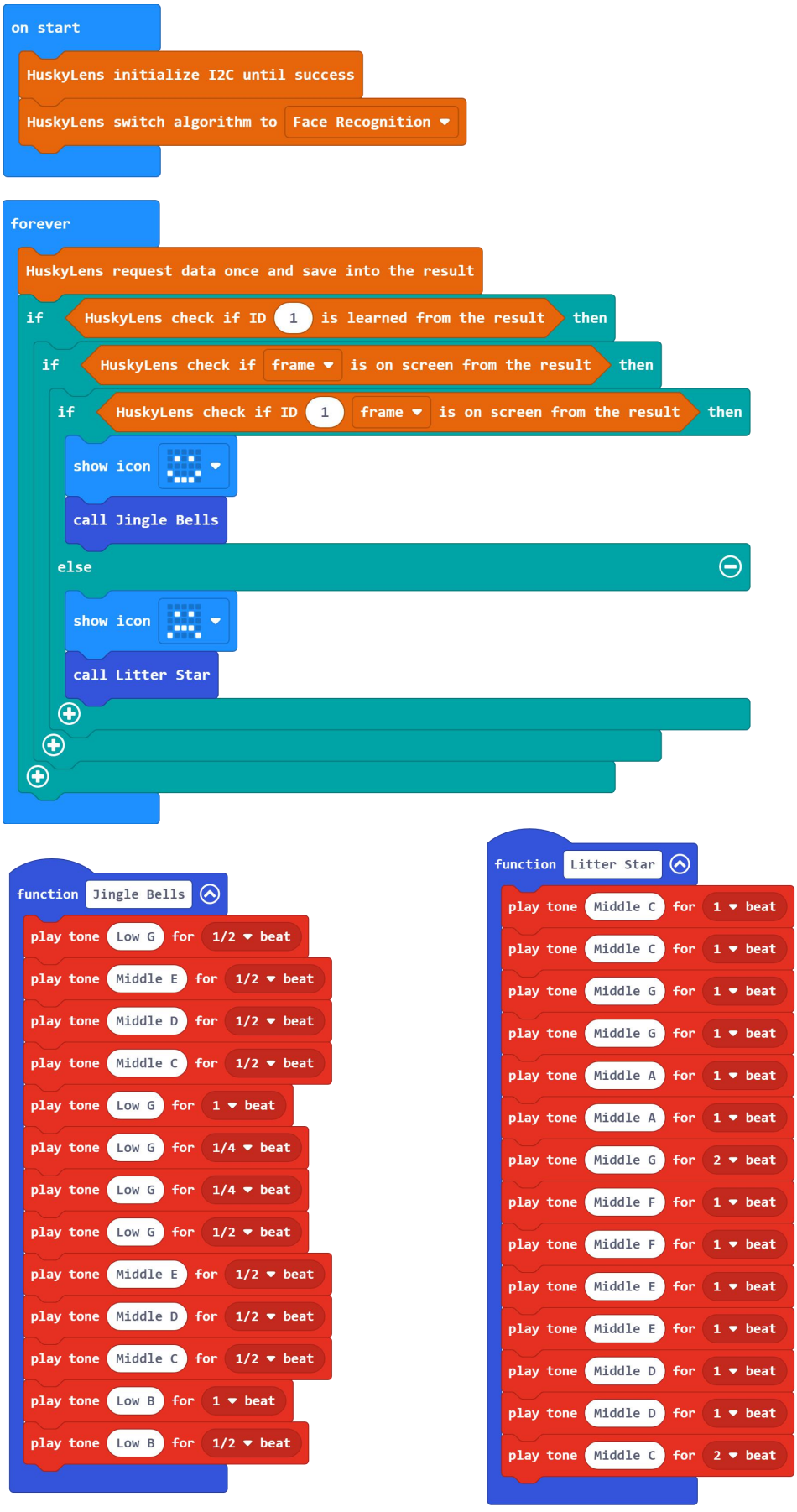
How to match music scores with instructions? Here is a simple identification method, take note 2 as an example, as shown in the following table.

Bass	$\underset{\cdot}{2}$	One beat	2
Alto	2	Half beat	$\underline{2}$
Treble	$\dot{2}$	Quarter beat	$\underline{\underline{2}}$

Program Flowchart:


Start
If learned faces?
No
Yes
If there is a face on screen?
No
Yes
If ID1 appears on screen?
No
Yes
Display a smiling face
Play the music Jingle Bells
Display a crying face
Play the music Litter Star

Program example:



Operating Effect

Run the program, when the camera detects a face that has been learned, it will play the music "Dashing thr'o the snow, In a one horso open sleigh." If the camera detects other faces, it will play the music "Twinkle Twinkle little star, how I wonder what you are!". Isn't it very interesting?

At last, you can also try to let it learn your face, find a small mirror, hide the hardware behind the mirror, expose the camera to collect face information, and build a musical magic mirror that can recognize you!



Project Review

Through the musical magic mirror project, we not only learned the working principle of face recognition, but also learned the instruction application of HUSKYLENS face recognition algorithm.

In the field of artificial intelligence visual recognition, face recognition is an indispensable part, and there are also variable application scenarios. Let's brainstorm together, think about what other face recognition applications that we can make.

Project Development

In this project, we combined face recognition with music to make a fun musical magic mirror. In fact, when you think about it, HUSKYLENS will make a sound when it recognizes an unknown face. Isn't this an access control alarm system?

Lesson 2 Face Recognition Access Control

With the development of technology, the application of face recognition in access control has become more and more widely used. Compared with traditional keys and access control cards, face recognition access control does make face your best pass.

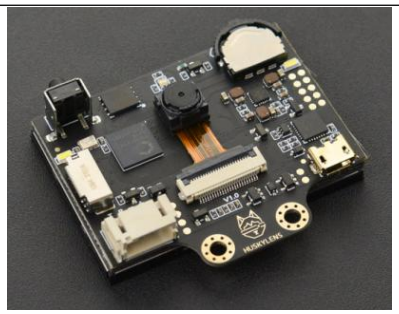
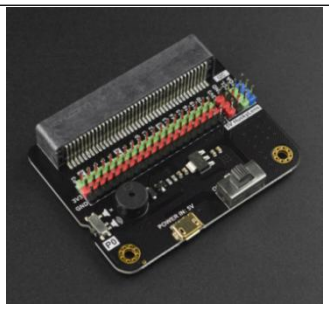
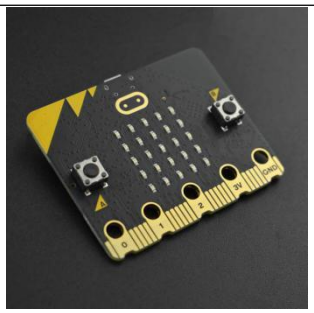


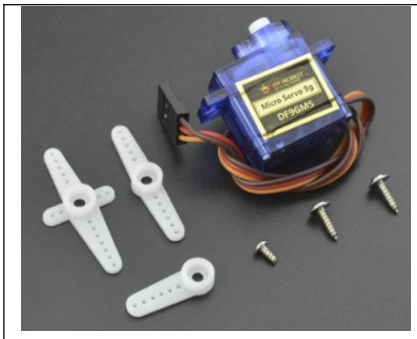
This project uses the function of HUSKYLENS to recognize multiple faces, and implements face recognition access control by performing different operations based on different recognition results. When a stranger is recognized, an alarm will sound; when a family member is recognized, the door will open automatically.

Learning Objectives

1. Think about the drawbacks of traditional keys and access control cards
2. Learn the advantages of face recognition access control
3. Use HUSKYLENS to make a face recognition access control

Preparation

		
HUSKYLENS ×1	IO Extender for micro:bit V2.0 ×1	micro:bit v2 ×1

		
DF9GMS 180° micro servo × 1	Character paper cards	

Learning Content

How many steps are needed to open a door with a key or access card?



1. Open the bakckpak and find the keys



2. Distinguish the key and open the door



3. Pull the key out and get in

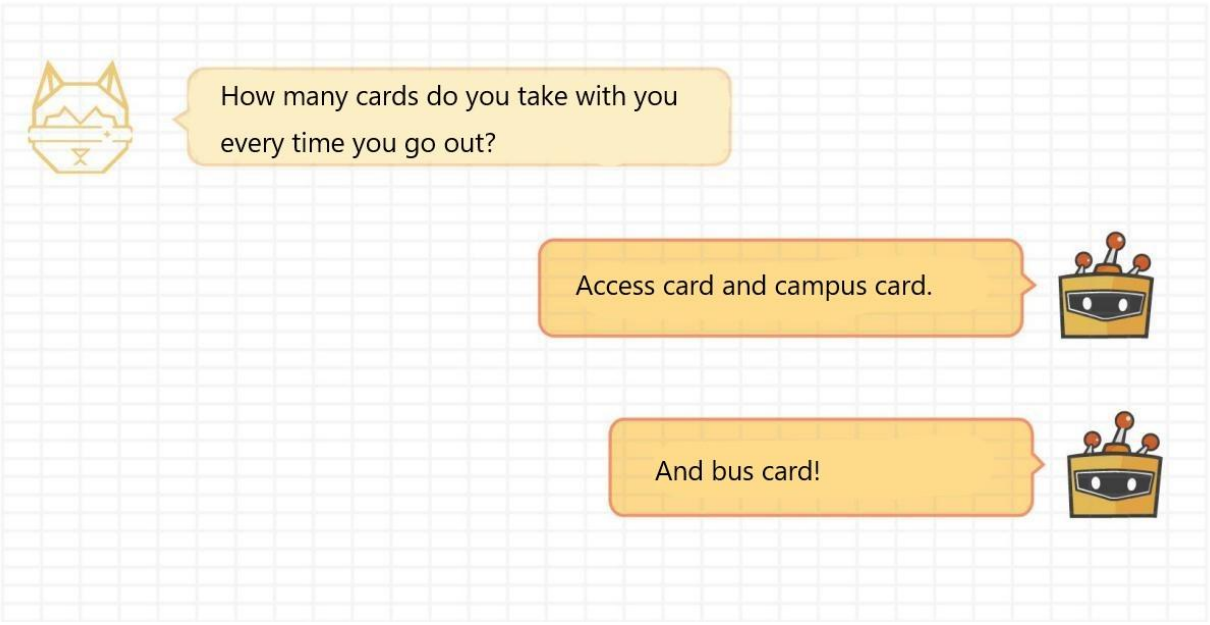
The door opening action that we are accustomed to, actually needs to be divided into 3 steps. Moreover, this situation is based on the absence of accidents such as losing the key or forgetting the key. If there is an unexpected situation, door opening needs far more than these steps.

The Disadvantages of Traditional Keys And Access Cards

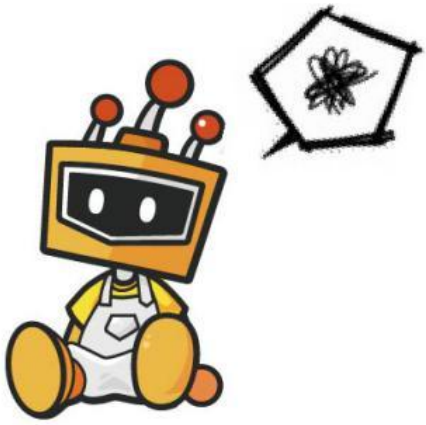
1. Inconvenient to open the door when your hands are occupied

After a huge purchase on the weekend, your hands must be full of “trophies”. When standing in front of the door with your hands occupied, you have no choice but put all your “trophies” down to free your hands. Then you rummage around in your pockets and bags for your keys. Finally, you get the door open. But when seeing all your “trophies” scattered all over the place, you must feel a headache.

2. It is difficult to sort out various cards



After staying at home for a long time, we can finally go out~. But we may find that every time we go out, we need to sort out and check the various "cards" that we need to bring, travel card, ID card, credit card, etc. With so many cards, we may worry about messing around, rummaging when using, forgetting to bring, or losing it. Life is so “hard”！



In your daily life, do you encounter door-opening problems that make you have a headache?

The Advantages of Face Recognition Access Control

Compared with the traditional key and card access control, face recognition access control has the following advantages:

1. Easy to use

Face recognition access control achieves a non-inductive operation. Just stand in front of the door, glance at it, you will be automatically recognized, and you can get in/out then. It liberates your hands so you can open the door easily even when your hands are full of things.

2. Accurate recognition

Simply input the face, and all recognition tasks are handed over to the algorithm. No longer have to find the key corresponding to the door in a bunch of similar keys.

3. High security

What we fear most is the loss of keys and cards. Loss means that they may be used fraudulently. At the same time, the key and the door card are easy to be copied, which would be a potential safety hazard. The face recognition access control can make the door open only for you, and people without permission cannot enter.

HUSKYLENS Face Recognition Function Demonstration

If you want to use HUSKYLENS to achieve face recognition access control, you must first let HUSKYLENS learn the faces of family members and let it know the corresponding members.

In Lesson 1, we learned that the HUSKYLENS face recognition function is set to learn and recognize only one face by default, but there must be more than one person in a family. So, it needs to be set to a state that can learn and recognize multiple faces.

1. Operating and Setting—Learn Multiple Faces

1. Dial the function button to the left until the word "Face recognition" is displayed at the top of the screen.
2. Long press the function button to enter the parameter setting of the face recognition function.
3. Dial the function button until "Learn Multiple" is displayed, then short press the function button, and dial to the right to turn on the "Learn Multiple" switch, that is, the progress bar turns blue and the square icon on the progress bar moves to the right. Then short press the function button to confirm this parameter.



4. Dial the function button to the left until "Save & Return" shows. And the screen prompts "Do you want to save the parameters?" Select "Yes" in default, now short-press the function button to save the parameters and return automatically.

2. Learning and Detection

（1）Face Detection

Point the HuskyLens at any faces. When a face is detected, it will be automatically selected by a white frame with the words "Face" on the screen. Adjust the angle and distance between HUSKYLENS and the face, so that the white frame can contain the entire face as much as possible.



（2）Face Learning

After the face is detected, press the "learning button" to learn the first face, and then release the "learning button" to end the learning. A message will be displayed on the screen: "Press again to continue, press other buttons to end". If you want to continue learning the next face, press the "learn button" before the countdown ends. If you no longer need to learn other faces, just press the "function button" before the countdown ends, or do not operate any buttons and wait for the end of the countdown. The face ID displayed by HUSKYLENS is consistent with the order of entering the face, that is: ID will be marked as "ID1", "ID2", "ID3" and so on in order, and different face IDs correspond to different color frames.



（3）Face Recognition

When HUSKYLENS recognizes a learned face, there will be a colored frame on the screen to select the face and display the ID of the face. The size of the frame changes with the angle of the face, and the frame will automatically track the face. Multiple different faces can be recognized and tracked at the same time, and the colors of the frames corresponding to different faces are also different.



Project Practice

After learning the operation of HUSKYLENS, let's complete the production of face recognition access control together!

First of all, the function to be realized is that when HUSKYLENS recognizes human faces, it can detect family members from strangers in the program terminal. The second is to add a servo, using its rotation angle to control the opening and closing of the door, so as to realize the automation of opening and closing the door. Now, we are going to complete the project of face recognition access control in two tasks.

Task1: Face Distinguishing

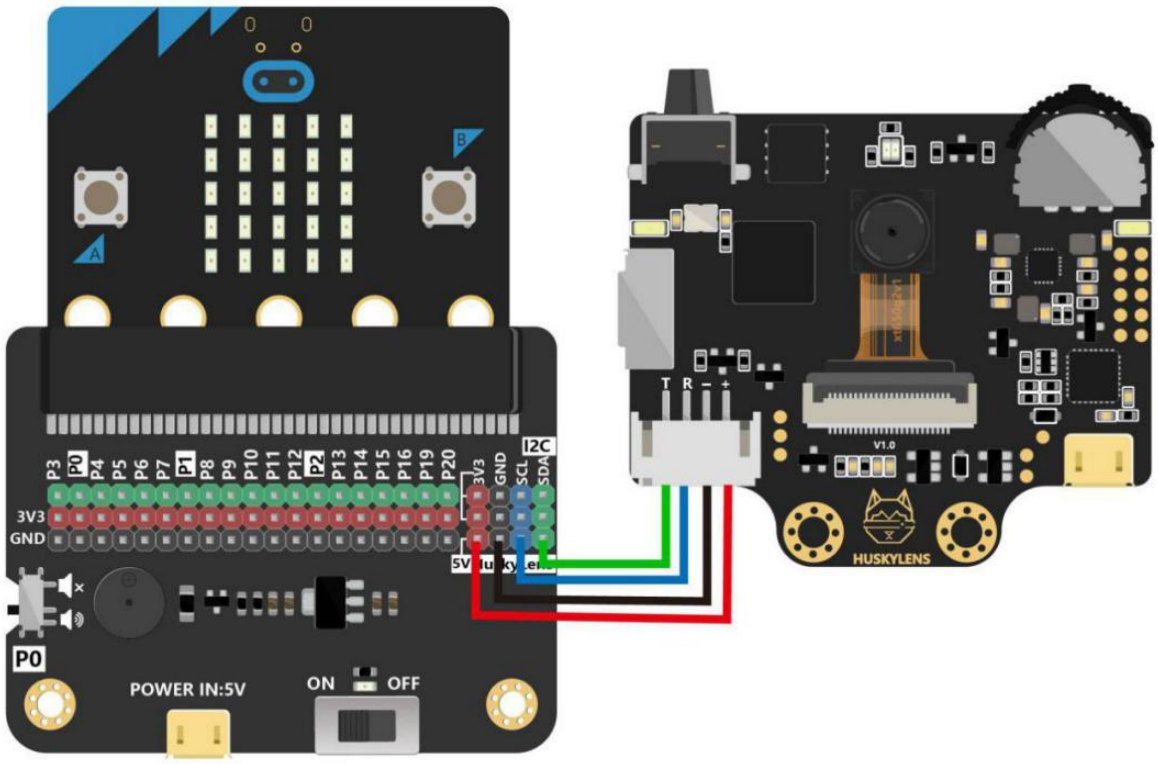
In the beginning, we need to let HUSKYLENS recognize multiple faces, distinguish faces, and give corresponding feedback

Task2: Smart Access Control

After learning how to distinguish faces, we can add more functions on this basis. For example, when a family member is recognized, the door will be opened automatically; when a stranger is recognized, the door not only does not open, but also makes an alarm.

Task 1: Face Distinguishing

Hardware Connection

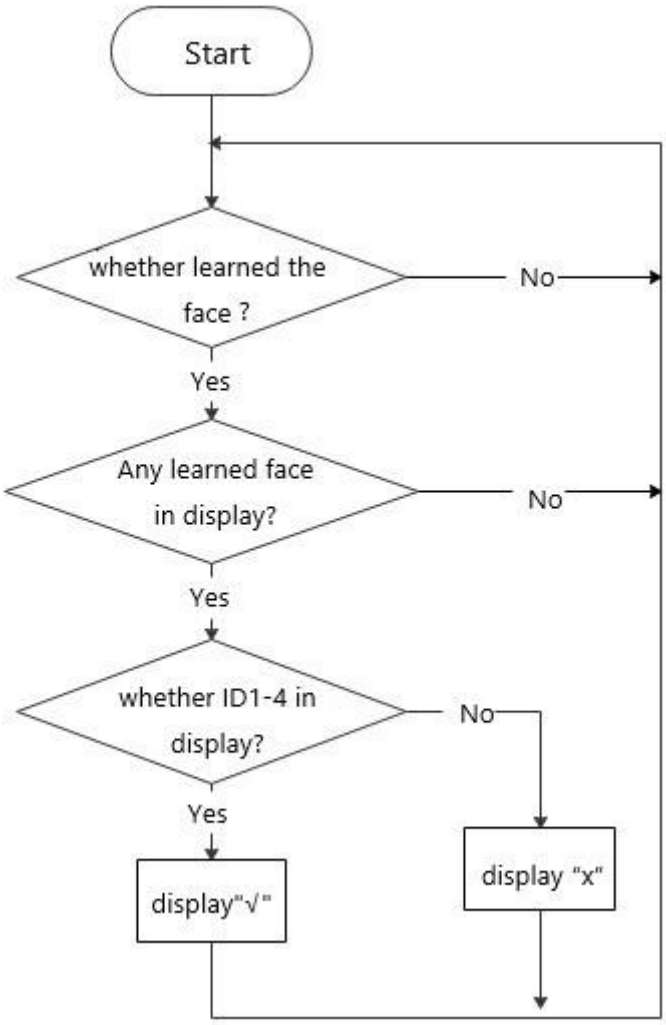


Program Design

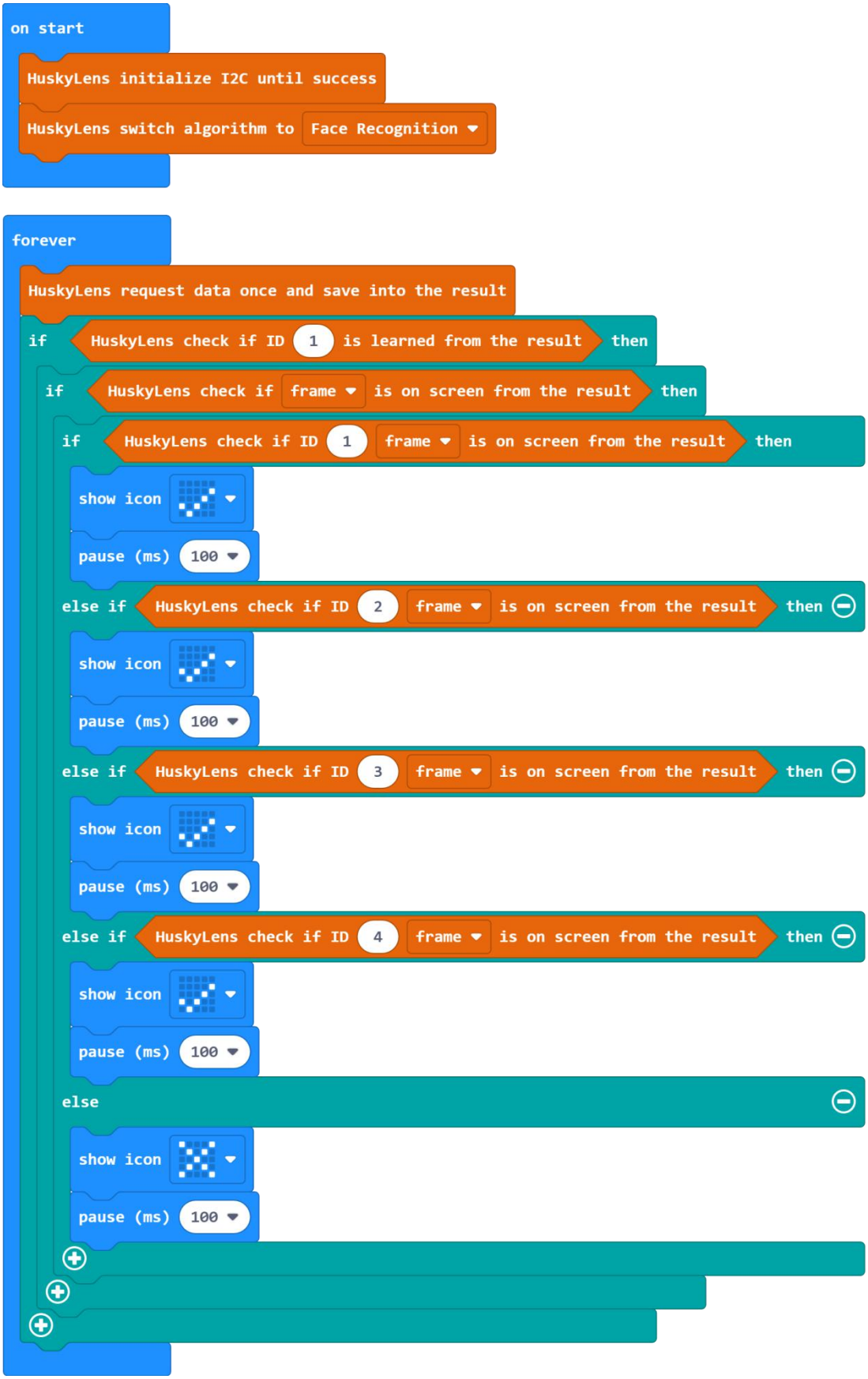
Function Instruction:

Here we take it by default that the HUSKYLENS has learned the faces of family members already (for the specific learning process, please refer to the demonstration of HUSKYLENS face recognition function in this lesson), when the camera detects the faces, it will determine whether the face belongs to the family that has been learned. In order to make the recognition process more intuitive, when a "√" is displayed on the main control board, it means that the recognized face belongs to the family; when a "×" is displayed on the main control board, it means that the recognized face is a stranger.

Flowchart Analysis:

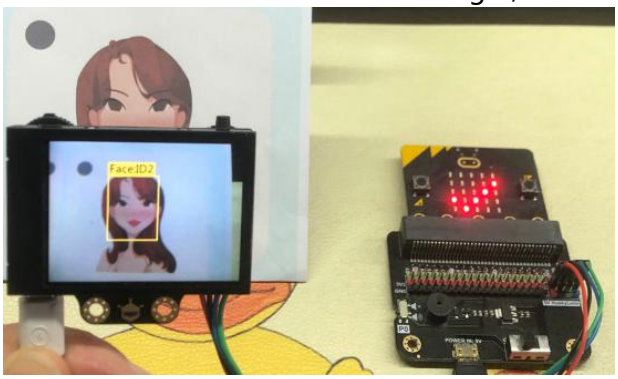


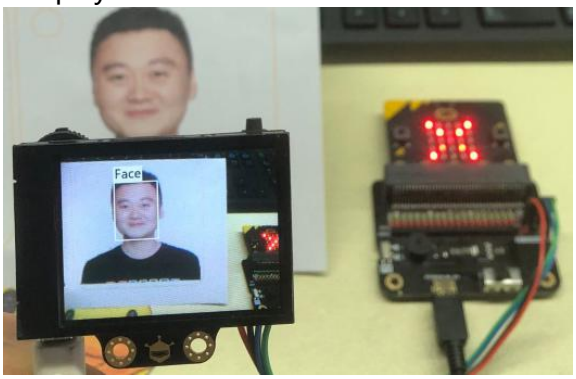
The Sample Program



Operating Effect

When HUSKYLENS detects the face belonging to the family, “ √ ” is displayed on the micro:bit. When HUSKYLENS detects the face of a stranger, the micro:bit displays "×".

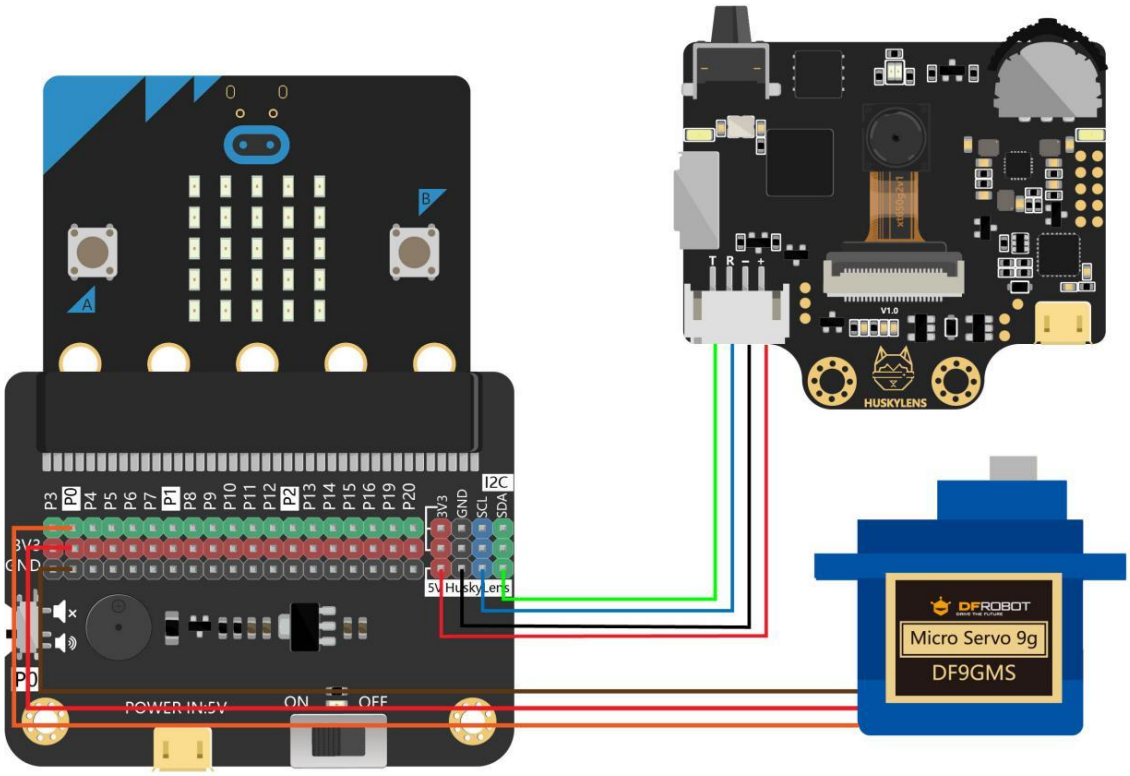




Task 2: Smart Access Control

Connect HUSKYLENS to the I2C interface of the expansion board through the 4Pin cable as the terminal input device of the smart access control system, and then the servo should be connected to the P0 port of the expansion board according to the correct line sequence.

Hardware Connection

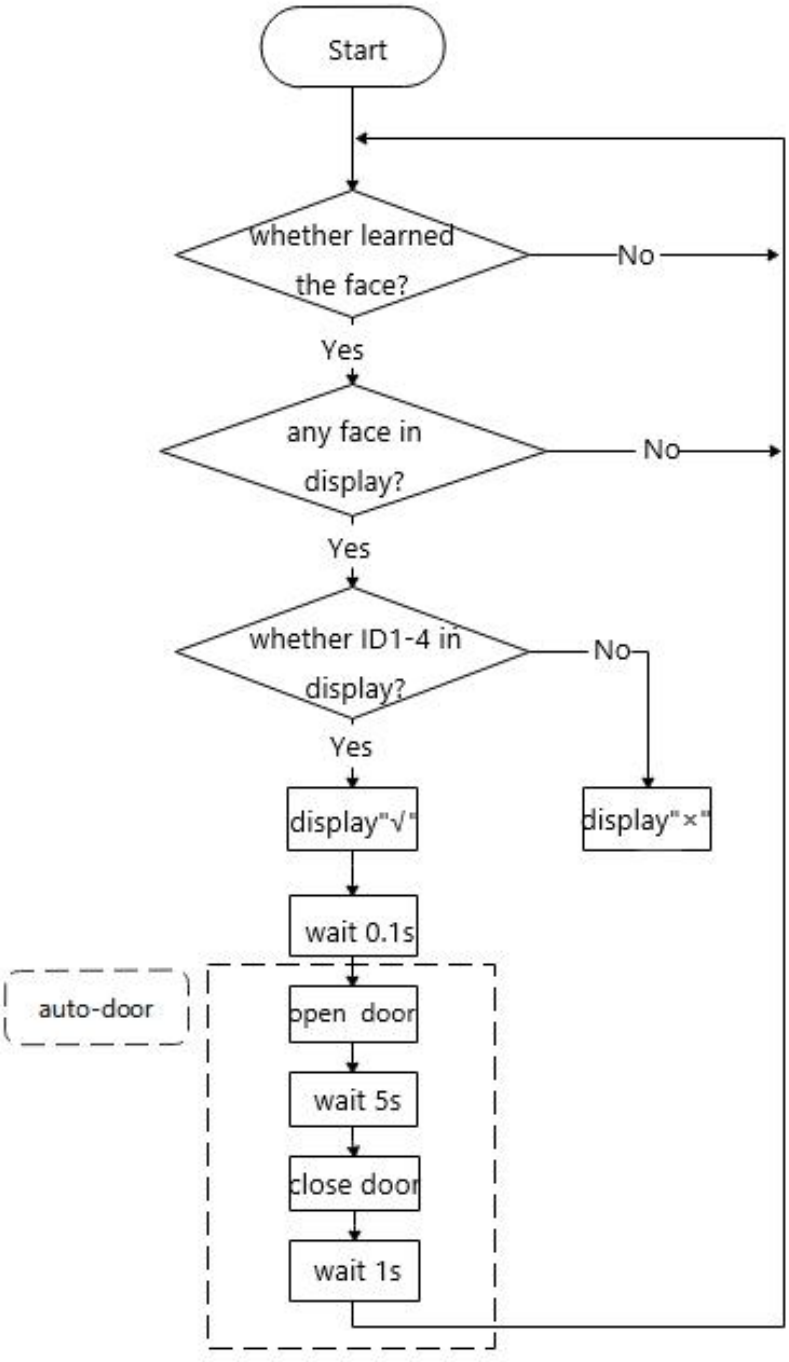


Program Design

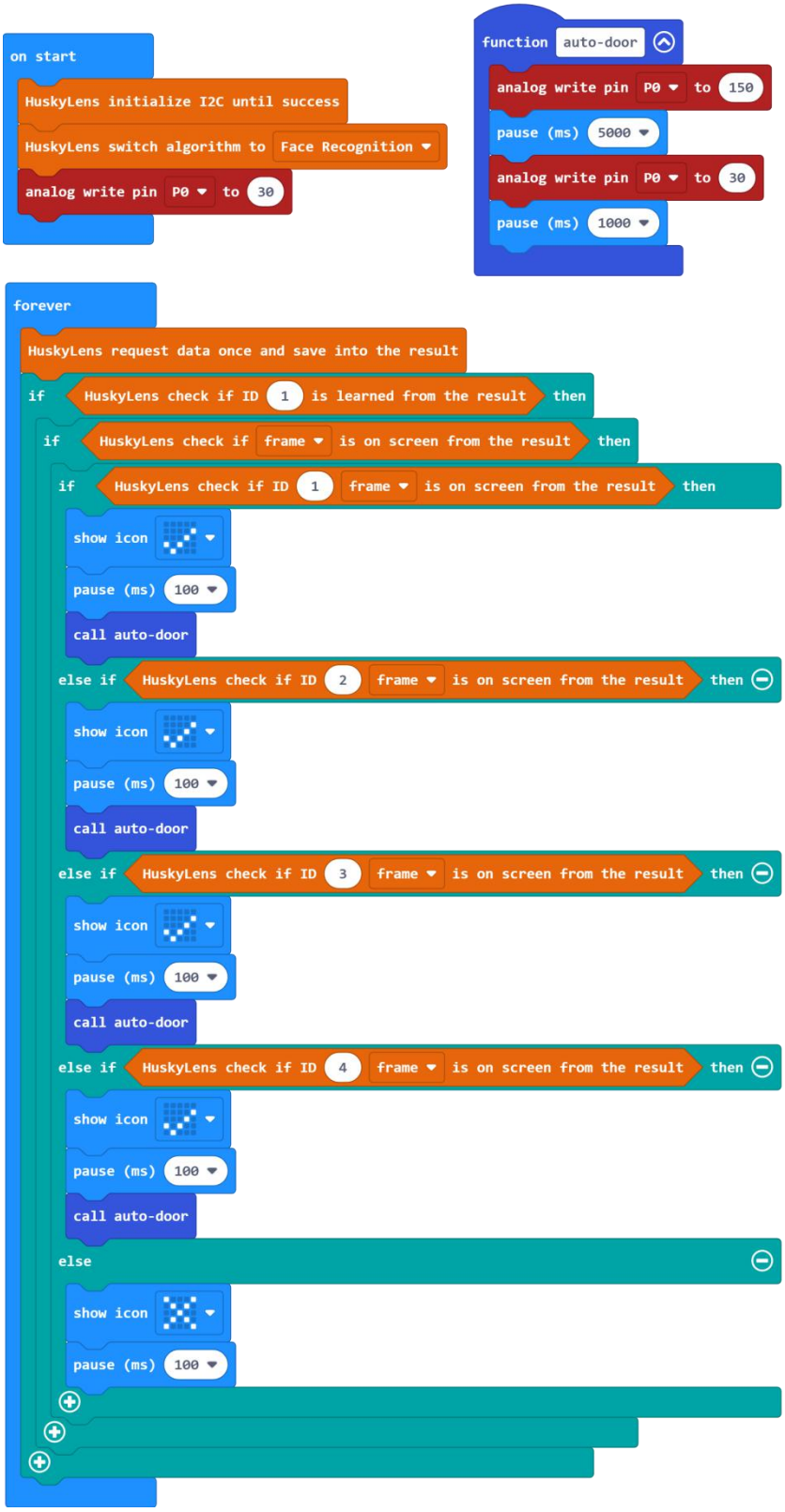
Function Instruction:

Next, add the actuator-the servo, and simulate the process of opening and closing the door by controlling the angle of the servo.

Flowchart Analysis:



The Sample Program:



Note: It is recommended to operate the servo within 10° -170° to avoid hardware damage.

Operating Effect

When HUSKYLENS detects a face belonging to the family, it displays "√" on the dot matrix screen, opens the door, and automatically closes the door after 5 seconds. When HUSKYLENS an unlearned stranger, "×" is displayed on the dot matrix screen, and the servo does not perform any operation.

Project Review

Through the face recognition access control project, we have further understood the principle of face recognition and the operation of HUSKYLENS to learn and recognize multiple faces.
Nowadays, face recognition systems can be seen everywhere in our lives, What problems can we solve by using HUSKYLENS's face recognition function in our daily life?

Project Development

Campus security needs to be consolidated from human defense, technical defense, and security defense. With the development of intelligent security, smart methods such as the application of face recognition systems have gradually been applied to campus security.
The application of face recognition on campus, on the one hand, can bring an intelligent experience to the teachers and students, on the other hand, it can prohibit strangers from entering and leaving at will, making our school safer.
Think about how to apply face recognition to campus management? To better save manpower and material resources, and make campus management more efficient and intelligent, which places should we put HUSKYLENS at?

Lesson 3 Colorful Piano

Have you ever had a musical dream? Imagine playing the piano as gracefully as a pianist. Piano, the "King of Musical Instruments", interprets perfect sound effects and romantic feelings with its smooth notes, bringing pure enjoyment to people. But for various reasons, maybe you have never learned piano or failed to own a piano.



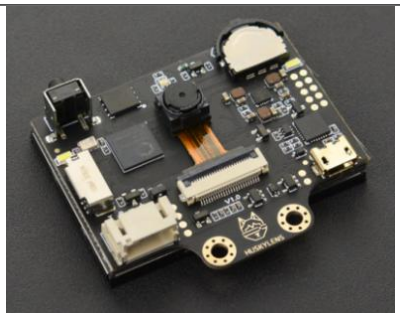
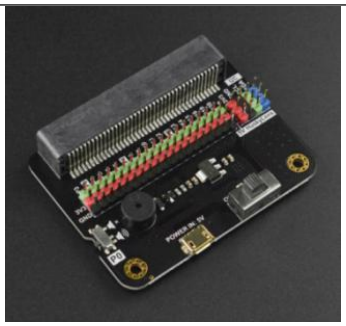
Now, with HUSKYLENS, we can make a color piano for ourselves, to realize the musical dream. Let’ s use the color keys to play graceful notes.

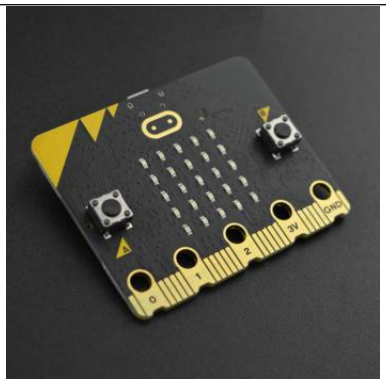
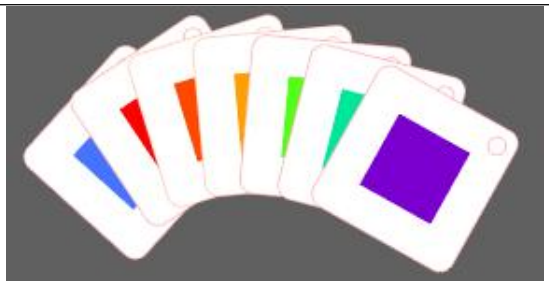
This project uses the color recognition function of HUSKYLENS to recognize different color keys and play different notes, so that your "playing" will be beautiful and pleasant, and also with absolutely wonderful stage effects.

Learning Objectives

1. Learn the operating principles and application fields of color recognition.
2. Learn to use the color recognition function of HUSKYLENS.
3. Learn to make a color piano using HUSKYLENS.

Preparation

	
HUSKYLENS ×1	IO Extender for micro:bit V2.0 ×1

	
micro:bit V2 ×1	Color Papercard

Learning Content

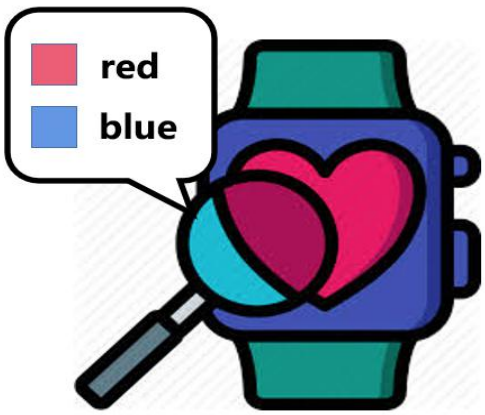
In today's society, automated production has become a social development trend. Machine vision, as the eyes of "robots", is particularly important.

As one of the important technical directions, color recognition has undergone multiple generations of technology upgrades. And this project is to use the color recognition function of the HUSKYLENS sensor to distinguish and recognize colors, and play the piano according to different colors.

Working principles and applications of color recognition

1. What is color recognition?

What is color recognition? First of all, we must understand what color is.

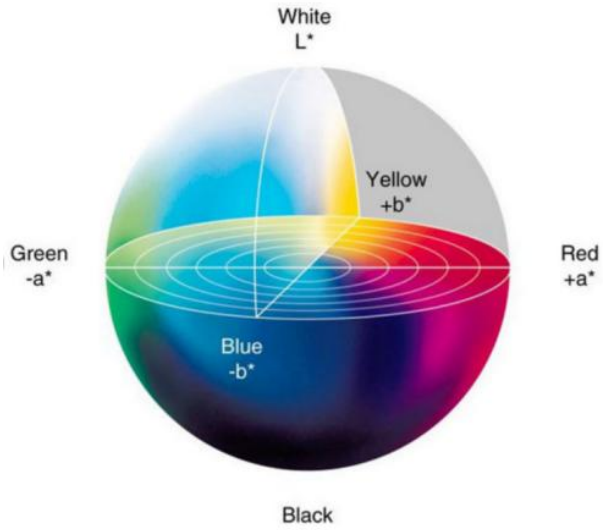


Color is a visual effect on light produced by our eyes, brain, and our life experience. The light we see with the naked eye is caused by a very narrow wavelength range produced by electromagnetic waves.

The electromagnetic waves of different wavelengths appear in different colors. Color recognition is based on the color attributes under different brightness to recognize and distinguish.

2. The principle of color recognition

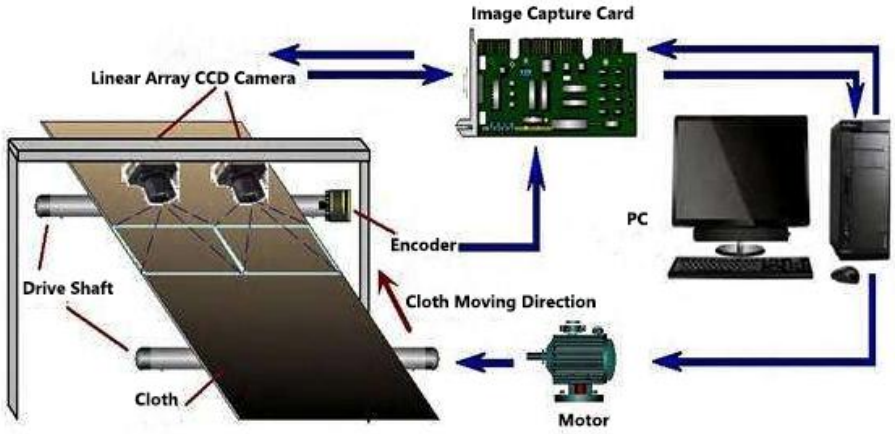
Color recognition is based on Lab color space, with dimension L representing brightness, a and b representing opposite dimensions of color, which is based on the CIE XYZ color space coordinates of nonlinear compression. We can regard these three parameters of Lab as the XYZ of the three-dimensional coordinate system. Compare the Lab parameters of the recognized and learned colors. When the two colors match within a certain error range, they are judged to be the same color.



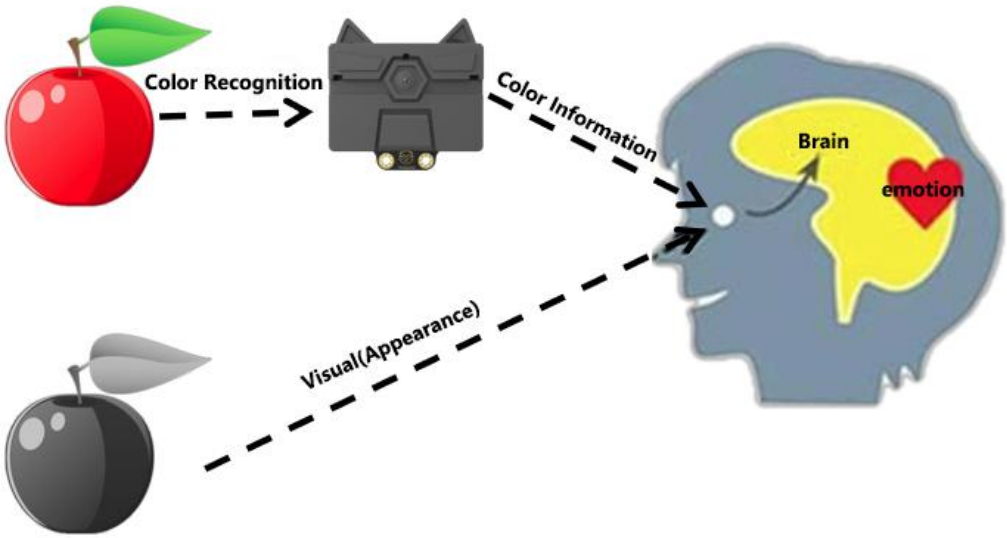
In our usual use of color recognition, the hue and saturation in the color attributes of the same module are fixed, but the brightness will change due to the influence of the ambient brightness, so you must try to ensure that when using the HUSKYLENS color recognition function, the environment brightness during learning and recognition is consistent with that during actual work.

3. Application Scenario of Color Recognition

1) Color recognition is now widely used in industries such as printing, coatings, and textiles for color monitoring and calibration.



2) Help people with color weakness or visual impairment to recognize and enhance their understanding of color.



HUSKYLENS Color Recognition Demonstration

If you want a color piano to play smoothly, first you should let the HUSKYLENS sensor learn the color of the keys, and let it know the notes corresponding to each color. The color recognition function in the HUSKYLENS sensor uses the sensor's built-in algorithm to identify the ID of different colors and feed them back to the main control board by learning and recording different colors.

The HUSKYLENS sensor is set by default to learn, recognize and track only one color, but there is not only one key on the piano, so we need to set it to recognize multiple colors.

1. Operational Settings - Learn Multiple:

1. Dial the function button to the right or left until the word "Color Recognition" is displayed at the top of the screen.
2. Long press the function button to enter the parameter setting of the color recognition function.
3. Dial the function button until "Learn Multiple" is displayed, then short press the function button, and dial to the right to turn on the "Learn Multiple" switch, that is, the progress bar turns blue and the square icon on the progress bar moves to the right. Then short press the function button to confirm this parameter.



4. Dial the function button to the left until "Save & Return" shows. And the screen prompts "Do you want to save the parameters?" Select "Yes" in default, now short-press the function button to save the parameters and return automatically.

In this way, the multiple learning function is set up.

2. Learning and Recognition

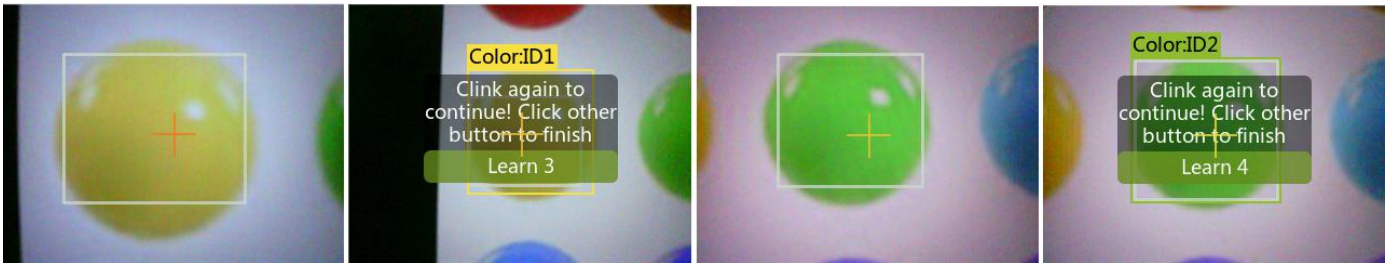
1）Color detecting

Point the icon "+" in the center of the HuskyLens screen to the target color block, and a white frame will appear on the screen, which selects the target color block automatically. Adjust the angle and distance of the HuskyLens to the color block so that the white frame can include the entire target color block as far as possible.



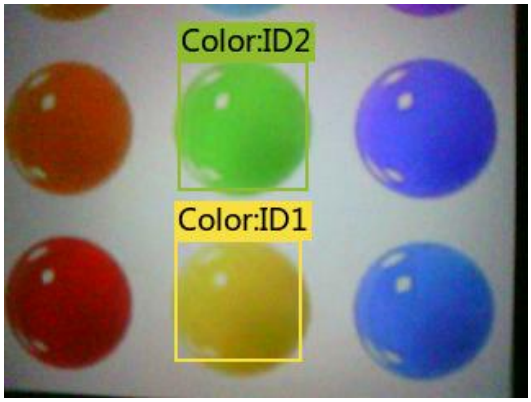
2）Color Learning

When detecting colors, press the “learning button” to learn the first color, and release it to finish learning. Meanwhile, a message “Click again to continue! Click other buttons to finish” will be displayed. Please short press the "learning button" before the countdown ends if you want to learn other colors. If not, short press the "function button" before the countdown ends, or do not press any button to let the countdown ends. The color ID displayed is consistent with the order of learning, that is, ID will be displayed as “ID1”, “ID2”, “ID3”, and so on, and different colors correspond to different colors of frames.

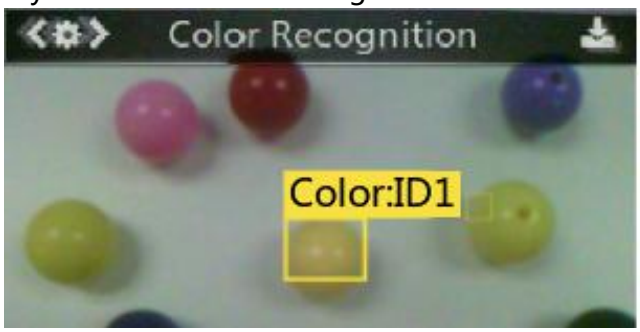


3）Color recognition

When encountering the same or similar color blocks, a color frame with an ID will be automatically displayed on the screen, and the size of the blue frame is the same as the size of the color blocks. Multiple colors can be recognized and tracked at the same time and different colors correspond to different colors of frames.



When multiple blocks of the same color appear, the separated color blocks cannot be recognized at the same time, and only one block can be recognized at one time.



Note: Color recognition is greatly affected by ambient light. Sometimes HuskyLens may misidentify similar colors. Please try to keep the ambient light unchanged. In the process of using color paper card, you can put a piece of white paper under the paper card to reduce the influence of ambient light on color recognition.

Project Practice

We will complete the task with two steps. First, we will learn to use the color recognition function of HuskyLens and the output of the recognized color ID. Then we can play their corresponding sound according to the output color ID, so that we can complete our color piano.

Task 1: Multiple Color Learning

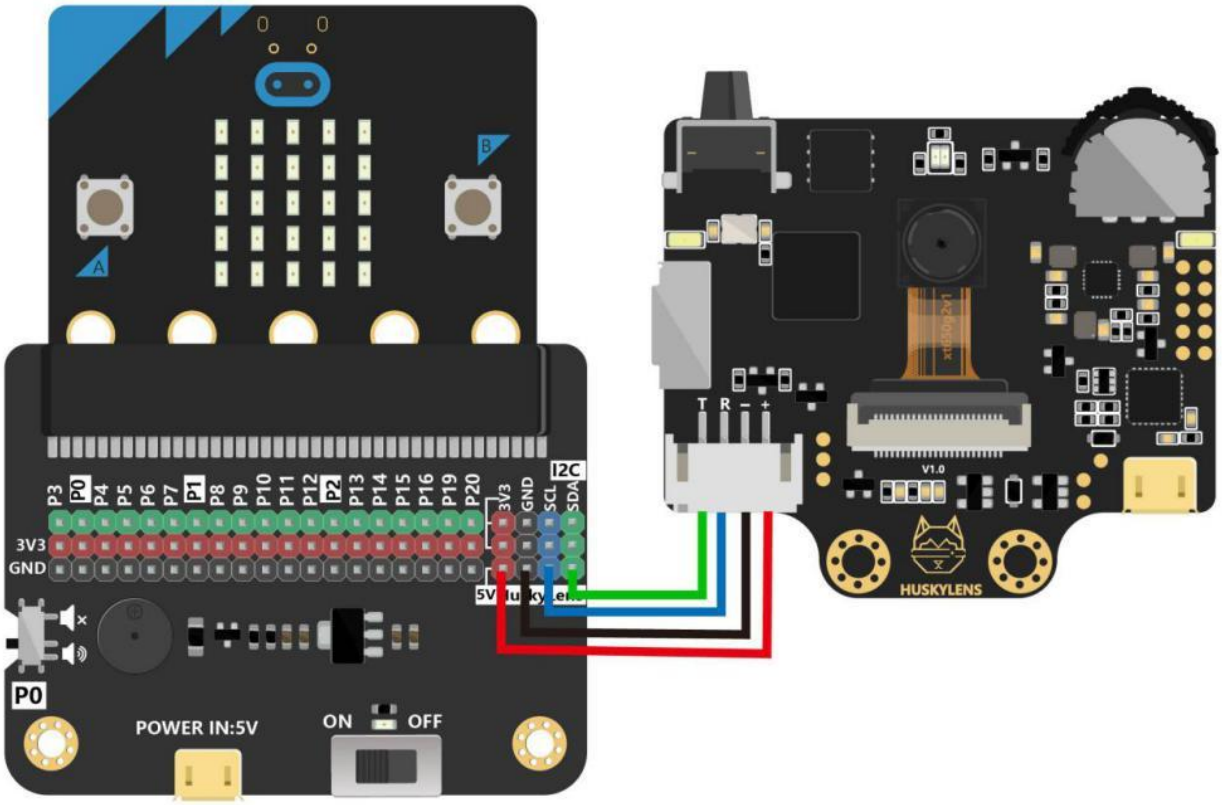
At the very beginning, we need to allow the HuskyLens camera to recognize multiple colors, distinguish the differences in these colors, and be able to give feedback so that we can add notes later.

Task 2: Define Notes for Each Color

After being able to accurately identify each color, we can define a sound for each color, so that they can be played according to a certain rule, so that a color piano can be realized.

Task1: Multiple Color Recognition

Hardware Connection



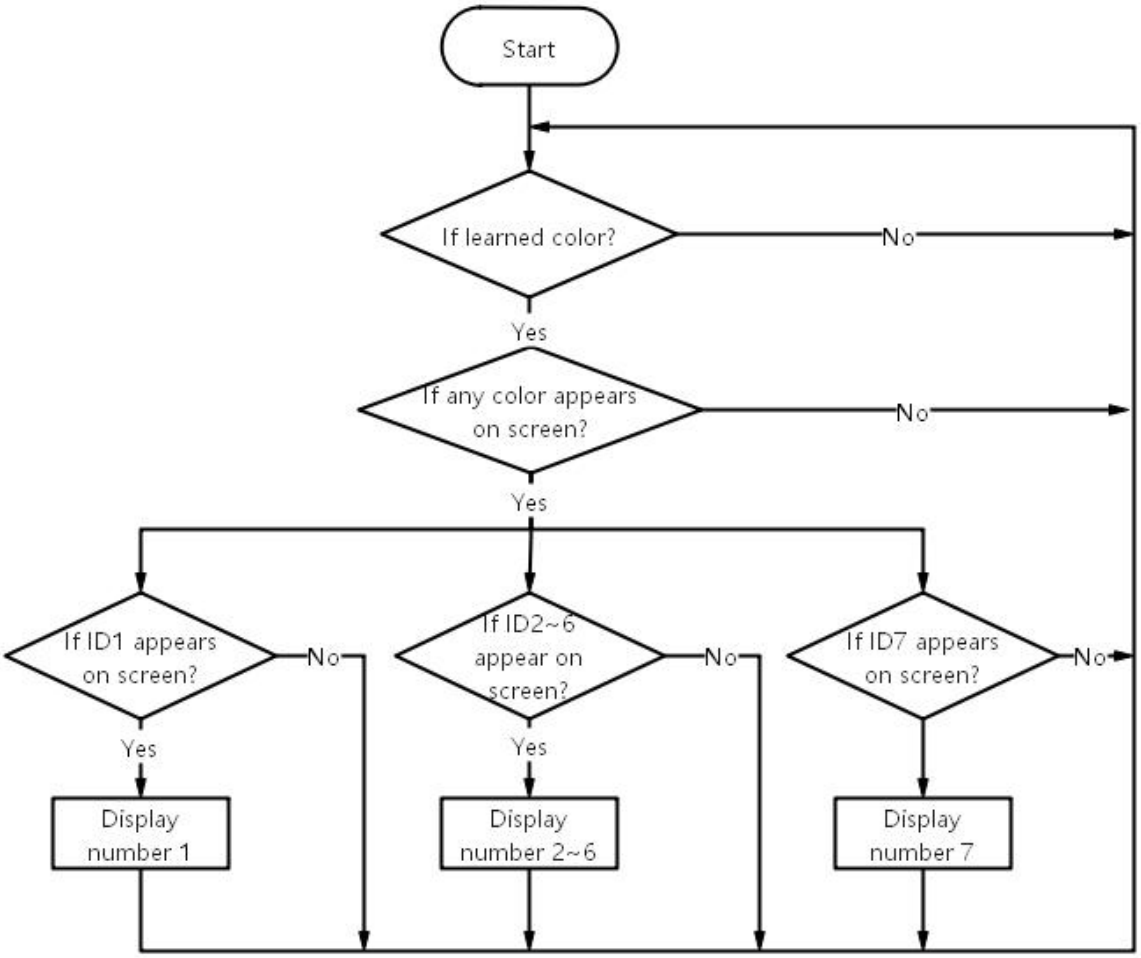
Program Design

Function Instruction:

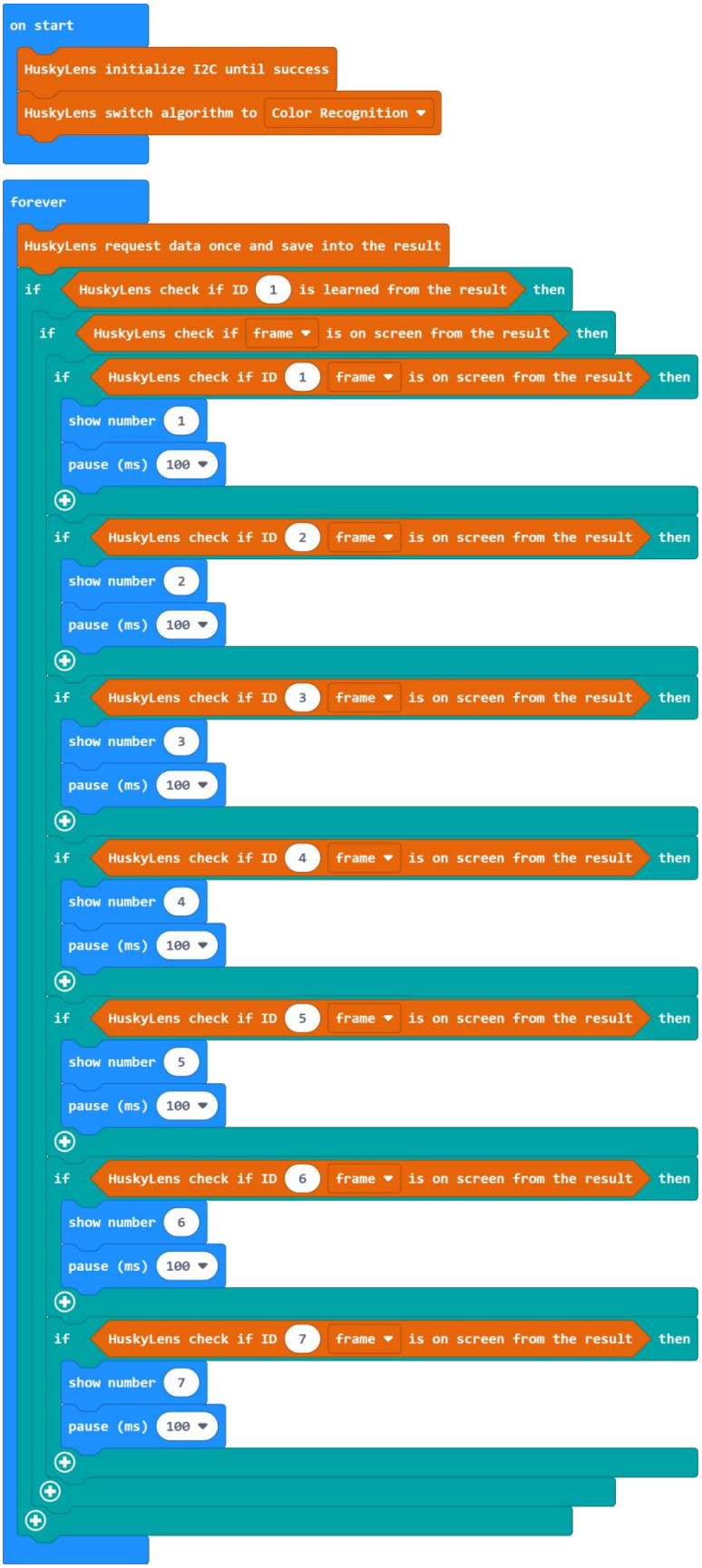
Here we need to let the HuskyLens sensor learn the color of each key and be able to output the color ID so that we can play the corresponding sound with the corresponding color. Such as: when it recognizes red (ID1), play the note Do.

Note: You need to enable the learning function first.

Flowchart Analysis:

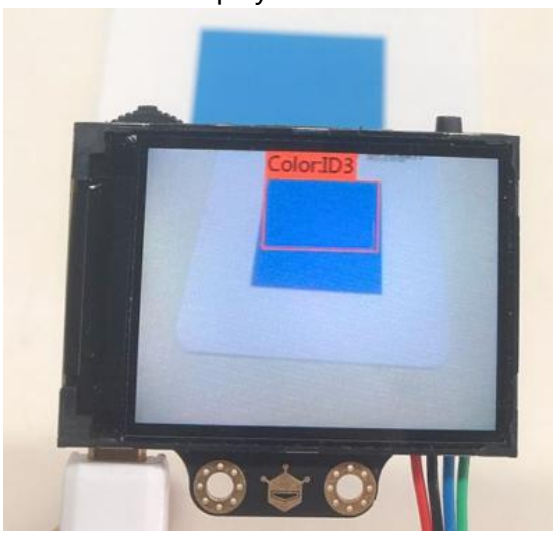


The Sample Program:



Operating Effect

When the ID of the corresponding color is recognized in the HUSKYLENS sensor, the corresponding number will be displayed on the micro:bit.





Task 2: Define Notes for Each Color

Program Design

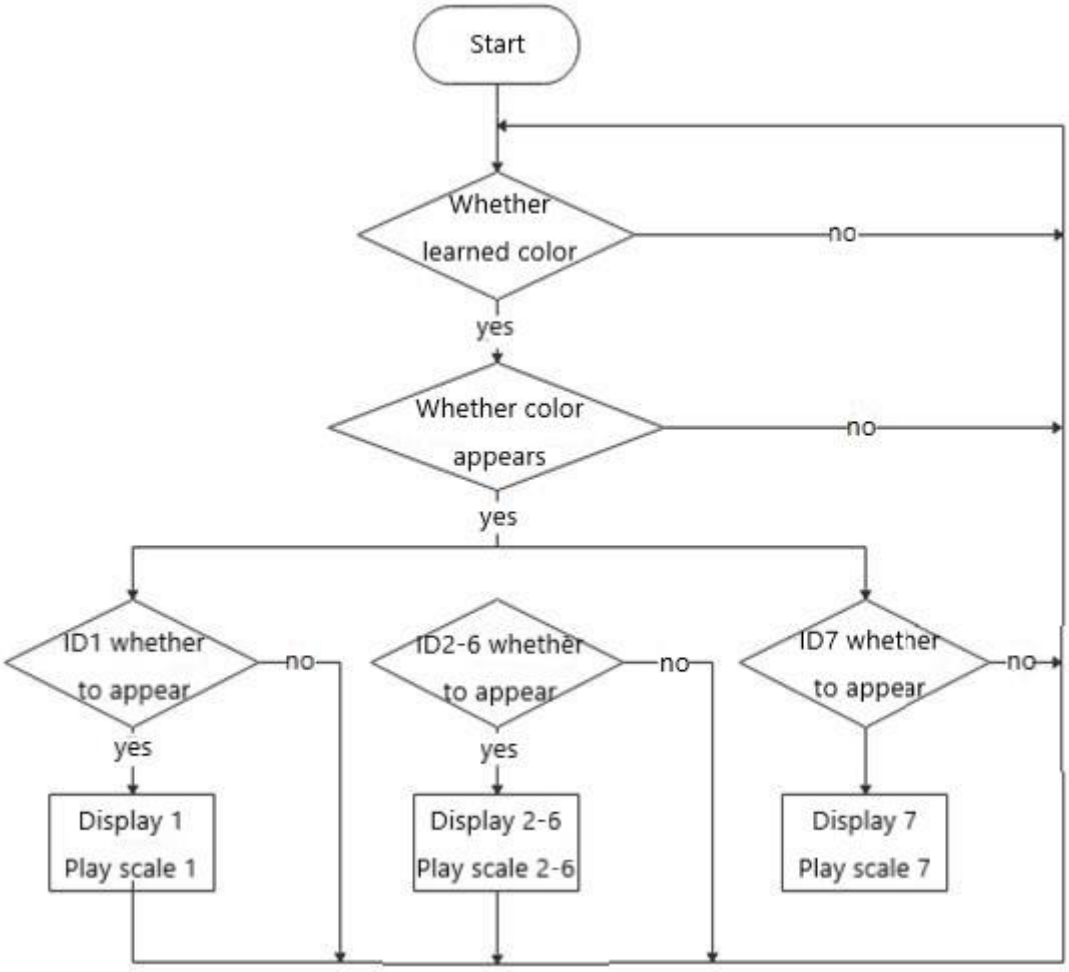
Function Instruction:

How to write a program? In Mind+, it comes with instructions for playing notes, which are divided into low, mid, and high notes, as well as various beats. We just need to add the corresponding note to the corresponding ID.



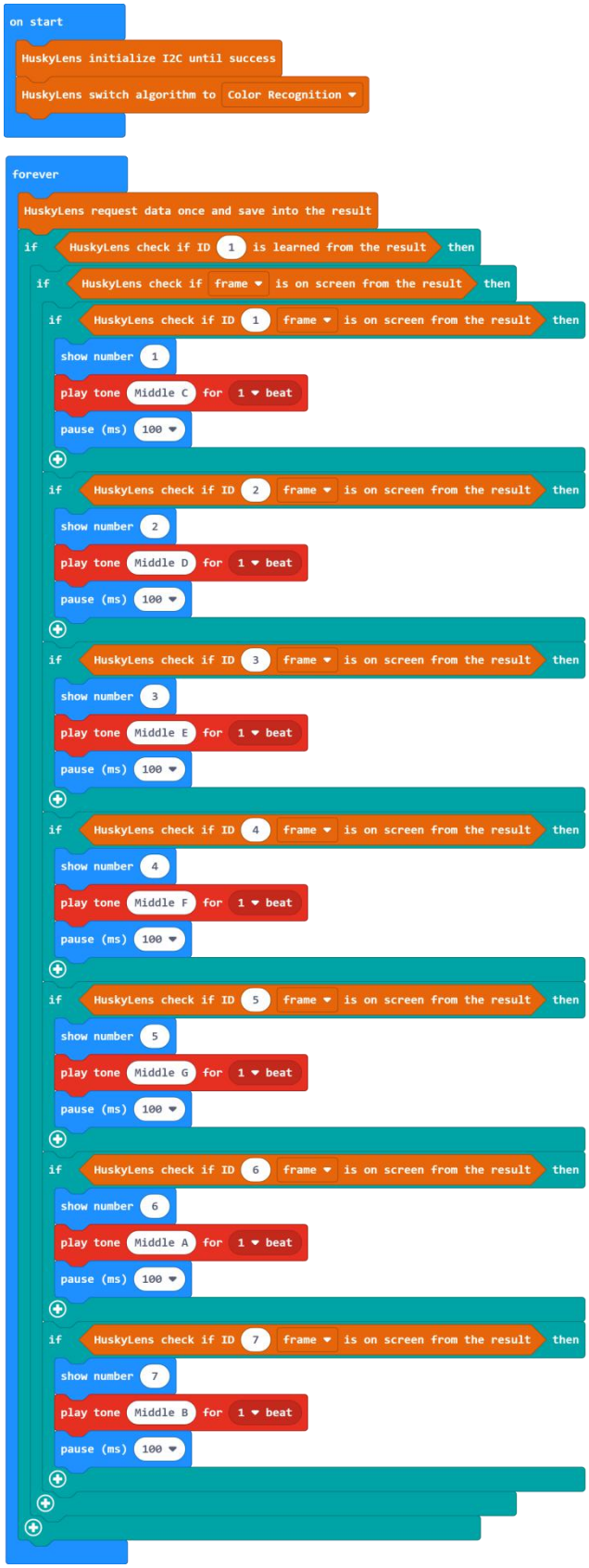


Flowchart Analysis:



The Sample Program

Add the note modules to the completed task 1 program.


on start
HuskyLens initialize I2C until success
HuskyLens switch algorithm to Color Recognition
forever
HuskyLens request data once and save into the result
if HuskyLens check if ID 1 is learned from the result then
if HuskyLens check if frame is on screen from the result then
if HuskyLens check if ID 1 frame is on screen from the result then
show number 1
play tone Middle C for 1 beat
pause (ms) 100
if HuskyLens check if ID 2 frame is on screen from the result then
show number 2
play tone Middle D for 1 beat
pause (ms) 100
if HuskyLens check if ID 3 frame is on screen from the result then
show number 3
play tone Middle E for 1 beat
pause (ms) 100
if HuskyLens check if ID 4 frame is on screen from the result then
show number 4
play tone Middle F for 1 beat
pause (ms) 100
if HuskyLens check if ID 5 frame is on screen from the result then
show number 5
play tone Middle G for 1 beat
pause (ms) 100
if HuskyLens check if ID 6 frame is on screen from the result then
show number 6
play tone Middle A for 1 beat
pause (ms) 100
if HuskyLens check if ID 7 frame is on screen from the result then
show number 7
play tone Middle B for 1 beat
pause (ms) 100

Operating Effect

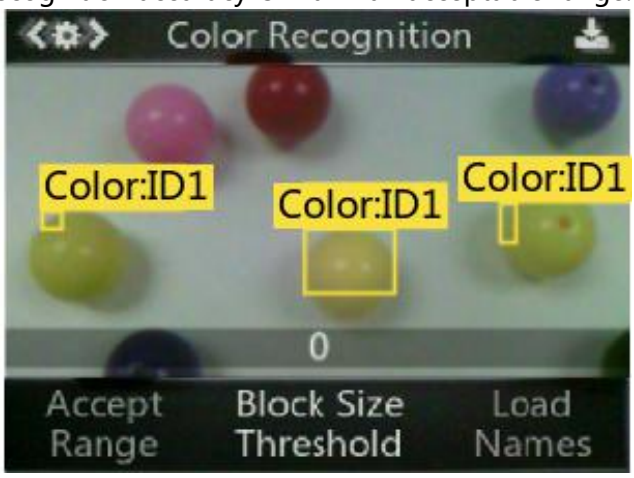
Run the program and find a music score, HUSKYLENS will play the corresponding notes according to the recognized colors.

Question 1: In the process, you must have discovered that when HUSKYLENS detects multiple colors, the notes produced by the color piano may not the notes we want. Do you have a good way to solve this problem?

Question 2: In the process, the color recognition will consider the similar color as the same color. How should we solve this problem?

Tip: Adjust the threshold of the recognition frame. In the submenu parameter setting interface of the color recognition function, there is a "**Recognition Frame Threshold**" parameter. The lower the value of this parameter, the lower the accuracy, but the more similar color blocks are recognized. As shown in the figure below, when the threshold is 20, only one yellow ball can be recognized; when the threshold is 0, all three yellow balls can be recognized. The threshold of the recognition frame can be adjusted according to the actual effect, so that the recognition accuracy is within an acceptable range.





Project Review

In this lesson, we learned about the working principle of color recognition, learned how to make a color piano, and learned the color recognition function.

Color recognition is a very important function in artificial intelligence visual recognition, and it has been widely used in the industry. Think about what other interesting functions can be achieved by color recognition?

Project Development

After completing the color piano, we will find a problem that the number of keys is relatively small. If we want to increase the number of keys, as the color increases, there will be many keys with similar colors, which may cause misidentification. If there are more keys, the camera's recognition range will not cover them all. So, is there any way we can broaden the range of our color piano?

(tips: We can use the A and B button on the micro:bit to realize the upgrade and downgrade functions.)

Lesson4: Vending Machine for Stray Cats & Dogs

We may often see abandoned cats or dogs wandering around the city and searching for food in trash bins. As animal lovers, we feel sad that these little guys are forced to suffer for whatever reason. All animals deserve to be loved and nurtured. Can we do anything for them?





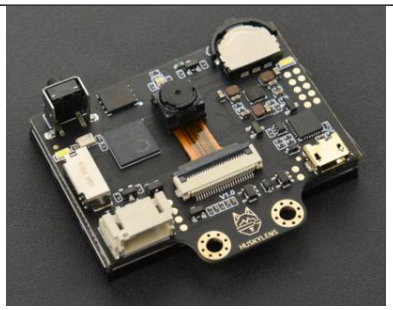
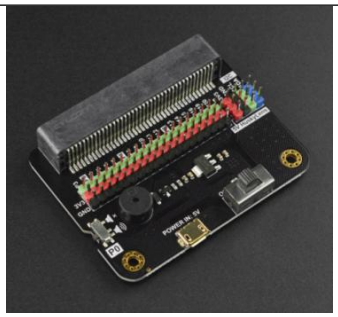
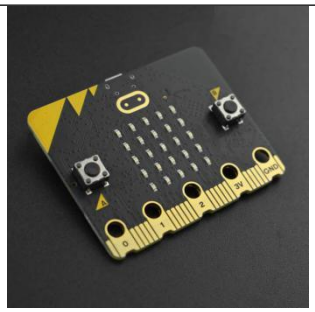
If there is a feeding machine that determines whether it is a cat or a puppy in front and feed it with corresponding food, just like a vending machine in stray animal world, wouldn’t it be great? Let’s use HuskyLens to make one!

In this project, we will use object recognition function of HUSKYLENS to distinguish cats and dogs through machine learning. Then a micro:bit will be used to process the result and control the servo to open the valve, and deliver corresponding food to cats and dogs.

Learning Objectives

1. Learn the working principle and application field of object recognition.
2. Learn to use the object recognition function.
3. Use HuskyLens to make a vending machine for stray animals

Preparation

		
HUSKYLENS ×1	IO Extender for micro:bit V2.0 ×1	micro:bit v2 ×1
		
DF9GMS 180° Micro Servo ×1	Animal paper cards	

Learning Content

The object recognition function of HUSKYLENS belongs to image recognition. Image recognition technology is an important field of artificial intelligence. It refers to the ability of a computer powered camera to identify and detect objects or features in a digital image or video. In this project we will use the image recognition of HUSKYLENS to distinguish and recognize cats and dogs.

Working Principle and Application Field of Image Recognition

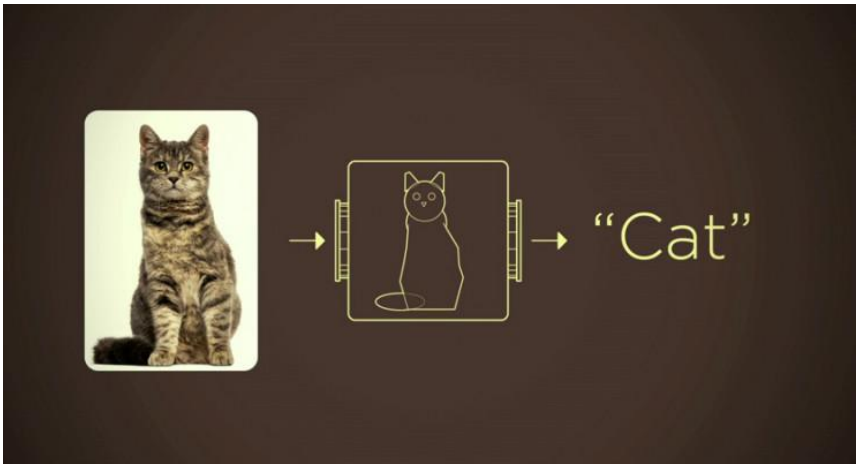
1. What is image recognition

Image recognition, a practical application of deep learning algorithm, refers to the processing, analysis, and understanding of images by computer so as to recognize targets and objects in different modes. It is divided into face recognition and product recognition currently. The former is mainly used in security checks, identification, and mobile payment, while the latter can be usually found in the process

of commodity circulation, especially unmanned shop, intelligent retail counter, and other unmanned sales fields.

2. Working principle

Four steps for traditional image recognition:



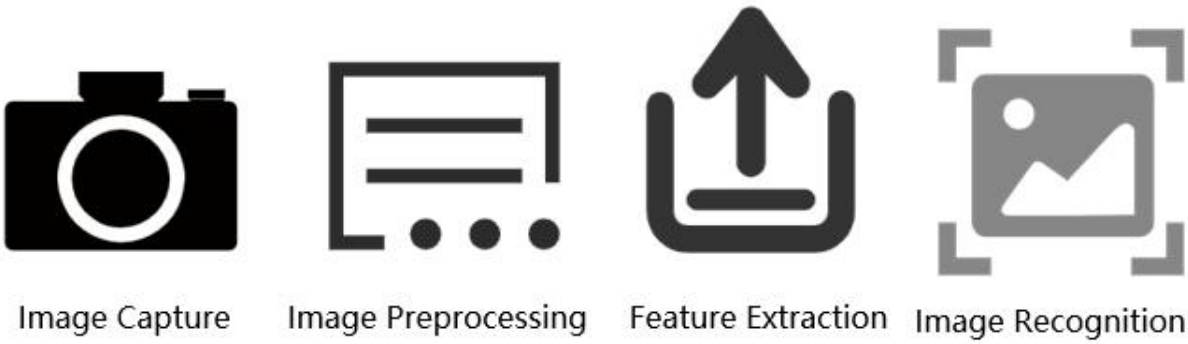
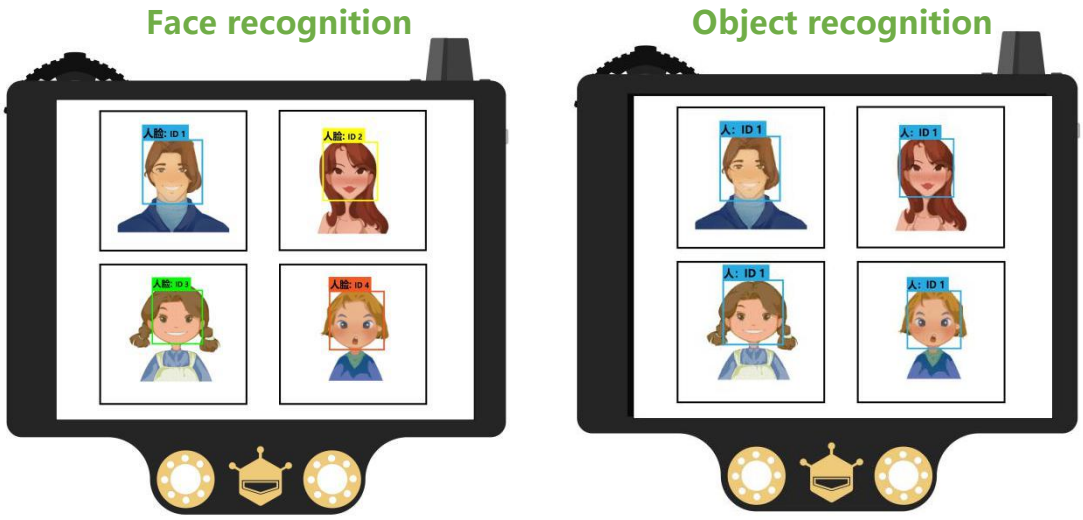


Image Capture: capture the image by the camera, and prepare for later recognition.
Image Preprocessing: analyze and process the images through a series of algorithms.
Feature Extraction: according to the information processed in the previous step, extract the key information, such as color, outline, etc.
Image Recognition: compare the information extracted with the sample base, the HUSKYLENS sensor image recognition includes a built-in sample library and it can be enriched by learning.

3. Similarities and differences between image recognition and other recognition:

We have already learned a lot of functions about camera recognition, such as face recognition and color recognition. What are the differences between them?
We can infer that face recognition, as one of the image recognition, is specifically used for distinguishing human faces. Imagine this scenario: when a group of people passes the camera, the name of each person can be "called out" if the information has been input in advance, while the

image recognition can only offer the result human, human, human.... because it can only recognize objects but not distinguish individuals.



We may find image recognition similar to object tracking. Both of them are function of recognition, but technically, object tracking can only learn and track one object, while image recognition can recognize multiple objects because object tracking learns an object from different angles so that accurate tracking can be achieved while image recognition only learns with only one side and recognition cannot be achieved once from another angle.

Color recognition and QR code recognition are easy to distinguish since they are both specific function-oriented.

4. Application Scenarios

Biomedicine:

Image recognition is widely used in modern medicine because of its explicitness, non-invasive, safe, and convenience, especially in clinical diagnosis and pathological research. For example, during the period of COVID-19, AI is deployed to quickly review the CT of the patients.

Remote Sensing Image Recognition

Aerial remote sensing and satellite remote sensing images are usually processed with image recognition to extract useful information. This technology is mainly used for terrain and geological exploration, forest, water, marine, agricultural and other resource surveys, disaster prediction, environmental pollution monitoring, meteorological satellite cloud image processing, and ground military target recognition.

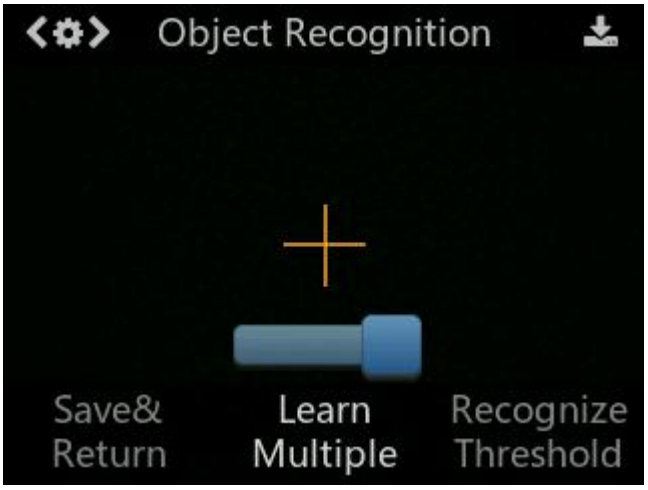
Demonstration of HUSKYLENS Object Recognition

This function is able to recognize what the object is and track it. 20 objects are supported: airplane, bicycle, bird, boat, bottle, bus, car, cat, chair, cow, dining table, dog, horse, motorbike, person, potted plant, sheep, sofa, train, television. The default setting is to frame and recognize one object. In this chapter, we are going to frame and recognize multiple objects as an example.

1. Operating Setting-Learn Multiple Objects

Dial the "function button" to the left or right until "face recognition" is displayed at the top of the screen. Long press the "function button" to enter the parameter setting interface of the submenu of object recognition.

Dial the "function button" to the left or right, select the "learn multiple". Then dial to the right and turn on the "Learn Multiple" switch, that is, progress bar turns blue and the square icon on the progress bar moves to the right. After that, short press the "function button" to confirm this parameter.



Dial the "function button" to the left and select "save & return", short press "function button". It will display "Do you want to save the parameters?", and "yes" is the default one. Short press the "function button", the data will be saved, and it will automatically return to the object recognition mode.

2. Learning and Recognition

1. Detecting Object

When detecting objects, HUSKYLENS will automatically recognize them, and the object will be displayed by the white frame with its name on the screen. At present, only 20 built-in objects can be recognized.

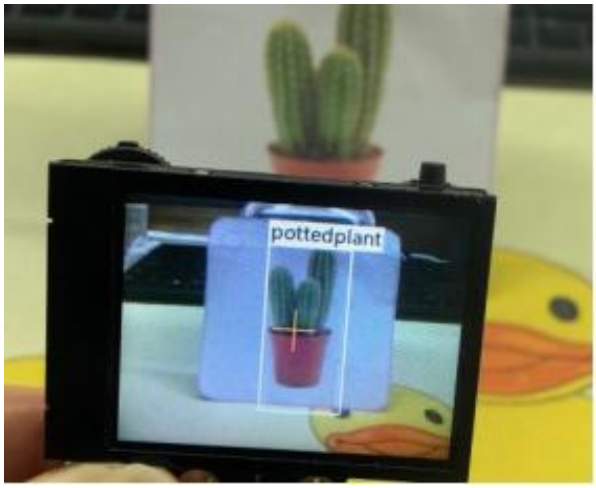




2. Marking Object

Point the “+” symbol at the object, then short press the “learning button”. When pressing, the color of the frame changes from white to blue, and the name of the object and its ID number will appear on the screen. There will be a notice:” Click again to continue! Click other buttons to finish”. Please short press the "learning button" before the countdown ends if you want to learn other objects. If not, short press the "function button" before the countdown ends, or just do not press any button to let the countdown ends.

The ID number is related to the order of marking objects. For example: the ID will be displayed as “ID1”, “ID2”, “ID3” in order, and different objects are matched with different colored frames.





3. Object Recognition

When encountering the learned objects, they will be selected by colorful frames, and the name and ID number will be displayed. It also supports simultaneous recognition of multiple types of objects, such as recognizing bottles and birds at the same time.





Tip: This function cannot distinguish the differences between objects of the same category. For example, it can only recognize that this is a cat, but cannot recognize what kind of cat it is.

Project Practice

We will complete the task in two steps. First, we will learn to use the image recognition function of HUSKYLENS and output the recognized results. Then, distribute the corresponding food according to the result.

Task 1: Distinguish cats and dogs

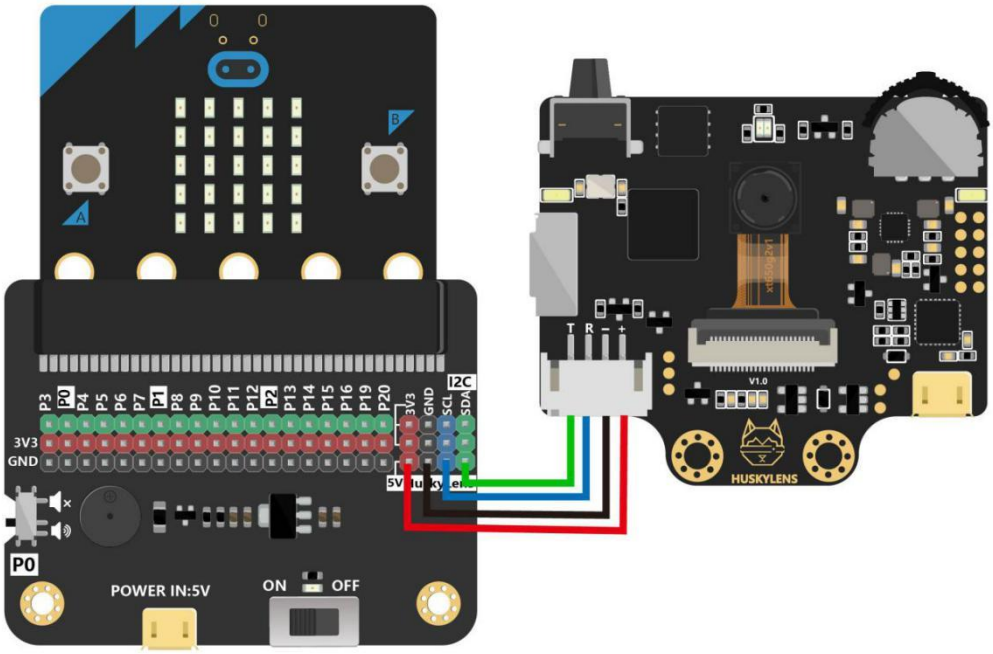
In this step, we need to enable the HUSKYLENS camera to recognize and distinguish cats and dogs and give feedback so that we can release the corresponding food in the next step.

Task 2: Add the function of dispensing food

In this step, you need to add the function of dispensing food based on the previous step, and make the corresponding structure.

Task1: Distinguish cats and dogs

Hardware Connection

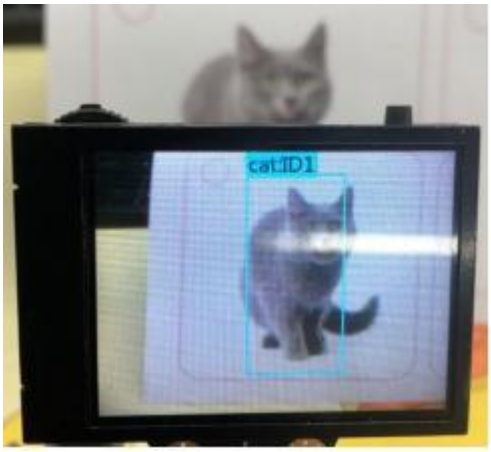


Program Design

Function instruction:

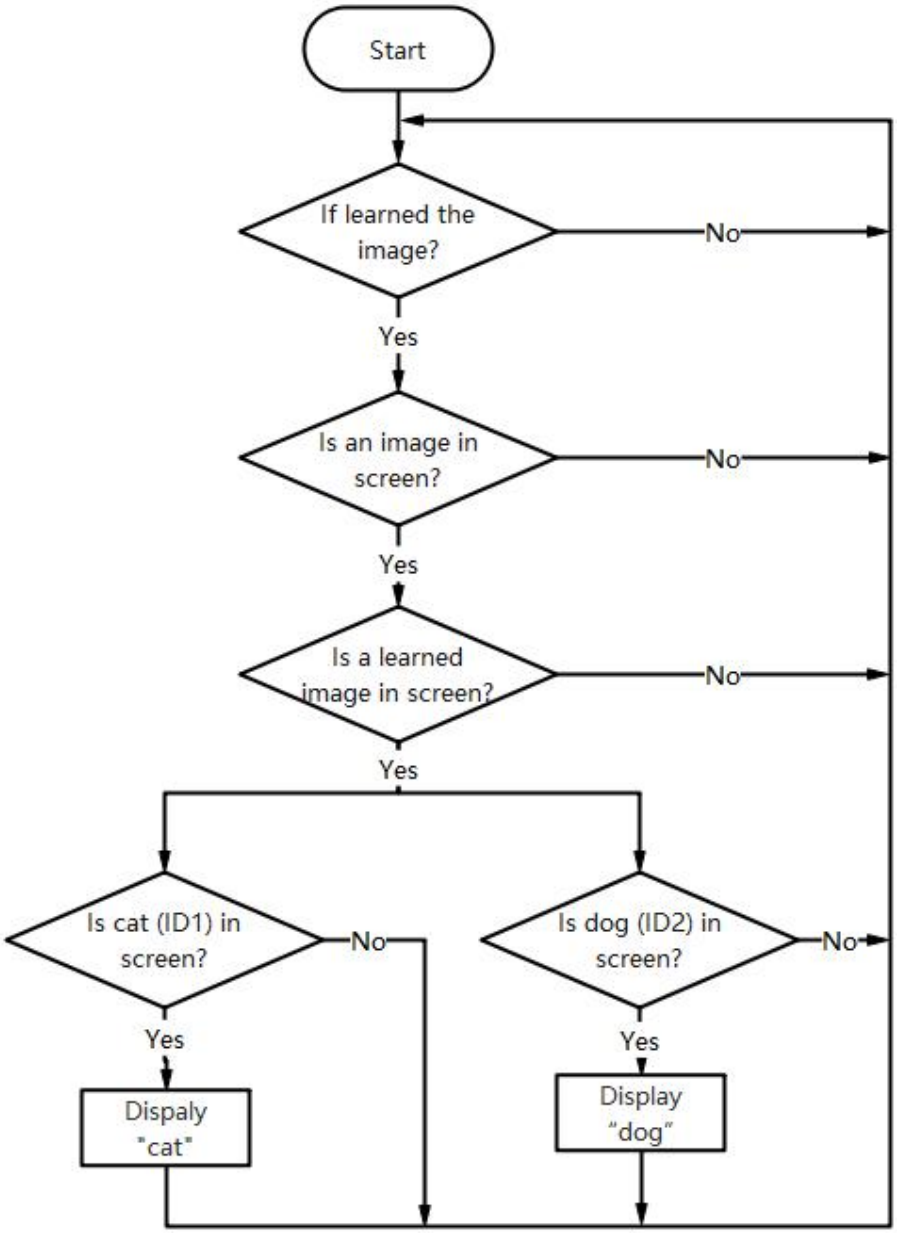
Here we need to let the HUSKYLENS sensor learn the images of cats and dogs, and output the result after recognition.

Tip: Turn on the Learn Multiple function in advance.

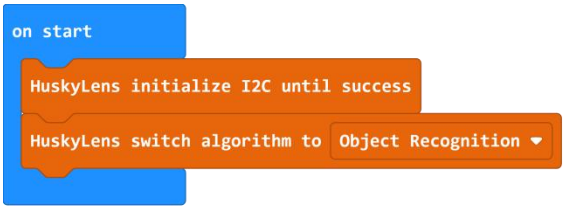


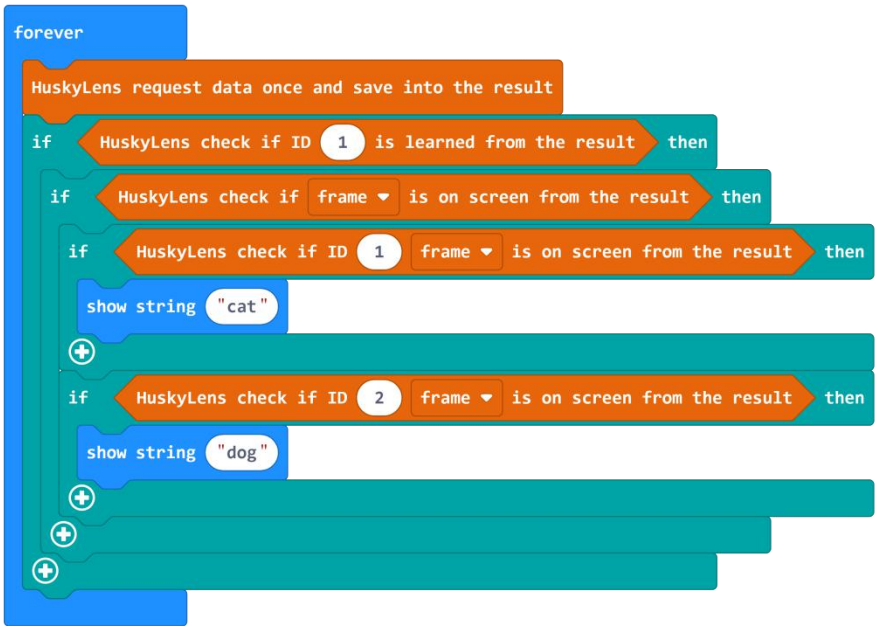


Flowchart Analysis



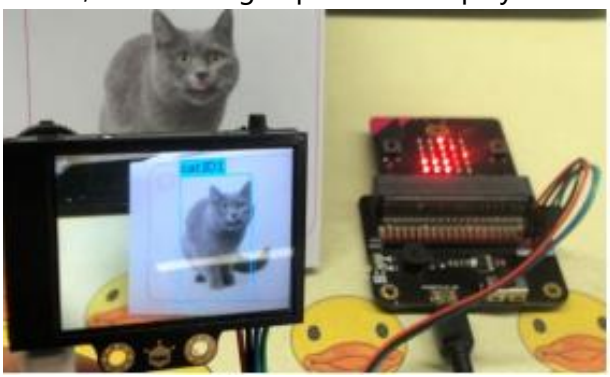
The Sample Program





Operating Effect

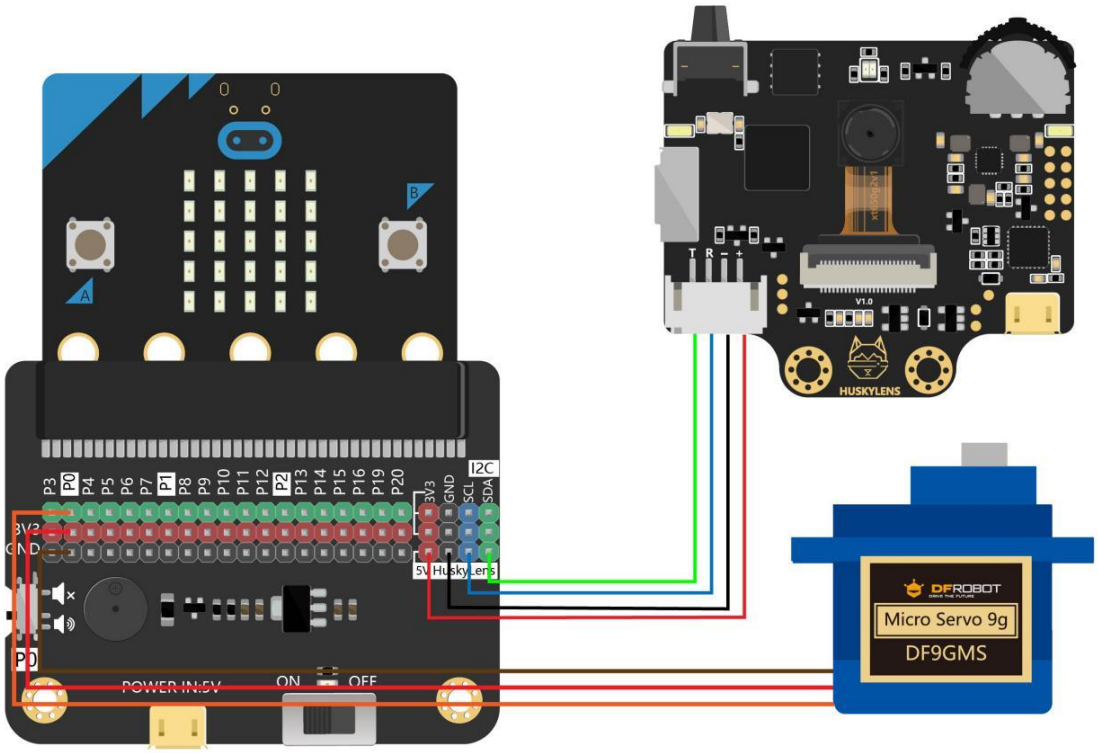
When a cat is recognized in the HUSKYLENS sensor, pattern for “cat” will be displayed on micro:bit matrix, while “dog” pattern is displayed for dogs.





Task2: Add the Function of Dispensing Food

Hardware Connection

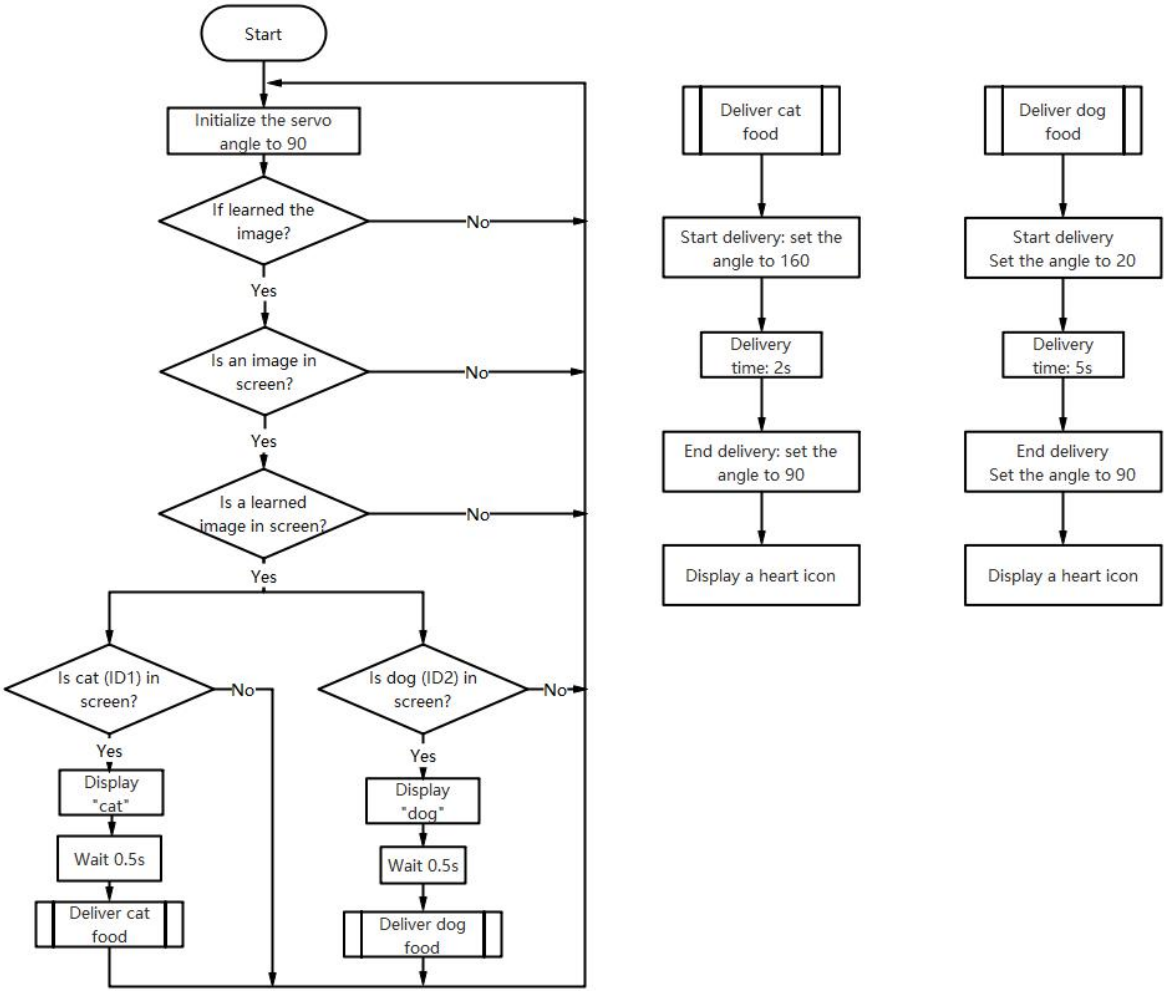


Program Design

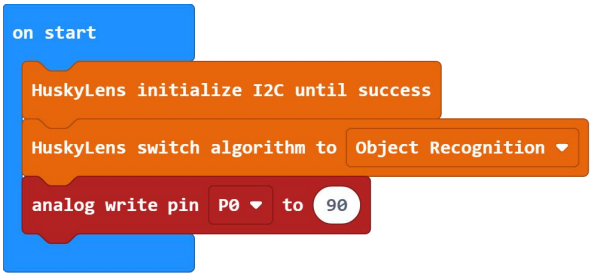
Function instruction:

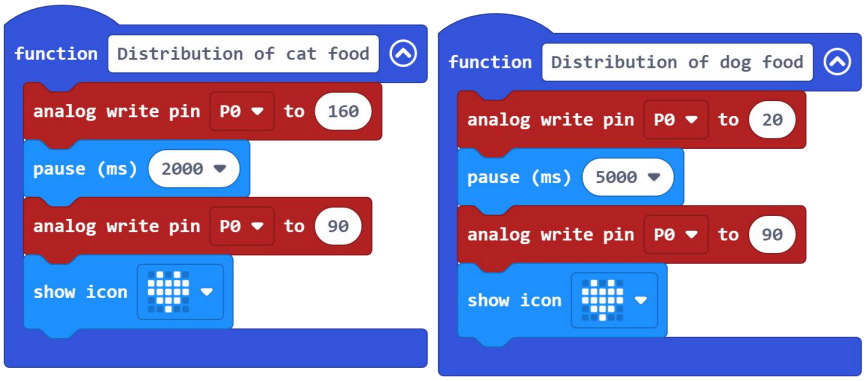
Through the rotation angle of the servo, it simulates the action of food delivery. Since cats and dogs have different food intakes, different delays need to be set to control the food dispensing.

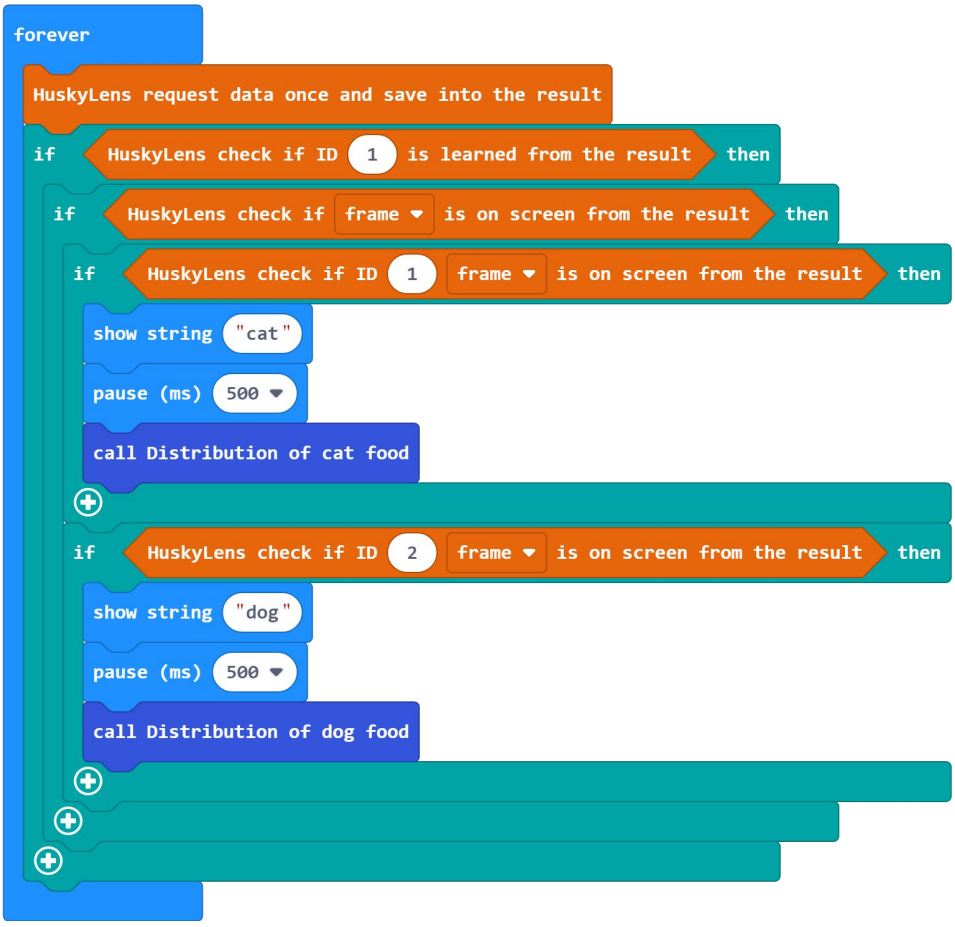
Flowchart Analysis



The Sample Program







Operating Effect

When a cat is recognized to be approaching, the micro:bit matrix screen will display “cat” pattern, and the cat food will be released. When completed, a heart image will be displayed. When a dog is recognized to be approaching, the matrix screen displays “dog” pattern and the dog food will be released. And also a heart will be displayed when done.

Question:

Although HUSKYLENS can distinguish between cats and dogs, the program can also perform corresponding food dispensing operations, but if some cats and dogs always appear within the recognizable range of HUSKYLENS, the automatic feeding machine will always release food. How can we solve this problem in the program?

Project Review

We use HUSKYLENS to complete a stray vending machine step by step. It is not only good news for stray animals, but also convenient for good-hearted people to add food independently, donate their love to help them.

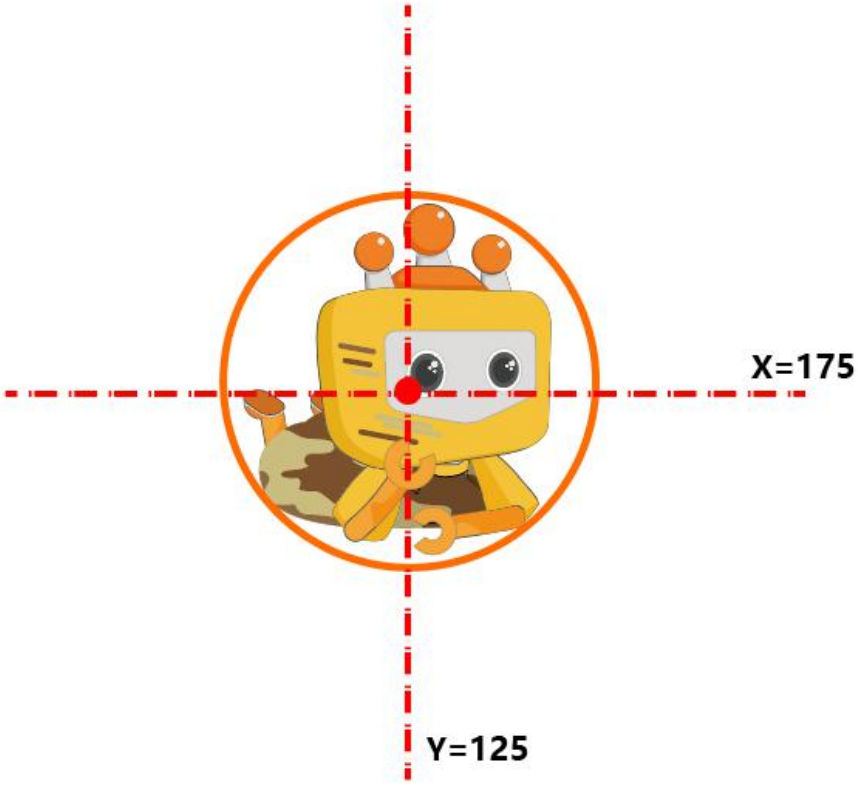
Project Development

After completing the vending machine, if you want to help more abandoned animals, can you add a statistical function to record the number of cats and dogs that come to the feeding machine to eat every day, so that the food can be rationed more reasonably.

Lesson 5: Object Tracker

Nowadays, shooting games are very popular. In the game, you need to judge the enemy's location and counterattack. The game requires a high degree of concentration and vigilance. The enemy's position is determined by "listening, watching, and observing ".

Can we make a machine that performs in-depth learning of the target? When the target appears in the visible range, it automatically tracks and feeds back the coordinates of the target; when the target disappears, it gives a notification, in this way, we don’ t have to worry about being attacked.

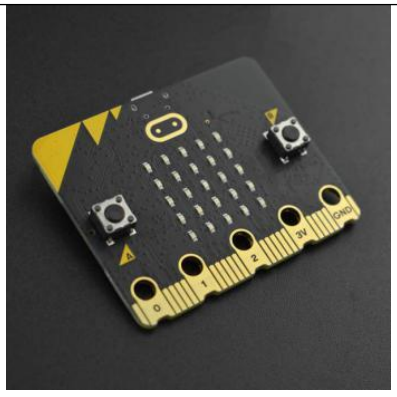


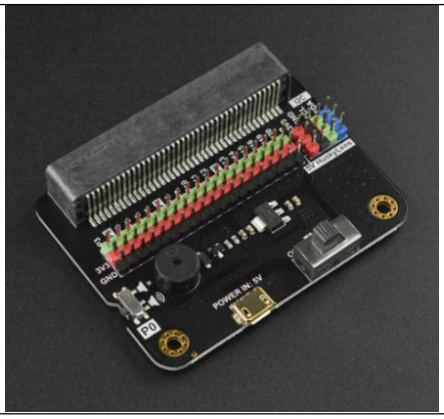
What is the relationship between HUSKYLENS and object tracking? This project mainly uses the object tracking function of HUSKYLENS to recognize and track the object through in-depth learning of the object (person) and feedback the position of the object to you in real-time. When the tracking object is lost, there will be a corresponding vibration notification.

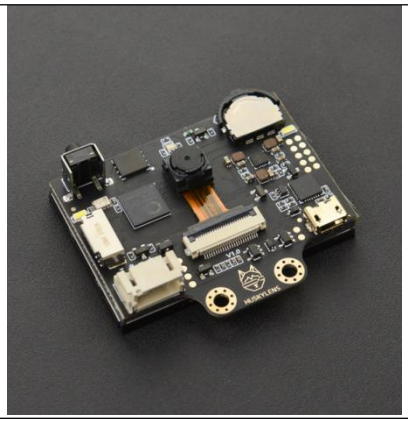
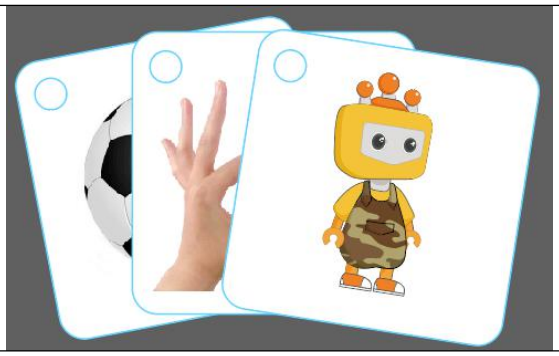
Learning Objectives

1. Learn the working principle and application field of object tracking.
2. Learn to use the object tracking function of HUSKYLENS.
3. Use HUSKYLENS to make with an object tracker.

Preparation





micro:bit V2 x 1	IO Extender for micro:bit V2.0 ×1
	
HUSKYLENS ×1	Micro vibration module ×1
	
Object tracking paper cards	

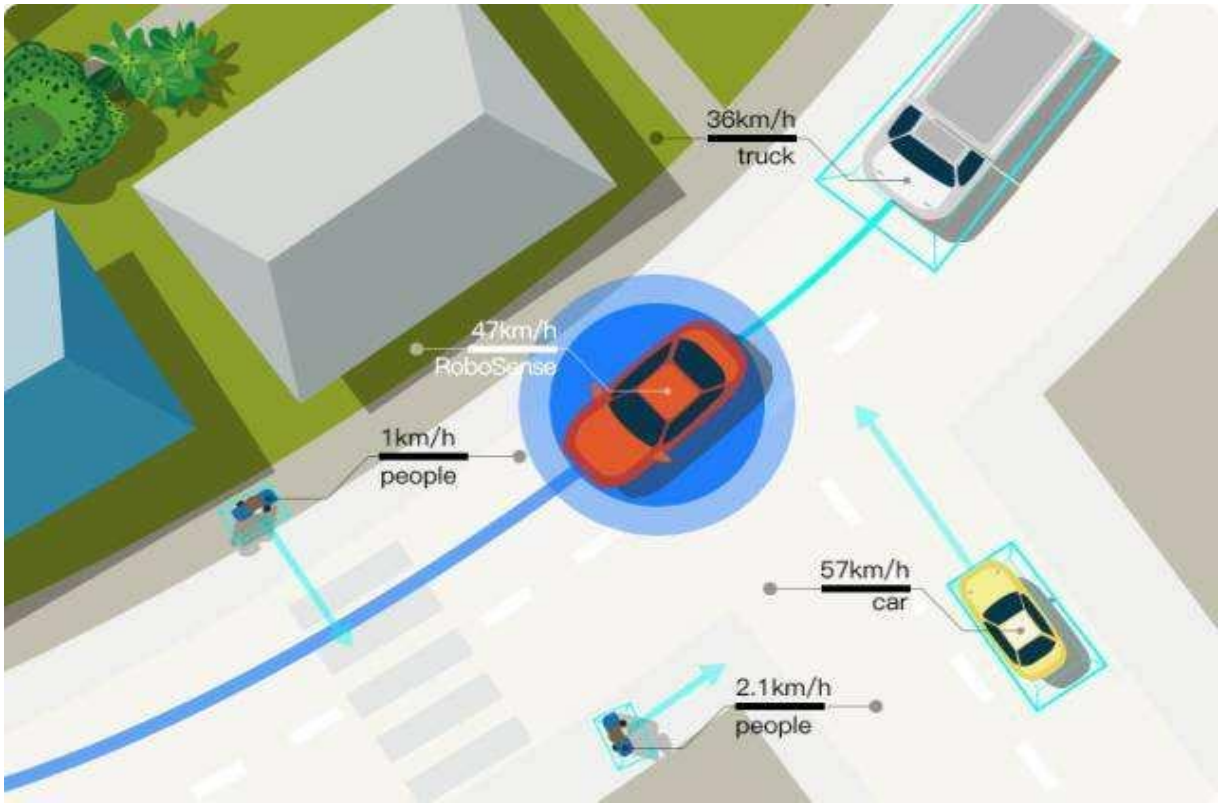
Learning Content

principle and application field of object tracking

When we are going to track an active object, visual object tracking is needed besides manual operation. This technology has already been widely used in our life, such as video monitoring, UAV following shot, etc. In this project, we make use of the object tracking of HUSKYLENS.

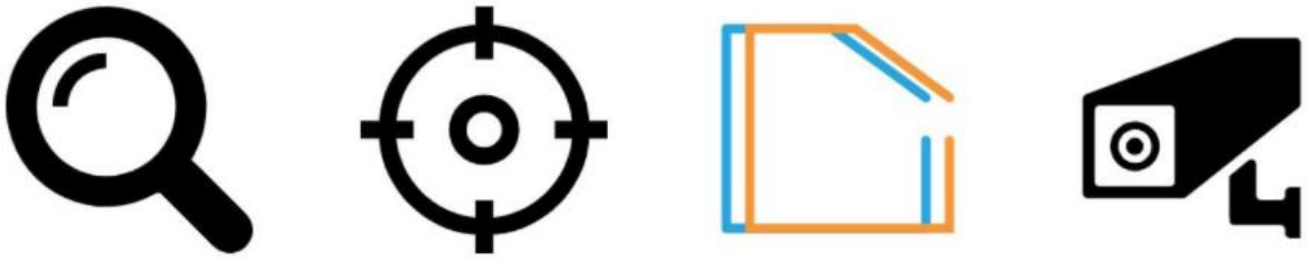
1. What is object tracking

As one of the vital functions of AI visual recognition, object tracking belongs to one type of object recognition. Object tracking is of vital importance in computer vision, referring to the process of making continuous inferences about target status in a video sequence, which can be simply regarded as recognizing and tracking the objects moving within the visual range of the camera.



2. Working principle of object tracking

The image information is collected by the camera and sent to the computer. After analysis and process, the computer can work out the relative position of the moving object. Meanwhile, it will make the camera rotate to carry out real-time tracking. The object tracking system includes four steps: object recognition, object tracking, movement prediction, camera controlling.



Object Recognition

Object recognition is to obtain accurate appearance information of the object through some image processing algorithms under a static background, then recognize and mark the shape of the object, as shown in the figure.



Object Tracking

Object tracking refers to tracking the subsequent image sequence through algorithms according to the appearance characteristics of the object obtained from the previous step, and carry out more in-depth learning in the subsequent tracking so as to make the tracking more and more accurate.

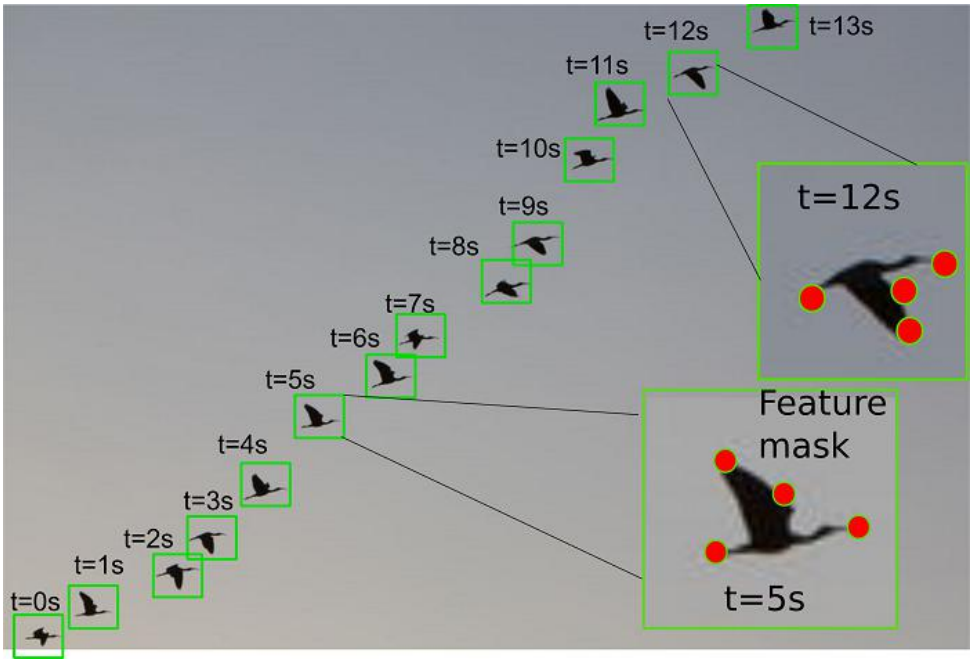






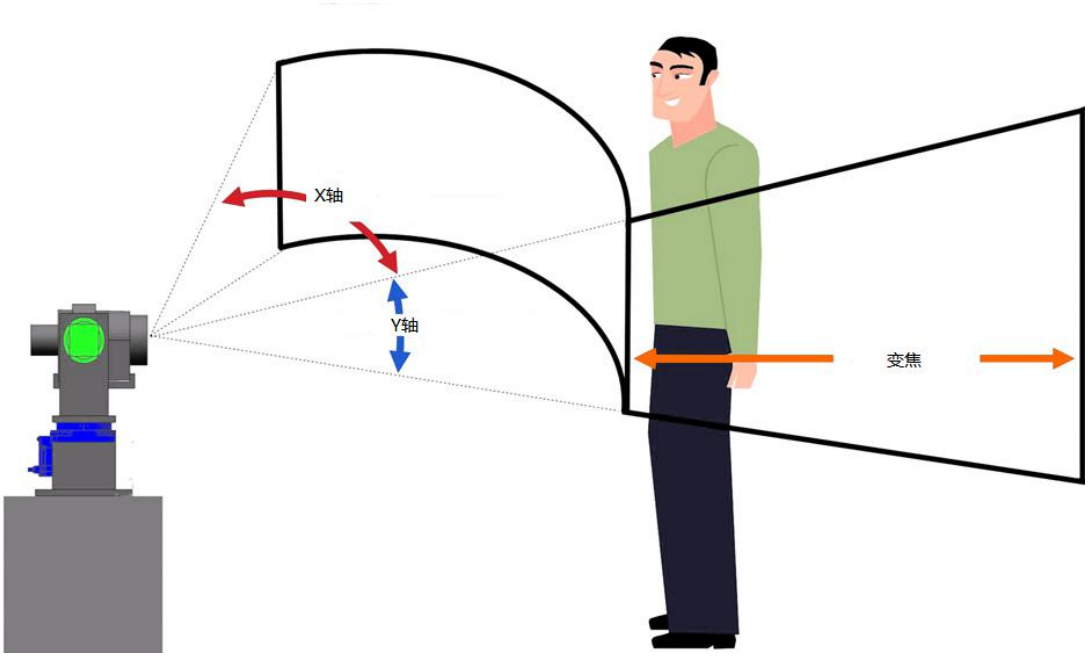
Object Motion Prediction

Motion prediction is using an algorithm to predict the image of a moving object in the next frame so that it can optimize the algorithm and improve efficiency. As the picture shows, the following movement path and action can be predicted by the bird’ s movement trend in the first few seconds.



Camera Controlling

Camera controlling is to move the camera according to the moving direction of the object while collecting the image information. It usually requires coordination with cloud platforms or other movement mechanisms.



3. Application field of object tracking

Smart Video Monitoring

Based on motion recognition (human recognition basing on footwork, automatic object detecting), automatic monitoring (monitor the suspicious acts), traffic monitoring (collecting the real-time traffic data to direct the traffic).



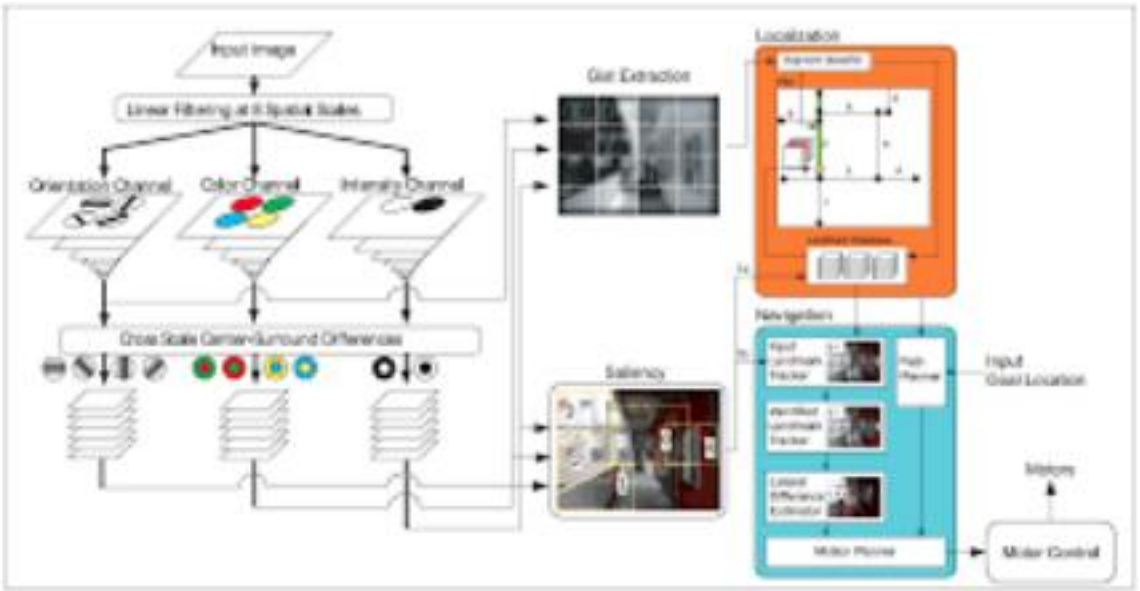
Human-computer Interaction

The traditional human-computer interaction is carried out by the keyboard and mouse of the computer. Tracking technology is the key point when a computer needs to be able to recognize and understand the posture, movement, and gesture.



Robot Visual Navigation

For smart robots, the tracking technology can be used to compute the motion trail of the object been shot.



VR

3D interaction and virtual character action simulation in the virtual environment directly benefit from the research results of video human motion analysis. They provide richer forms of interaction for the participants. And human tracking and analysis are the key technologies.



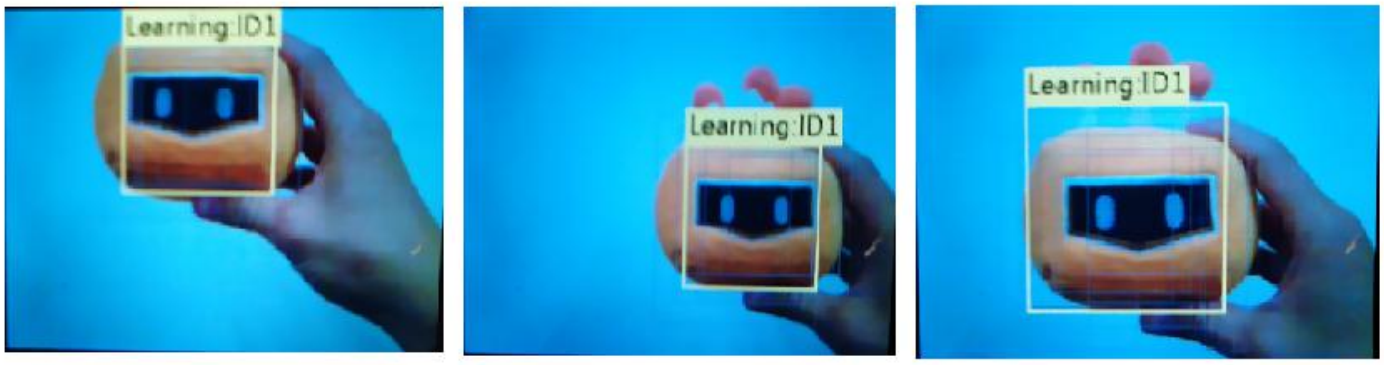
Demonstration of HuskyLens Object Tracking

The function is a built-in algorithm in the sensor, allowing it to learn the features of the object, track the position of an object on the screen and feedback the coordinate value of the position to main-controller. The vibration module is controlled by the obtained position value of the object to realize the tracking notification.

1. Object learning

Different from color recognition and face recognition, object tracking is able to completely learn and recognize an object (or human). Color recognition is only for color, while face recognition is only for a part of the human body. Object tracking is to learn the overall characteristics of the object to track.

Point HUSKYLENS to the target object, adjust the distance until the object is included in the yellow frame of the center of the screen. If the object is difficult to be completely contained in the frame, containing distinctive features is also okay. Then, long press the "learning button" to learn the object at various angles and distances. During the learning process, the frame with words the "Learning: ID1" will be displayed on the screen.



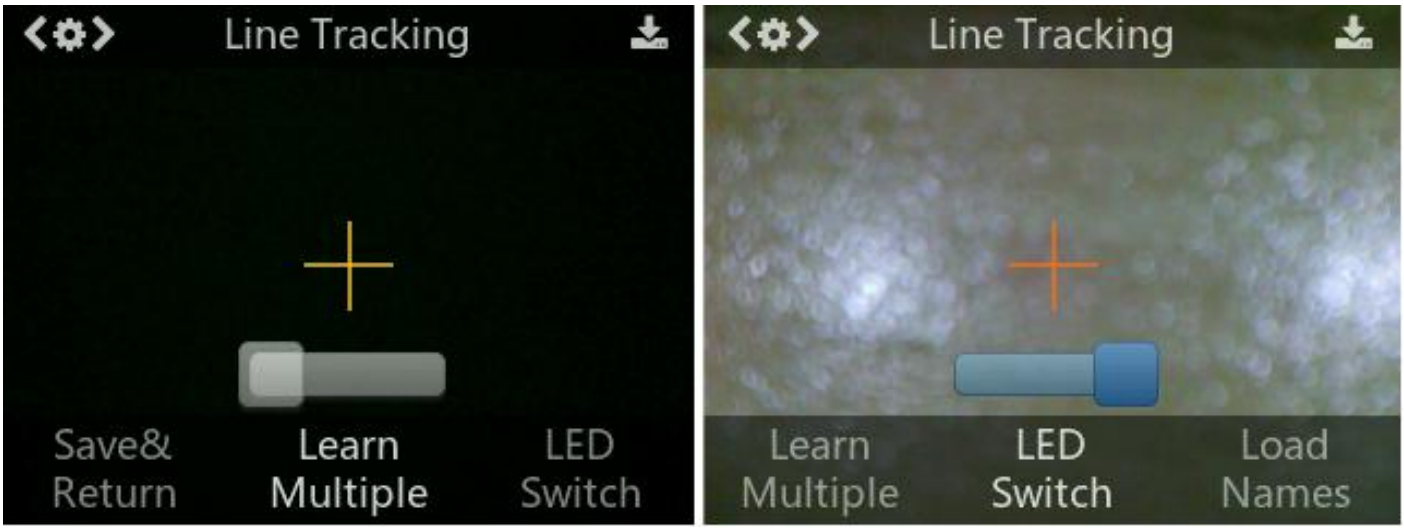
When HUSKYLENS can track the object at different angles and distances, you can release the "learning button" to end the learning.

Note: If there is no yellow frame in the center of the screen, it means that HUSKYLENS has learned an object before. Please let it forget the learned one and learn again.

2. Disable learning function

After the learning is completed, long press the "function button" to enter the parameter setting interface of the submenu of the object tracking function.

Dial the "function button" to the left or right, select "learning on", then short press the "function button", dial the "function button" to the left to turn off the "learning on" switch, that is: the progress bar turns white, the bar is located at the left side. Finally, save and return.



Project Practice

The project is divided into two tasks. First, learn to use the object tracking function of HUSKYLENS and read the coordinate data of the object. Then add a vibration module on this basis to achieve the final tracking notification function.

Task 1: Function of object tracking and the coordinate values

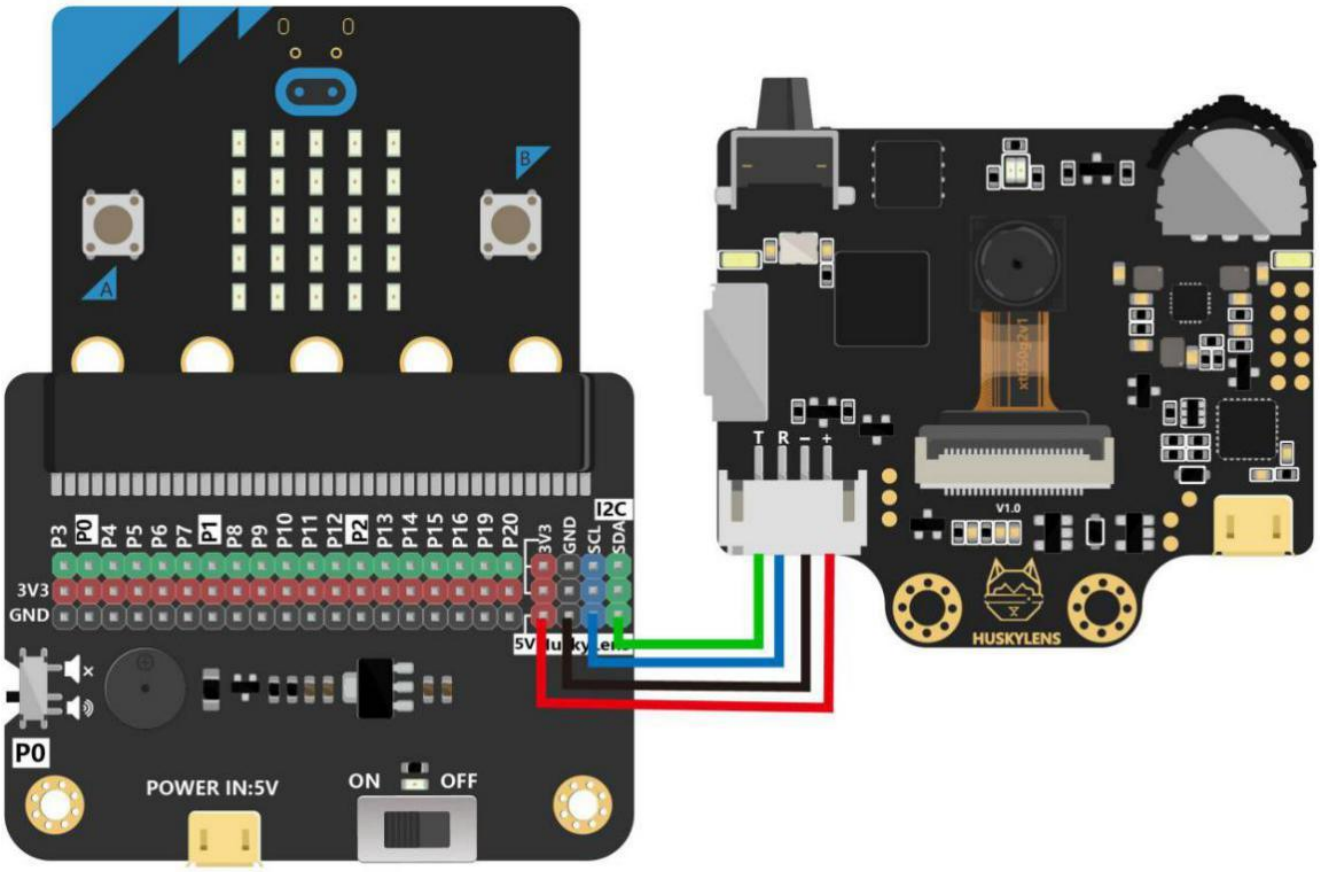
Obtain the coordinate value of the object on the screen through the HUSKYLENS sensor, and then judge the relative position of the object on the sensor through the coordinate value.

Task 2: Vibration feedback notification

When the HUSKYLENS sensor is tracking an object, if the target is lost, the vibration module is used to give a vibration feedback notification.

Task1: Function of Object Tracking and Coordinate Values

Hardware connection



Program Design

Function instruction

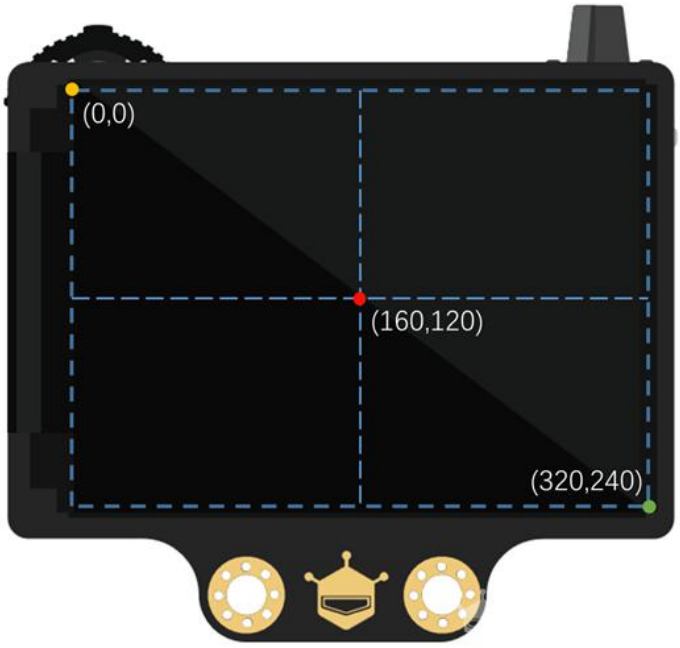
Before starting, let HUSKYLENS learn the target to be tracked——Mind+ in camouflage.



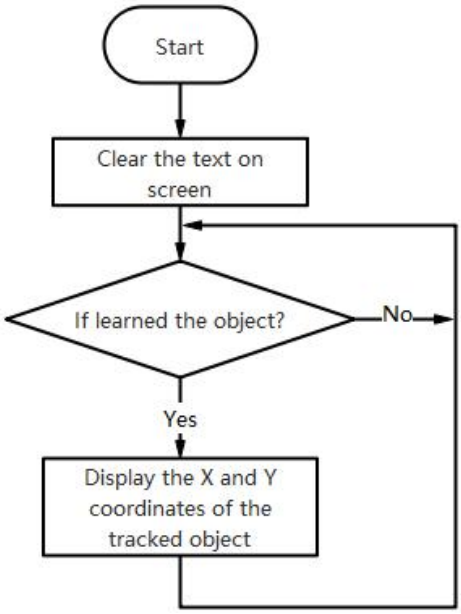


Coordinate analysis

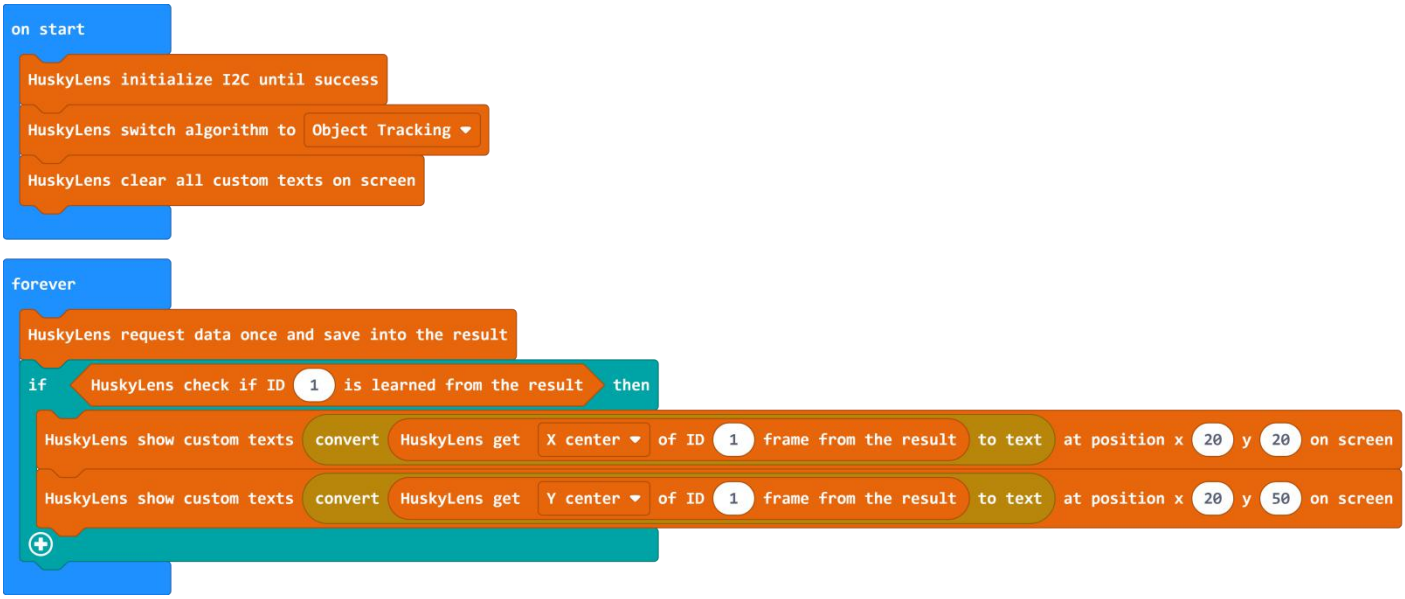
The screen resolution of HUSKYLENS is 320*240, as shown in the figure, the center coordinates of the object that need to be obtained through the program are also within this range. For example, if the obtained coordinate value is (160, 120), the tracked object is at the center of the screen.



Flowchart Analysis



Sample program

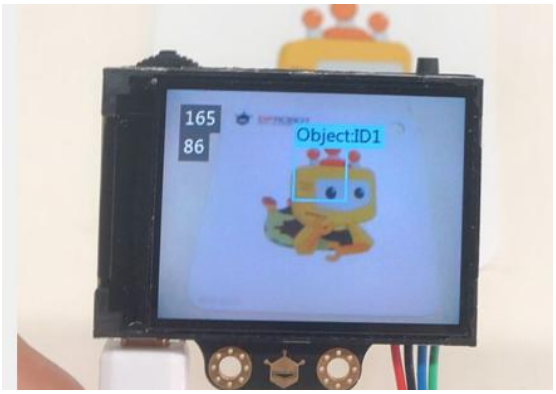


Operating Effect

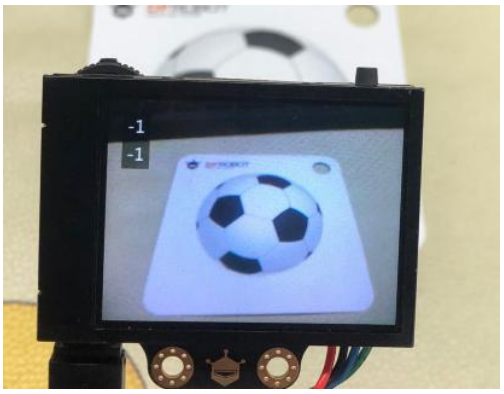
Run the program, HUSKYLENS will track the learned objects and display the coordinates of the objects on the HUSKYLENS screen.

Note: If -1 and -1 are displayed on the screen, the tracked object is lost.

Display X and Y coordinates

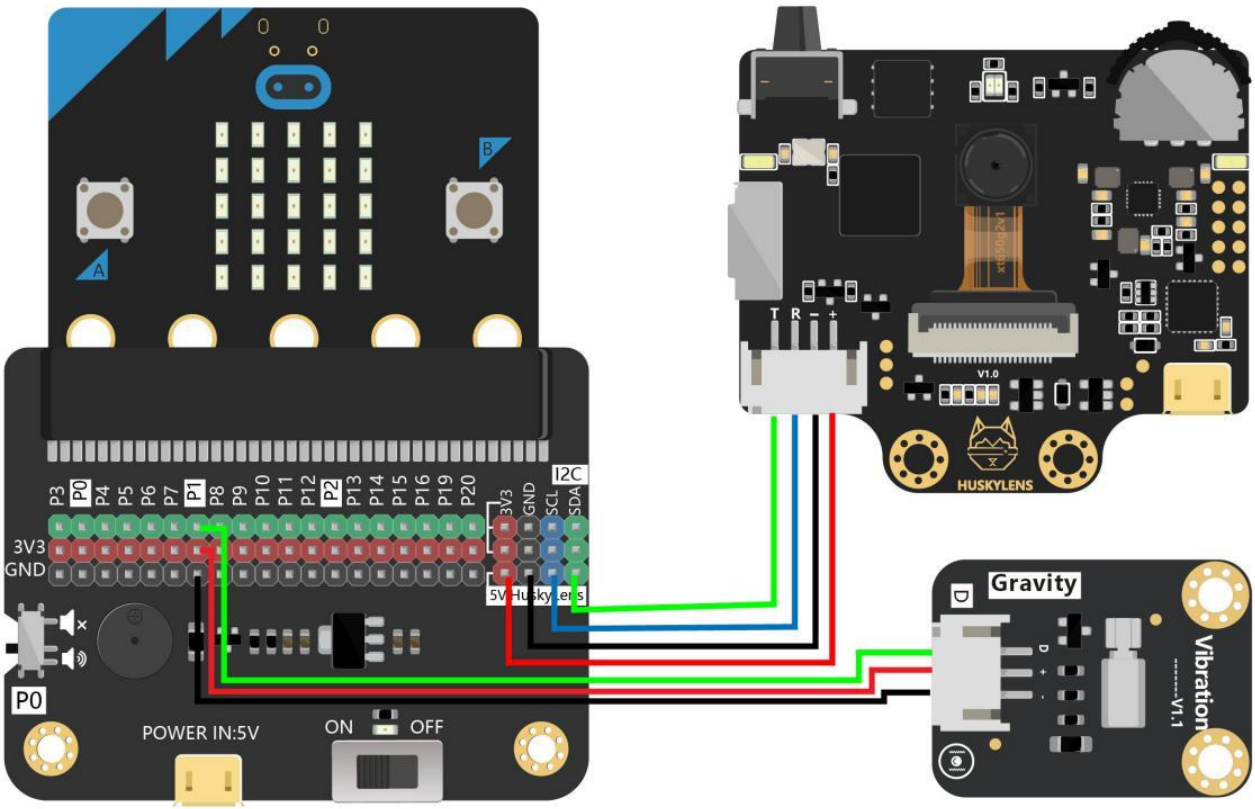


-1,-1 refers to lost



Task2: Vibration Feedback Notification

Hardware connection

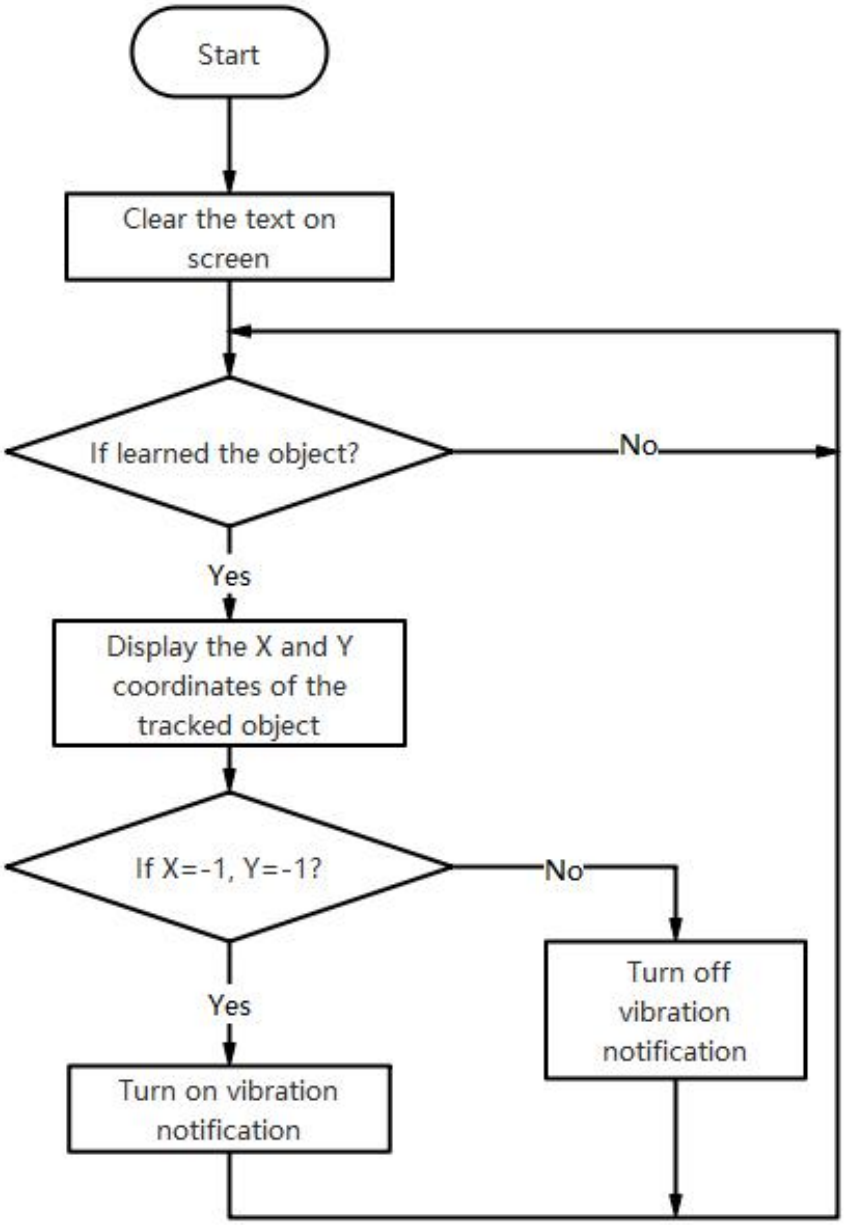


Program Design

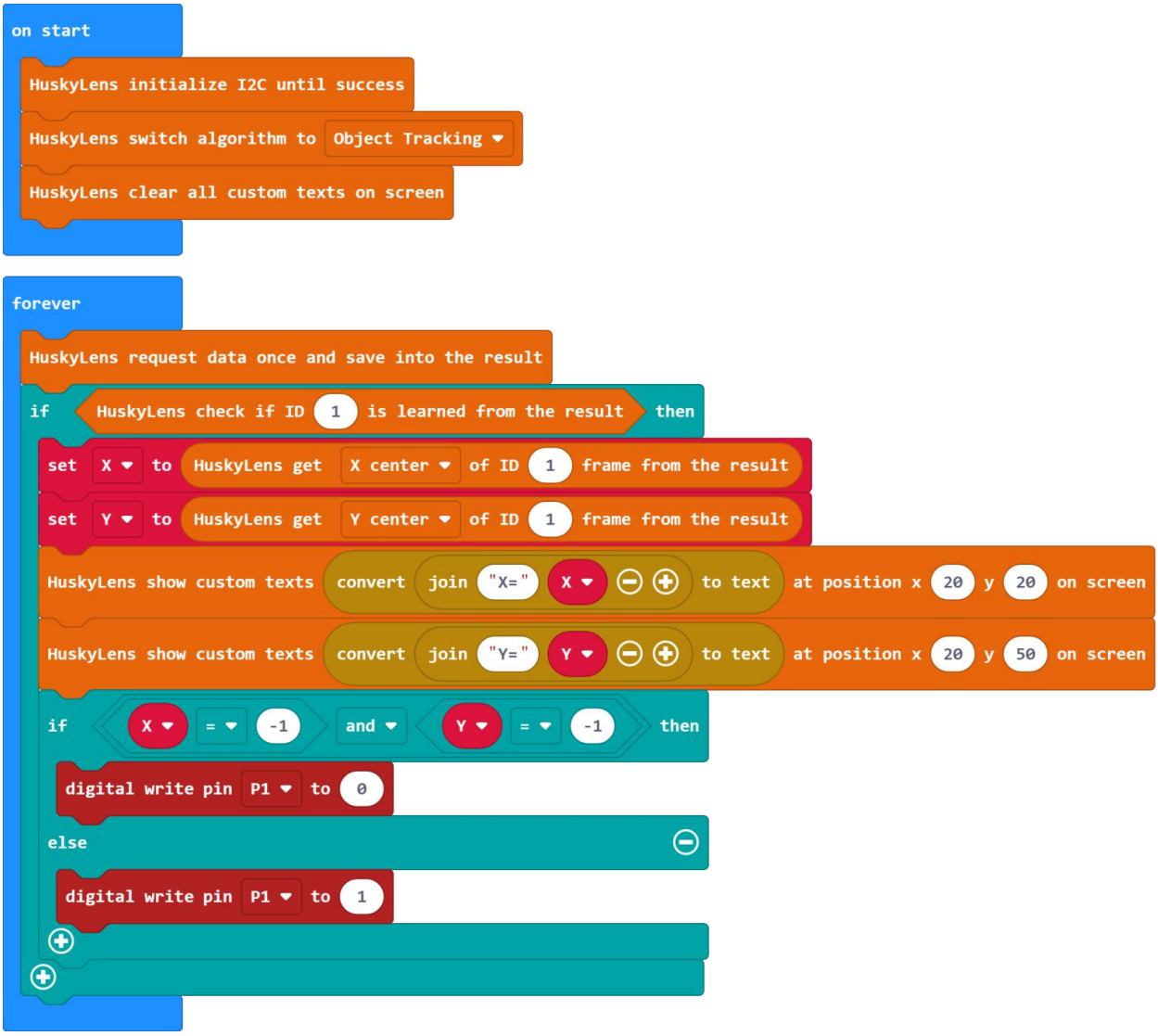
Function instruction:

When HUSKYLENS is tracking an object, X=-1 and Y=-1 displayed on the screen means that the tracked object is lost. The vibration module turns on the vibration notification, otherwise, turns off.

Flowchart Analysis



Sample program



Note: We can use high/low level to turn the vibration module on/off, or use a PWM signal to control its amplitude.

Operating Effect

When the tracked object is lost (X=-1, Y=-1), the vibration module turns on to send the vibration notification; when the tracked object appears on the screen, the X and Y coordinates of the tracked object are displayed, and the vibration module turns off.

Display the X and Y coordinates of the target, and the vibration is off.



Target is lost, vibration is on.



Project Review

In this lesson, we mainly learned to use the object tracking function of HUSKYLENS to find the specific coordinate value (X, Y) of the tracked object, and use the vibration module to notify when the target is lost. In real life, the object tracking function can be applied to more scenarios, such as the suspects tracking and reminding when children leave the visible range.

Project Development

Extend the project to realize functions like this: showing the location of the target person; when the person appears, the vibrating notification is turned on. Besides, can we do something to let the vibration amplitude change with the distance? For instance, the closer the target, the larger the amplitude; the farther the target, the smaller the amplitude.

Note: The amplitude needs to be controlled by PWM signal.

Lesson 6: Self-service Cashier

Every time you go on shopping sprees at a supermarket, the long queues at checkouts may cause you to become upset.



But now, thanks to technology development, the emergence of self-service cashiers in supermarkets brings us a lot of conveniences.

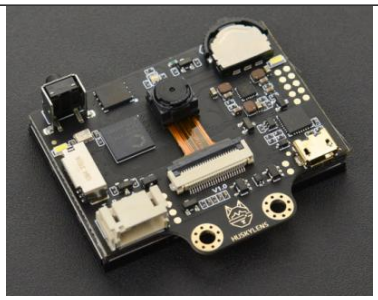
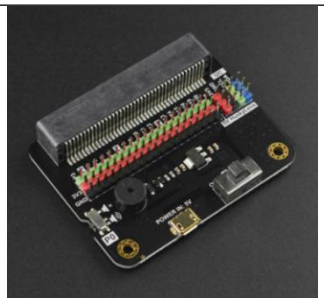
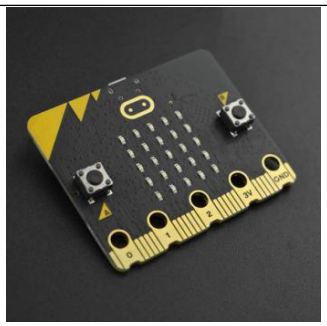


Can we make a self-service cashier with HuskyLens? What we need to identify each and every product accurately and estimate final bill?

Learning Objectives

1. Learn the working principle and application field of barcode.
2. Learn to use the barcode recognition function of HUSKYLENS.
3. Use the barcode recognition of HUSKYLENS to make a self-service supermarket cashier.

Preparation

	
HUSKYLENS ×1	IO Extender for micro:bit V2.0 ×1
	
micro:bit V2 ×1	Barcode Paper Card ×3
	
LED Switch-Blue ×1	

Learning Content

If we have a close-up view on the process of check-out, we will find whether it is a manual or self-service cashier, it' s all the same: scan the barcode of the product and calculate the bill. As the barcode of each product is different, we only need to find replacements for code scanner and barcode to realize our project.





Scanning devices----→ the tag recognition function of HUSKYLENS

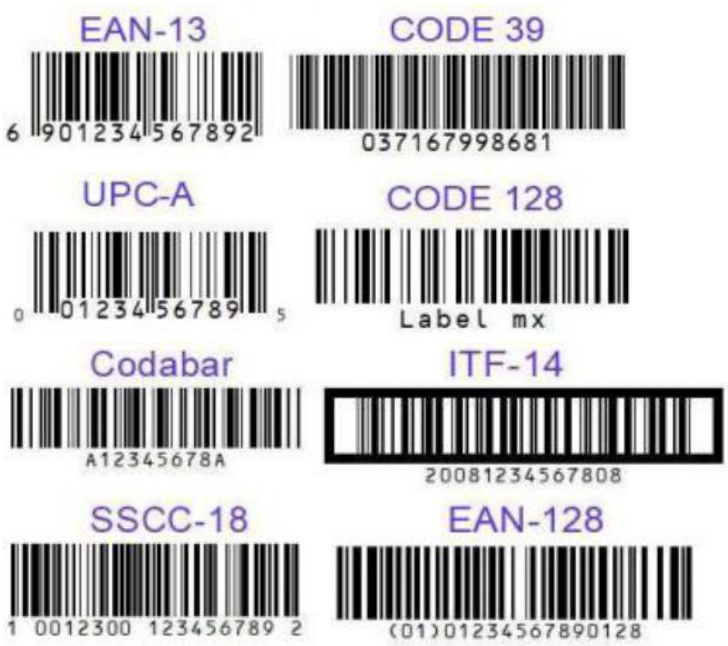
Barcodes-----→ Any Tags

working principle and application field of barcode

1. What is barcode technology?

Bar code is a symbol that expresses certain information composed of a group of bars, spaces and their corresponding characters arranged in a regular manner. Different code systems have different composition rules for bar code symbols. At present, the more commonly used code systems are EAN barcode, UPC barcode, 2/5 barcode, interleaved 2/5 barcode, Codabar, 93 barcode, 128 barcode and so on.

Barcode technology is an emerging technology that integrates coding, printing, recognition, data collection and processing developed on the basis of Computer Information Technology. The core content of bar code technology is to use photoelectric scanning equipment to read bar code symbols, so as to realize the automatic recognition of the machine, and quickly and accurately input the information into the computer for data processing, in order to achieve the purpose of automatic management.



2. Working principle of barcode technology

The barcodes compiled according to certain rules need to go through the processes of scanning and decoding to be converted into meaningful information. The color of an object is determined by the type of light it reflects. A white object can reflect various wavelengths of visible light, and a black object can absorb various wavelengths of visible light.

Therefore, after the light emitted by the light source of the barcode scanner is reflected by the barcode, the reflected lights shine over the photoelectric converter inside the barcode scanner, and the photoelectric converter converts the reflected light signal into corresponding electrical signal according to the different intensity of the reflected light signal.



3. Application field of barcode technology

Retail Industry

The most common use of barcodes is in the retail industry. EAN commodity barcodes build the foundation for the retail industry to apply barcodes. At present, most of the commodities sold in supermarkets use EAN barcodes. When selling, the cashier scans the EAN barcodes with a scanner. The

POS system finds the corresponding name, price and other information from the database, and counts the commodities purchased.

Library

Barcodes are also widely used in the circulation of books in libraries. Barcodes are affixed to books and library cards. When borrowing books, you only need to scan the barcode on the library card, and scan the barcode on the book, then the relevant information will be automatically recorded in the database.

Warehouse Management and Logistics Tracking

For the occasion of the mass flow of commodity goods, the application of barcode technology can realize fast and accurate recording of each item, and the various data collected can be processed by the computer system in real time. This enables various statistical data to accurately and timely reflect the status of items.

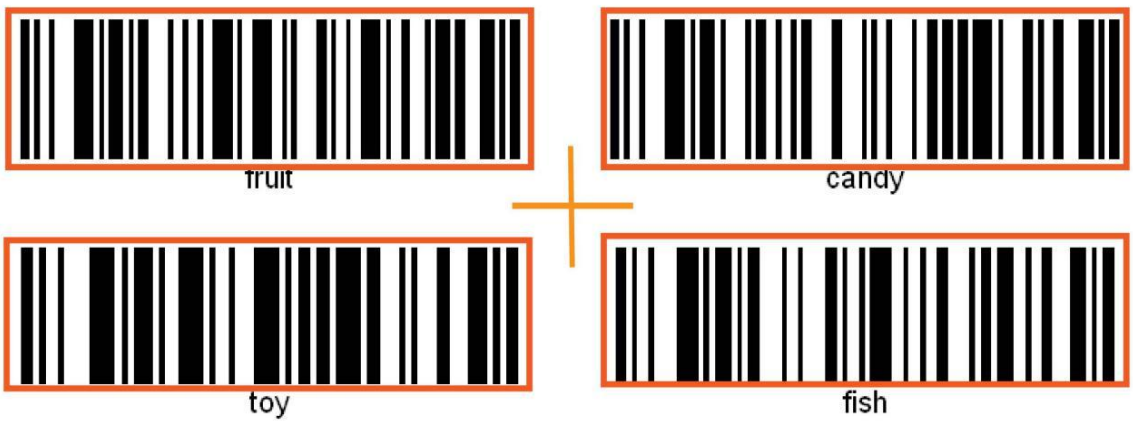
Can you find other applications of Barcode in daily life?

Demonstration of HUSKYLENS barcode recognition Function

1. Detect barcode

When detected by HuskyLens, all of the barcodes will be selected by orange frames on the screen automatically.

Note: When HUSKYLENS learns and recognizes barcodes, the ambient light may affect the recognition result. In order to improve accuracy, it is recommended to put a piece of white paper under the paper cards.



2. Learn barcodes

Now, learn and recognize the following three paper cards. (Note: The barcode on the front and back is the same).



Point the "+" symbol at the barcode, long or short press the "Learning button" to complete the first barcode learning. After releasing the "Learning Button", the screen will display: "Press the button again to continue! Press other buttons to end. " To continue learning the next barcode, press the "Learning button" before the countdown ends. If you no longer need to learn other barcodes, press the "Function button" before the countdown ends, or do not operate and wait for the end of the countdown.





In this project, we are going to learn multiple barcodes. Therefore, we need to short press the “Learning Button” before the countdown ends. Then, point the “+” at the second barcode, pressing or holding the “Learning Button” to complete this learning, and so on.

The tag ID is in the same order as the tag entered. That is, the learned tags will be tagged "Tag: ID1", "Tag: ID2", "Tag: ID3", and so on. Moreover, different tags correspond to different frame colors.

3. Recognize barcodes

When the HUSKYLENS encounters the learned barcodes again, there will be a colored frame to select them and display their IDs on the screen. The size of the frame can change with the size of barcodes and track barcodes automatically.







Project Practice

We will finish the project in two tasks. Firstly, use the barcode recognition function of HUSKYLENS to learn and recognize three products with barcodes, and then display the price on the HUSKYLENS screen.

Secondly, based on the previous step, add the start and end of the scanning code events to perform quantity statistics of customer’ s products. At last, add the total price settlement function to realize the supermarket self-service cashier function.

Task1: Price displaying

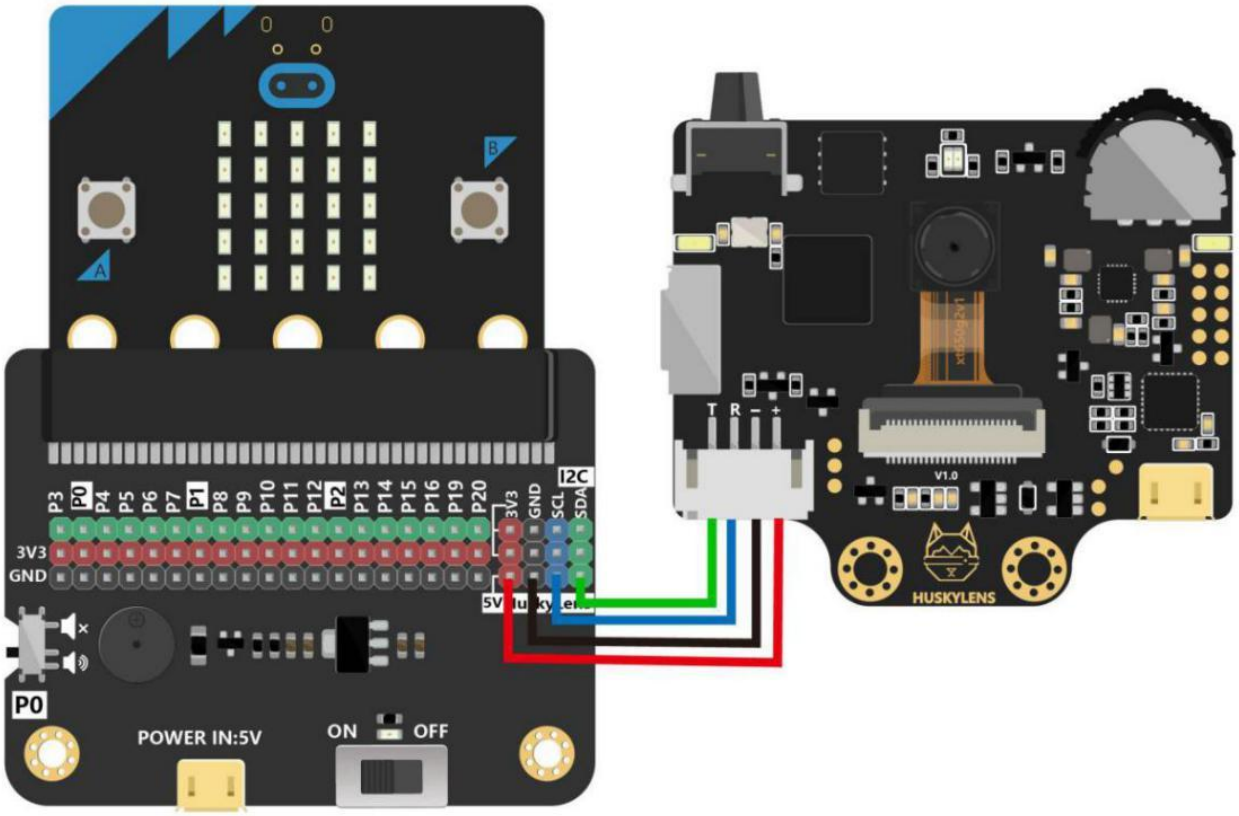
Let HUSKYLENS learn and recognize the barcodes of three different products. Write a program to make HUSKYLENS display the prices of the successfully recognized product on the screen.

Task 2: Price settlement

Based on task 1, add the product price settlement function. When any learned product is recognized, the price of a single product will be added to the total price.

Task1: Price displaying

Hardware connection

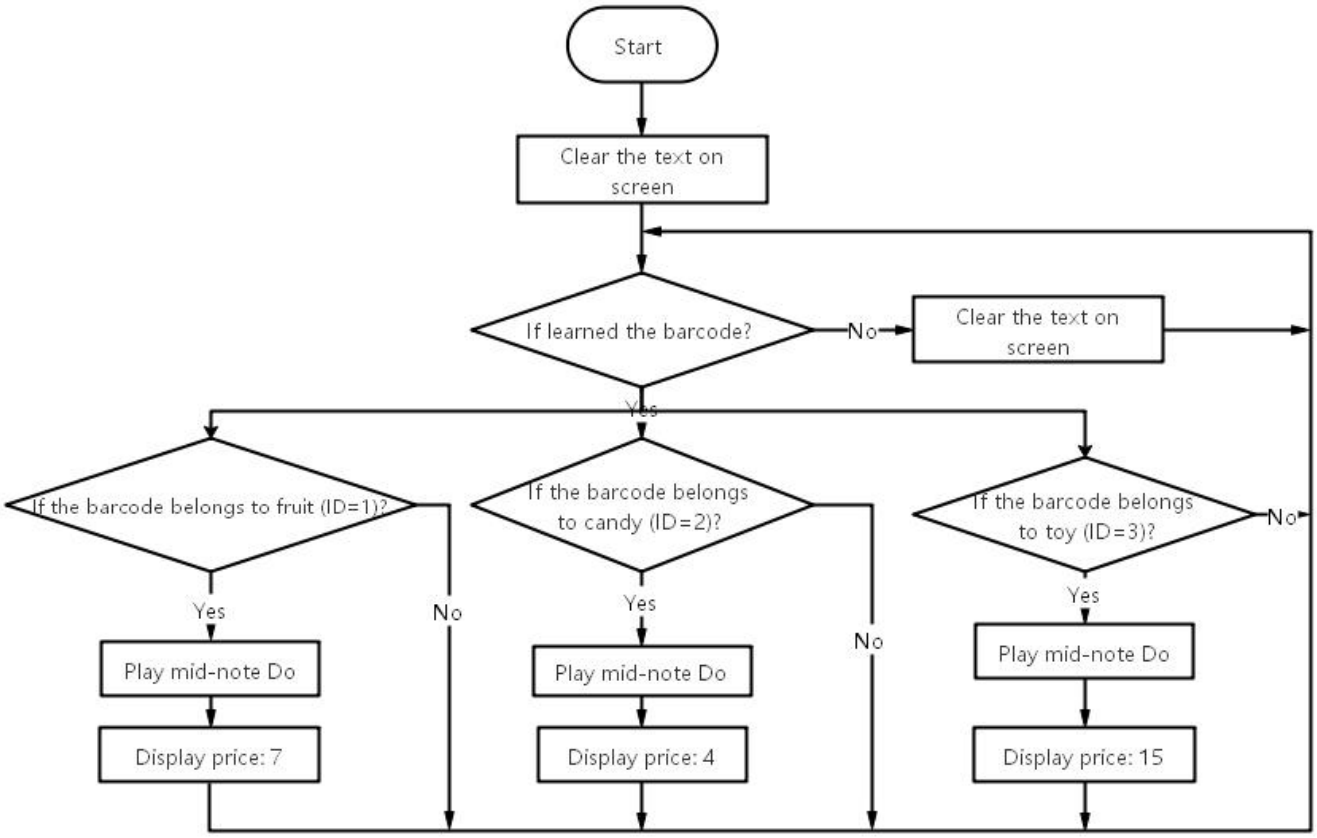


Program Design

Function instruction:

There are various products in the supermarket, and each one has a corresponding barcode and price. Now let HUSKYLENS learn three products in turn: fruit, candy, and toy. Assuming that the price of each product is: fruit (7 yuan), candy (4 yuan), toy (15 yuan), display the corresponding price of each product on the HUSKYLENS screen.

Flowchart analysis



Sample program



Operating Effect

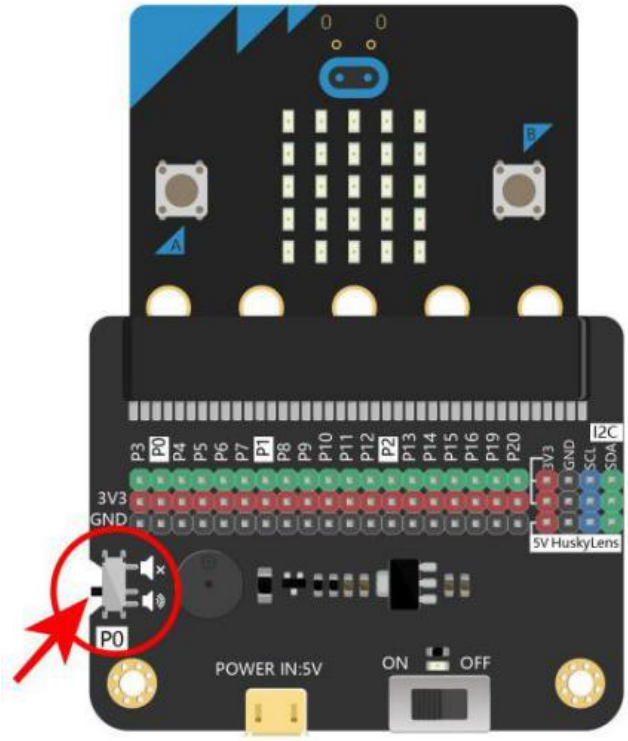
When the barcode is successfully recognized, the buzzer on the IO expansion board will emit a mid-tone Do, and the corresponding price of the product will be displayed on the HUSKYLENS screen.





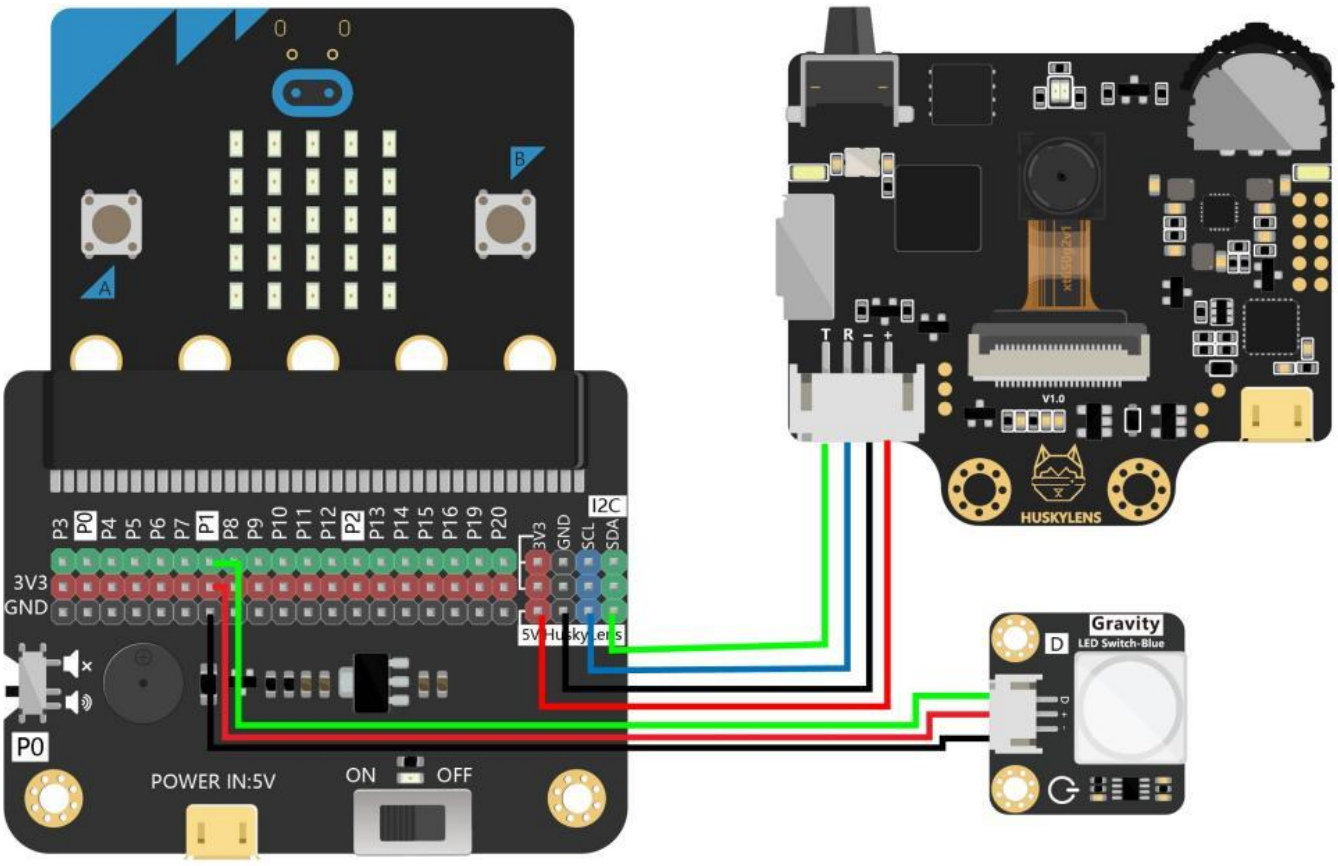


In this project, the buzzer that comes with the IO expansion board is used to indicate the success of the scanning, so you need to turn the switch of the buzzer to the playback position.



Task2: Price settlement

Hardware connection



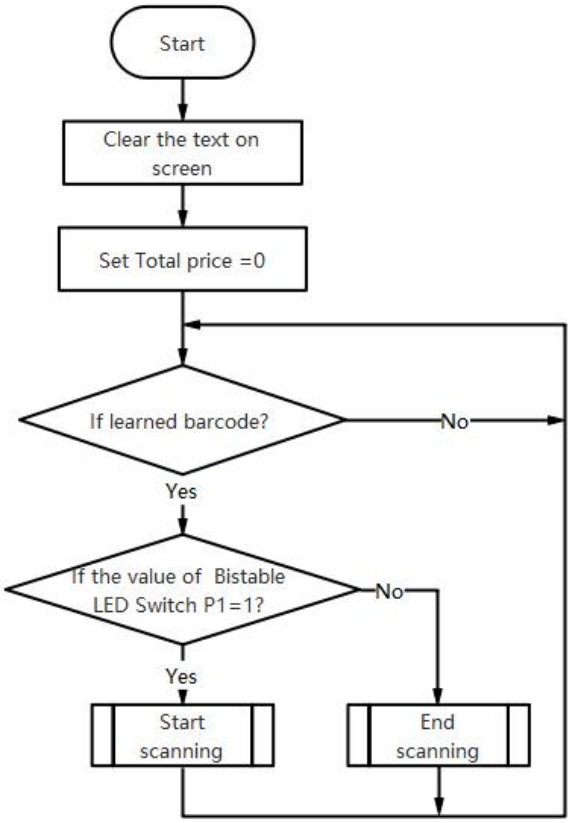
Program design

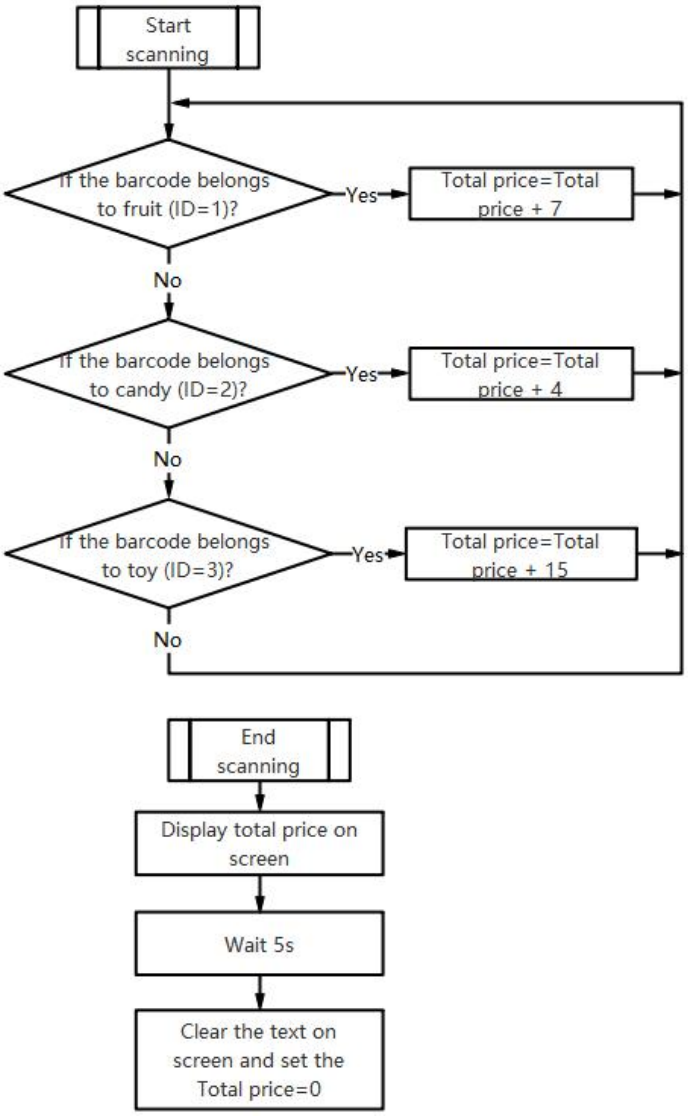
Function instruction:

How can we make sure that the total price is that of the goods we bought? Use the state of the Bistable LED Switch to control!

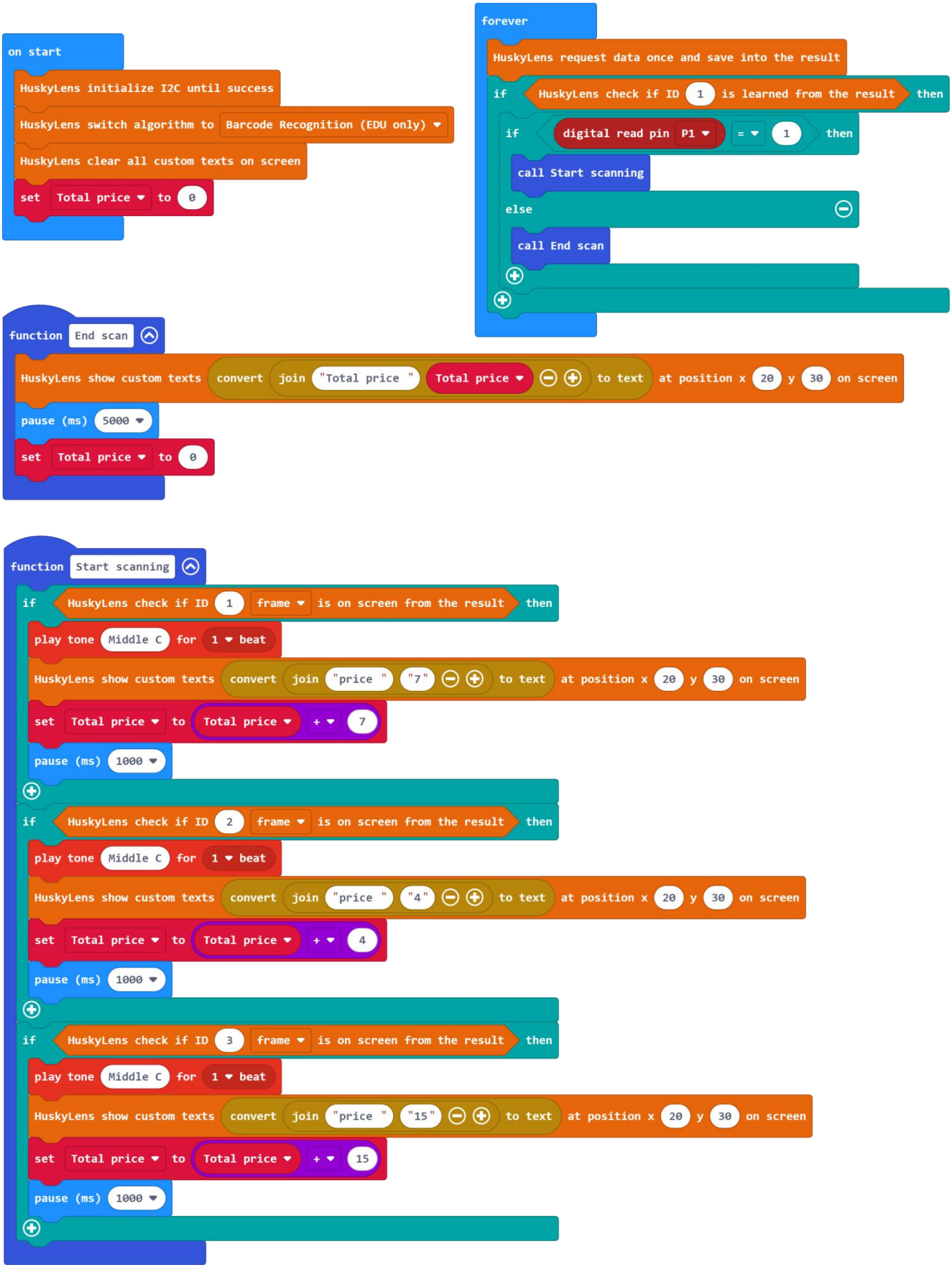
The customer presses the button for the first time (the state of the bistable switch is: 1), scanning starts, and the recognized product name is displayed on the HUSKYLENS screen; press the button again (the state of the bistable switch is: 0), When the scan is finished, the total price of the scanned product will be displayed on the HUSKYLENS screen. After the scan is completed, any product scanned by the same customer will not be displayed. The next customer presses the button to restart the scan.

Flowchart analysis





Sample program:



Note: The LED Switch is connected to the P1 pin of the IO expansion board in the project, so reading the digital value of the P1 pin is to get the state of the LED Switch.

Operating Effect

Press the switch, the LED light of the button is on, and you can start to scan the barcode of the products. After the barcode is successfully recognized, the corresponding product price will be displayed on the screen; press the switch again, the LED light of the button is off, and the scan ends, the total unit price of the scanned products will be displayed on the screen.



Project Review

This project mainly uses barcode recognition technology to recognize the corresponding product information and output a specific ID so as to display the product prices on the HUSKYLENS screen and realize the function of settlement. Now the supermarket self-service cashier system is basically done.

Project Development

Normally, there is a locker for free storage at the entrance of the supermarket. Before storing things, press the "Store" button, the locker will print a barcode, and the corresponding locker will

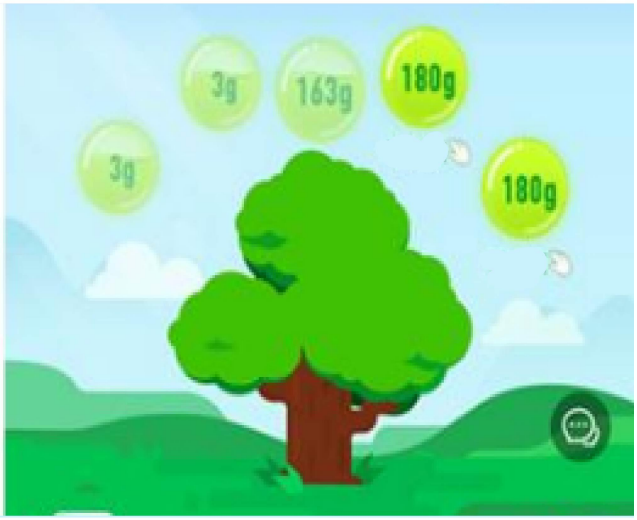
automatically open when you get the barcode. Can you use the HuskyLens to complete the locker project?



Lesson7: Energy Harvest

The Ant Forest in the Alipay app allows users to grow a virtual tree by collecting energy points. Different values of energy will be generated based on their real-life actions every day.





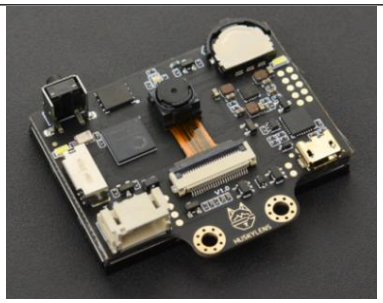
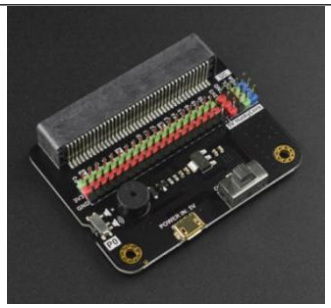
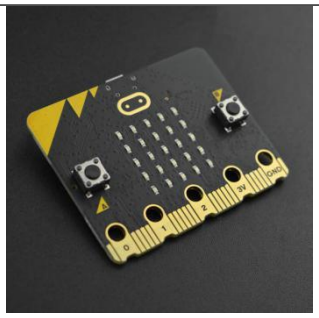
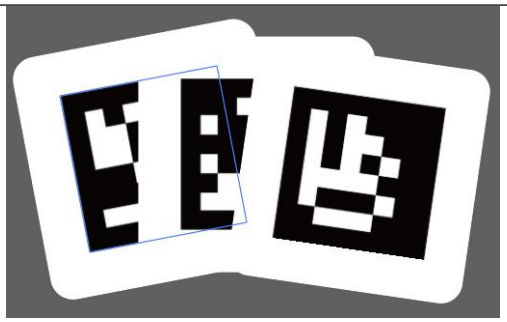
Can you guess what I am going to do now?

HuskyLens is really powerful, let’ s use the tag recognition function of HuskyLens to make an energy harvest game!

Learning Objectives

1. Learn the working principle of tag recognition.
2. Learn to use the tag recognition function of HuskyLens
3. Learn what a random number is
4. Use HuskyLens to make an energy harvest project

Preparation

	
HuskyLens ×1	IO Extender for micro:bit V2.0 ×1
	
micro:bit v2 ×1	Tag paper card ×3

Learning Content

Principle of Tag Recognition

1. What is tag recognition?

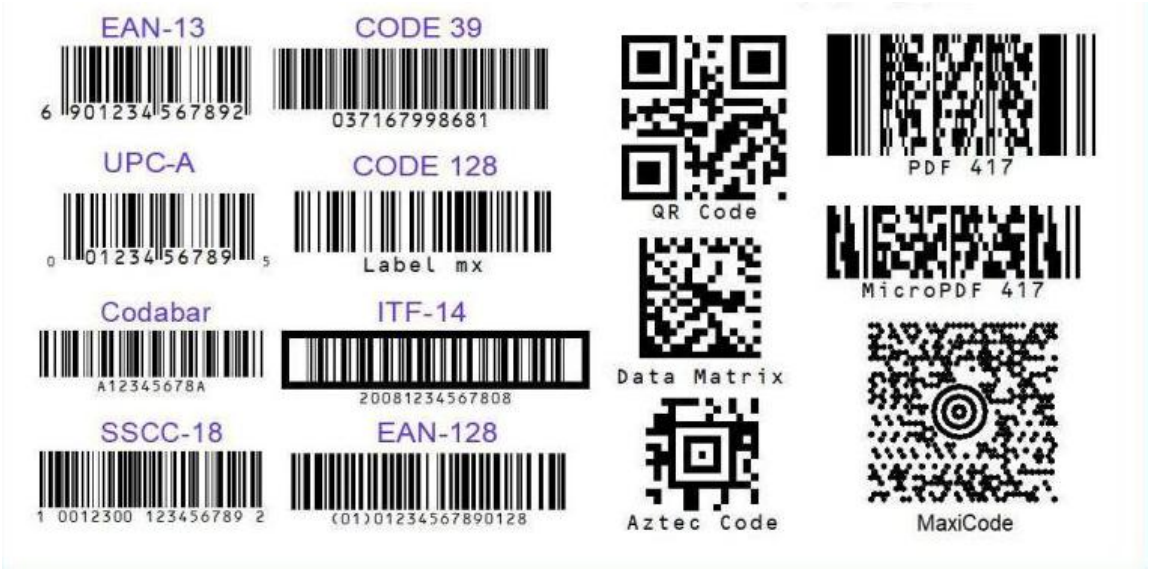
Tag recognition refers to the technology of effective and standardized coding and recognition of items, which is the basics of informatization. With people's increasing awareness of health and safety, the food industry has increasingly higher requirements on the quality and safety of products (from raw

materials, transportation, production, storage, traceability and management). Tag recognition also plays an important role in meeting the needs of companies for product tracking and tracing.

Tag recognition technology mainly includes barcode technology, IC card technology, radio-frequency recognition technology, optical symbol recognition technology, speech recognition technology, biometric recognition technology, remote sensing, robot intelligent perception and other technologies.

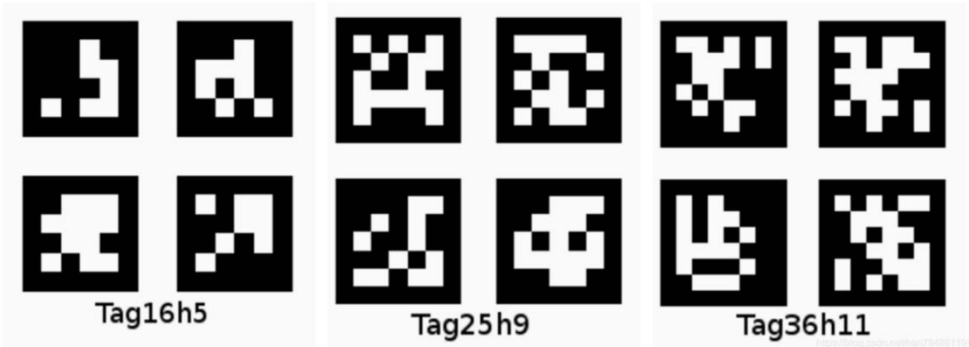
Types of 1D barcode

Types of 2D barcode



2. What is AprilTag?

AprilTags is a visual reference system from a UMich project team for AR, robotics and camera calibration. The tag acts as a barcode, storing a small amount of information (tag ID), while also providing a simple and accurate 6D (X, Y, Z, roll, pitch, yaw) pose estimation or the tag.



3. Differences between AprilTag, barcode and QR code

Compared with barcode and QR code, AprilTag is more useful in machine vision, because it can calculate the accurate relative 3D location, direction, and ID through AprilTag. The 3D location in

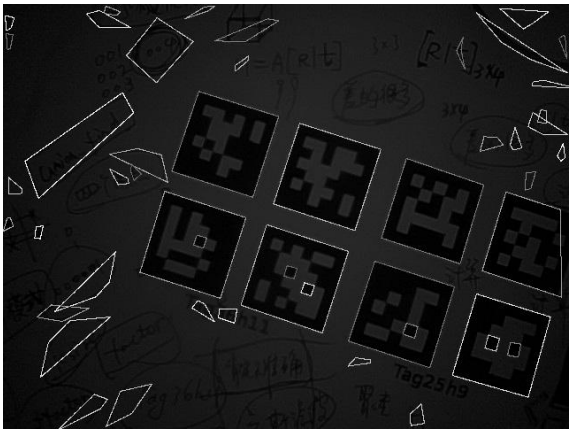
reality world is very useful for machine! AprilTag is usually used in fields like AR, robot, and camera calibration.

4. Working principle of tag recognition

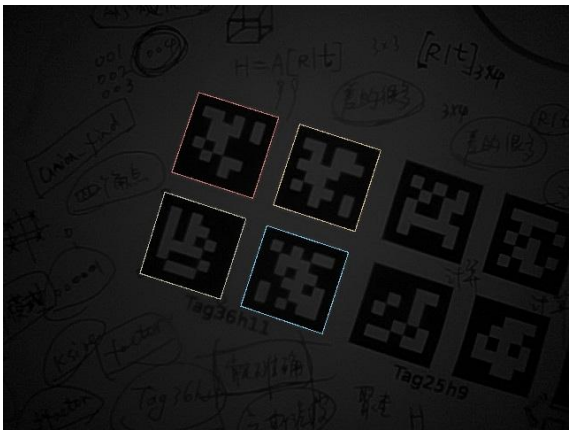
Edge detection: find out the edge contour in the image.



Quadrangle detection: find out the quadrangles in the contour



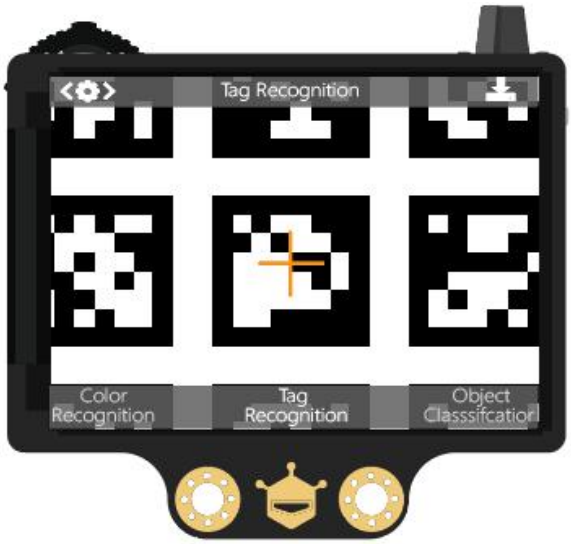
Decoding: match and check the quadrangles.



Demonstration of HuskyLens Tag Recognition Function

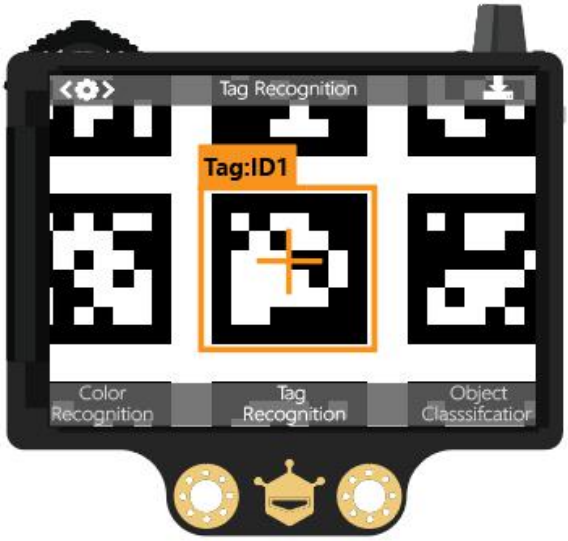
1. Tag detection

When HuskyLens detects a tag, the tag will be automatically selected by the white frame on the screen.



2. Tag learning

Point the "+" symbol at the tag, long or short press the "Learning button" to complete the first tag learning. After releasing the "Learning button", the screen will display: "Press the button again to continue! Press other buttons to end. " To continue learning the next tag, press the "Learning button" before the countdown ends. If you no longer need to learn other tags, press the "Function button" before the countdown ends, or do not operate and wait for the end of the countdown.

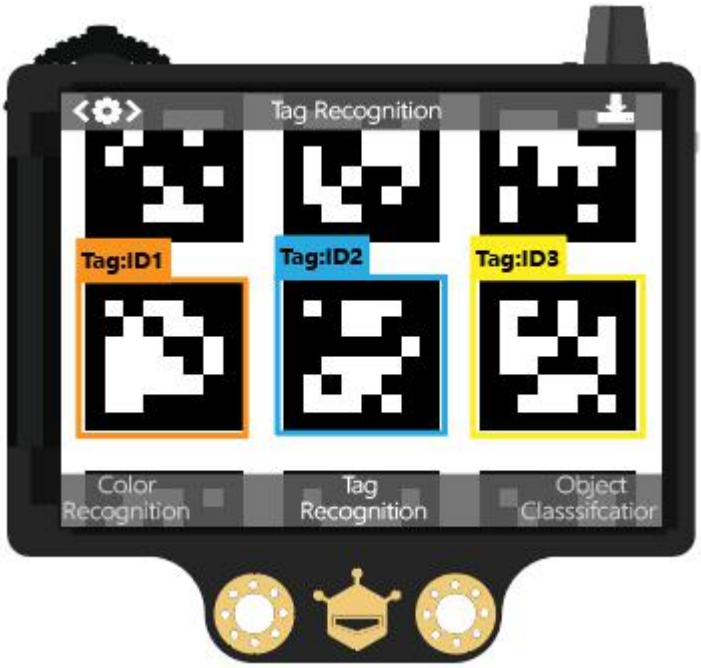


In this project, we are going to learn more tags, so press the "learning button" before the countdown ends, and then point the "+" in the center of the HuskyLens screen at the next tag that needs to be learned, short press or long press the "learning button" to complete the learning of the second tag. And so on.

The tag ID is consistent with the sequence of input, that is, the learned tags will be labeled as "tag: ID1", "tag: ID2", "tag: ID3", and so on, and the frame color corresponding to different tags is also different.

3. Tag recognition

When the HuskyLens encounters the learned tags, a colored frame with an ID will be automatically displayed on the screen. The size of the frame changes along with the size of the QR code tag and the frame will automatically track these QR code tags.



Project Practice

Next, the energy harvest game will be completed in two tasks. First, we must learn to use HuskyLens's tag recognition function, and output random numbers from the recognized results, and then determine whether the collected energy value can be used for energy synthesis by calculating the sum of the random numbers.

Task1: Generate energy

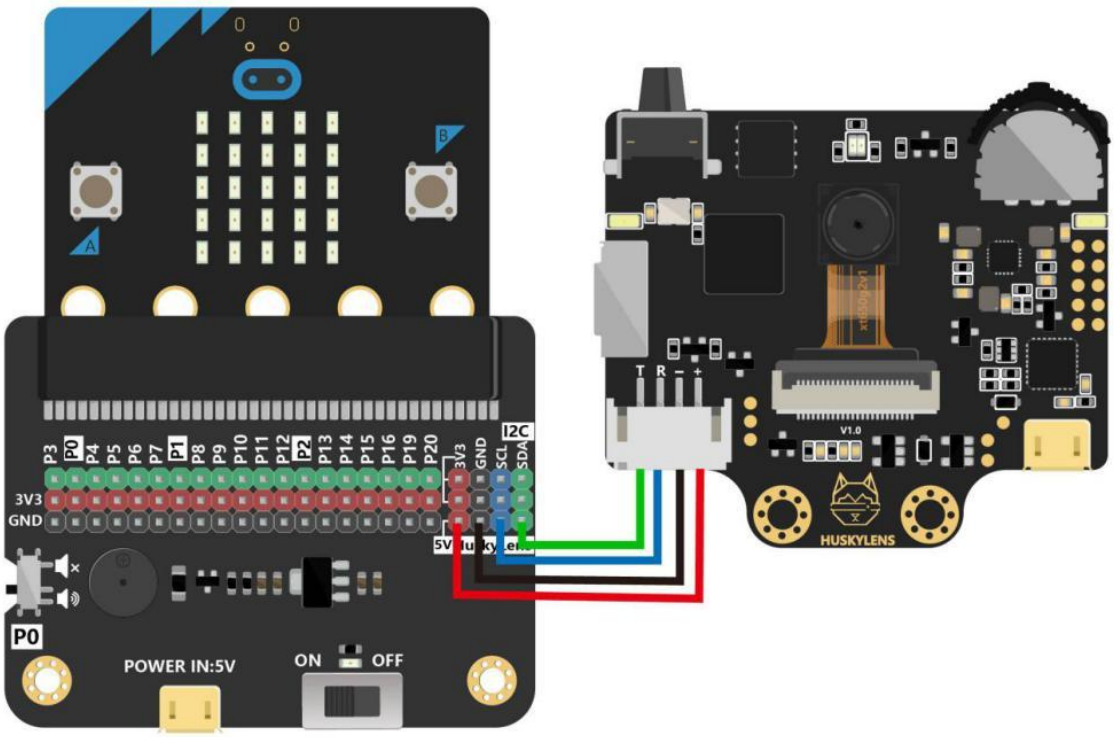
In the process of the game, the energy values collected are all randomly generated. Therefore, in this step, HuskyLens needs to recognize the AprilTag tag and generate a random number, also the collected energy value.

Task 2: Energy synthesis

Each AprilTag corresponds to a certain value. When the sum of all energy values is greater than 50, energy synthesis can be performed.

Task1: Generate energy

Hardware connection



Program design

Function instruction:

The HuskyLens sensor needs to learn multiple AprilTag tags and be able to display random numbers from 0 to 9 on the micro:bit matrix screen after the recognition is successful.

What is random number?

The **random number** is the number taken from the **random seed**.

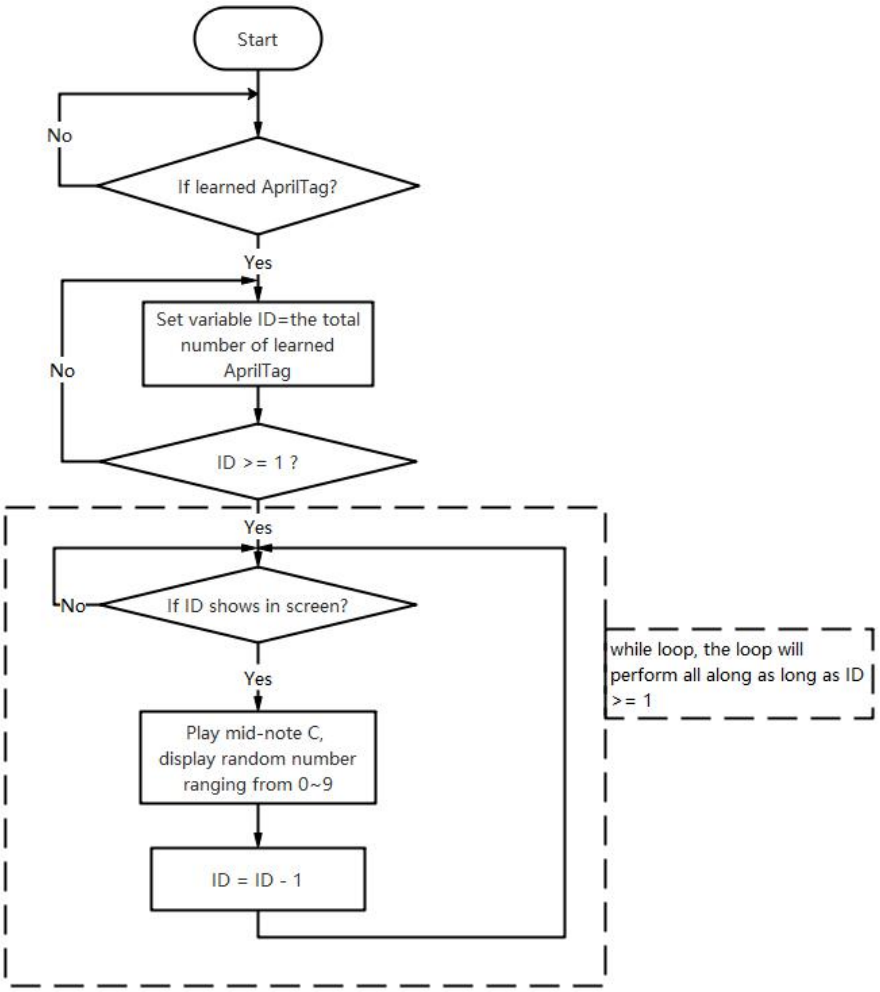
Next, we are going to use an simple example—throwing dice, to understand what a random number is.

The dice have 6 sides, and each side is marked with points: 1, 2, 3, 4, 5, 6. Suppose that we roll 6, then the number 6 is called a random number. The 6 numbers from 1 to 6 are called random seeds.

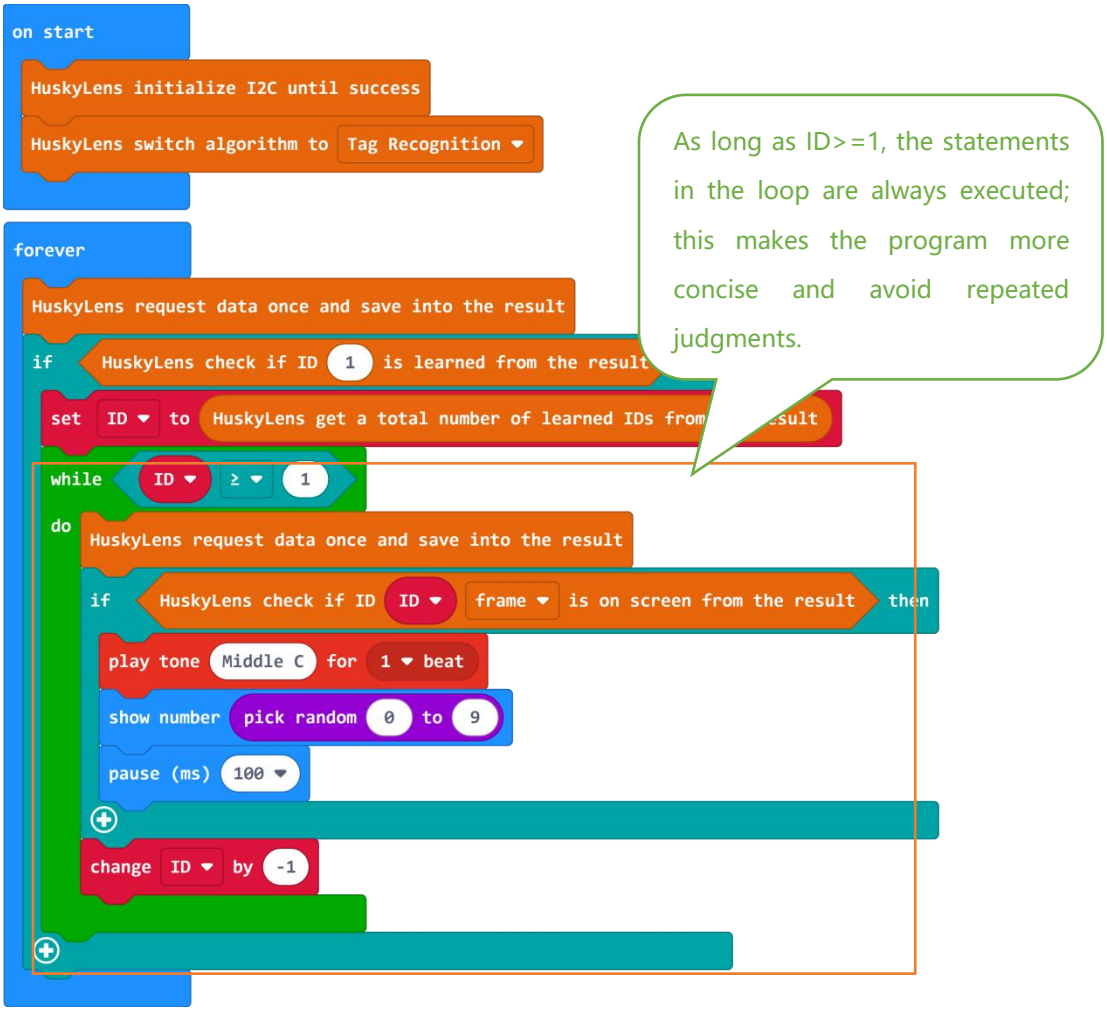
There is a random number block " 

" in MakeCode, where the range of 0~10 is the random number seed, and the number taken from 0~10 is the random number.

Flowchart Analysis

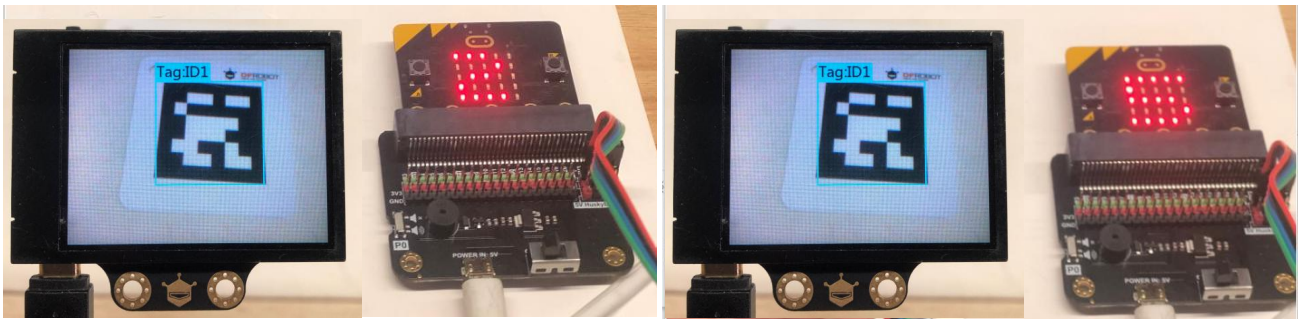


The Sample Program



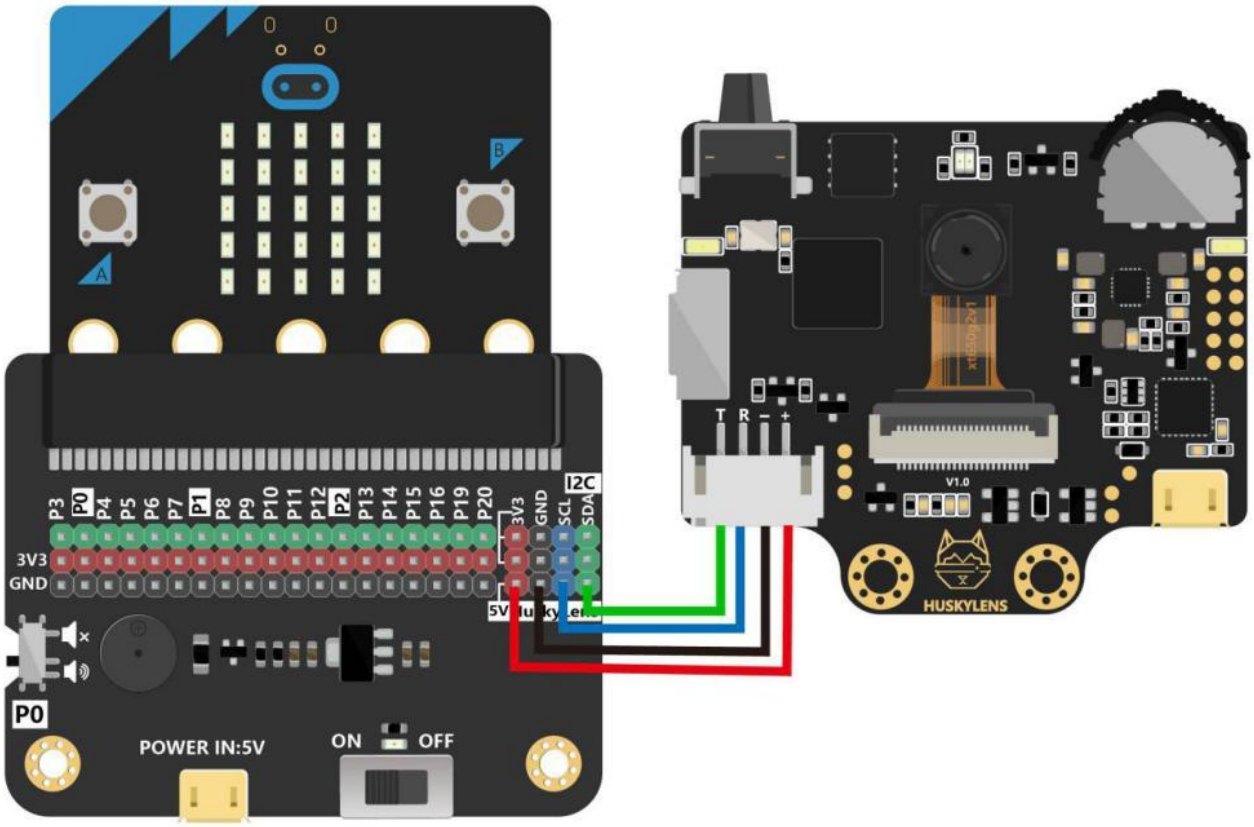
Operating Effect

In this project, at least one AprilTag needs to be learned, as long as HuskyLens recognizes the learned AprilTag, it will generate a random number ranging from 0 to 9. As shown in the figure, the random numbers generated by the same AprilTag tag are not the same.



Task2: Energy synthesis

Hardware connection

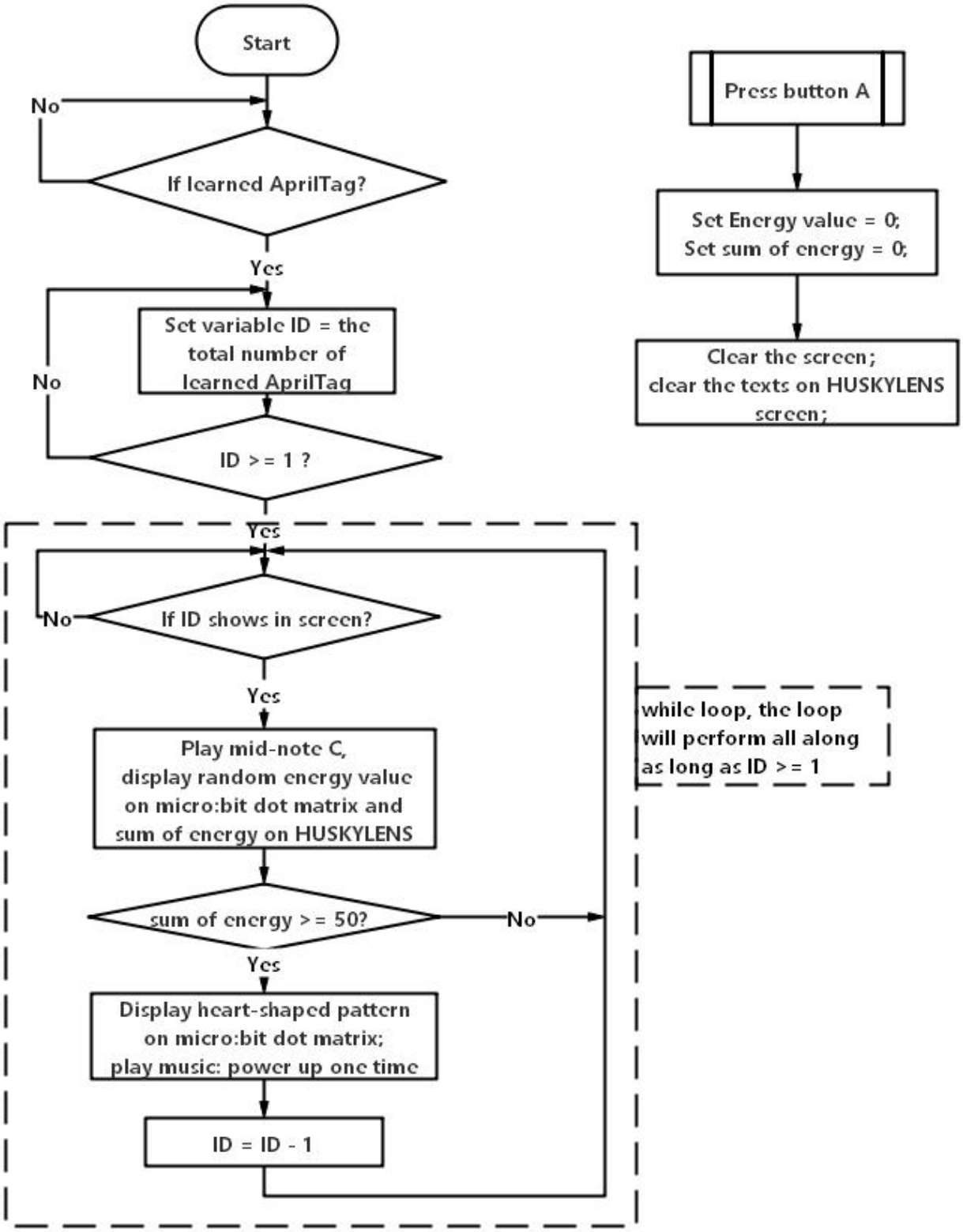


Program design

Function instruction:

On the basis of the previous project, the function of displaying the sum of energy (Sum of energy) has been added. When the sum of energy>=50, energy synthesis is performed. Meanwhile, the function of re-collecting is also added. Press button A on the micro:bit to clear the display and initialize the energy value and the total energy points.

Flowchart Analysis



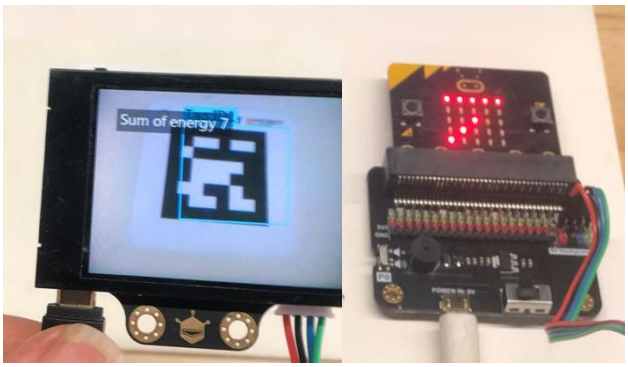
The Sample Program

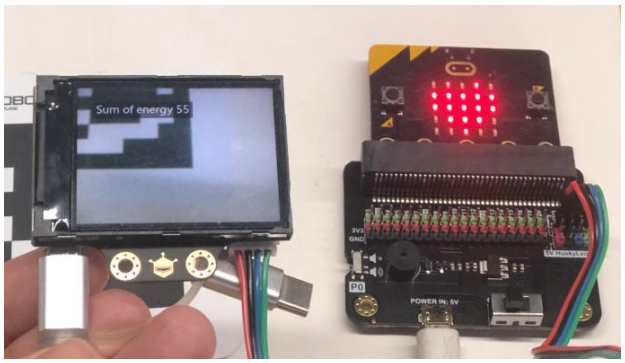


Operating Effect

Every time HuskyLens successfully recognizes a learned AprilTag tag, a random number will be generated and showed on the micro:bit dot matrix screen, and the sum of the calculated random numbers will be displayed on the HuskyLens screen; when the sum of random numbers is greater than 50, energy synthesis is performed, a "heart-shaped" pattern shows on the micro:bit, and the buzzer

plays music "power up"; press button A to clear all the displays and start a new round of energy harvesting.





Project Review

This project mainly uses the tag recognition function of HUSKYLENNS. Each time HuskyLens recognizes a learned tag, a random number ranging from 0~9 will be generated, then judge the sum of the random number and perform corresponding operations. Do you know where random number is used in daily life?

Project Development

The energy harvest is just a little game, it is funny when playing with friends. Now, call on your friends and have a competition with them!

Game rules:

1. The one using the tag recognition function of HuskyLens fewer times wins.
2. The one spending less time wins when the number of times using HuskyLens are the same.

Name	Number of times	Time

Lesson8: Smart Bus Ticketing Machine

In the past, bus tickets were sold by conductors on the bus. Whenever there is rush hour in the morning and evening, the bus is always crowded with people, and it becomes a hassle for the conductor to collect and change.



With the development of science and technology, the smart bus ticketing machine on the unmanned ticket vending vehicle has brought a lot of convenience to our lives. You can swipe your card or insert a coin directly. After the card payment is successful, you will hear the corresponding prompt "standard card", or "student card". Different cards correspond to different charging standards.

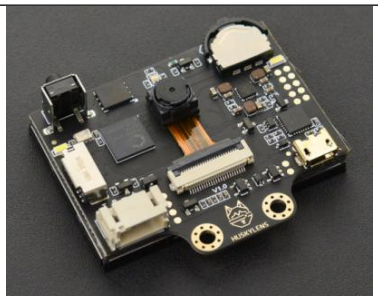
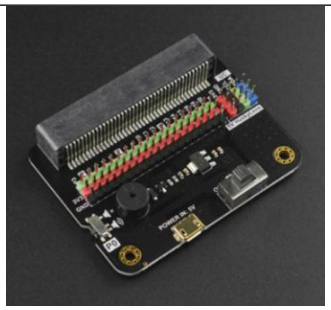
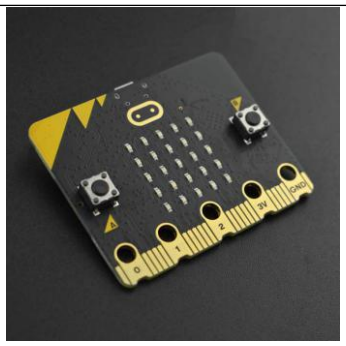


Now, let's make a smart bus card machine! How can we achieve the function of swiping a card and then deduce money from it?

Learning Objectives

1. Learn the working principle and application field of two-dimensional code technology
2. Learn the two-dimensional code recognition function of HUSKYLENS
3. Use the two-dimensional code recognition function of HUSKYLENS to make a smart bus ticketing machine.

Preparation

	
HUSKYLENS ×1	IO Extender for micro:bit V2.0 ×1
	
micro:bit v2 ×1	Gravity: Voice Recorder Module ×1
	
Stereo Enclosed Speaker ×1	Two-dimensional code cards ×3

Learning Content

Carefully observe the process of using the card, you will find that as long as the bus card is close to the machine, your money in the card will be automatically deducted. Different cards correspond to different charging standards. So as long as we find alternatives for card devices and cards, the project can be realized.





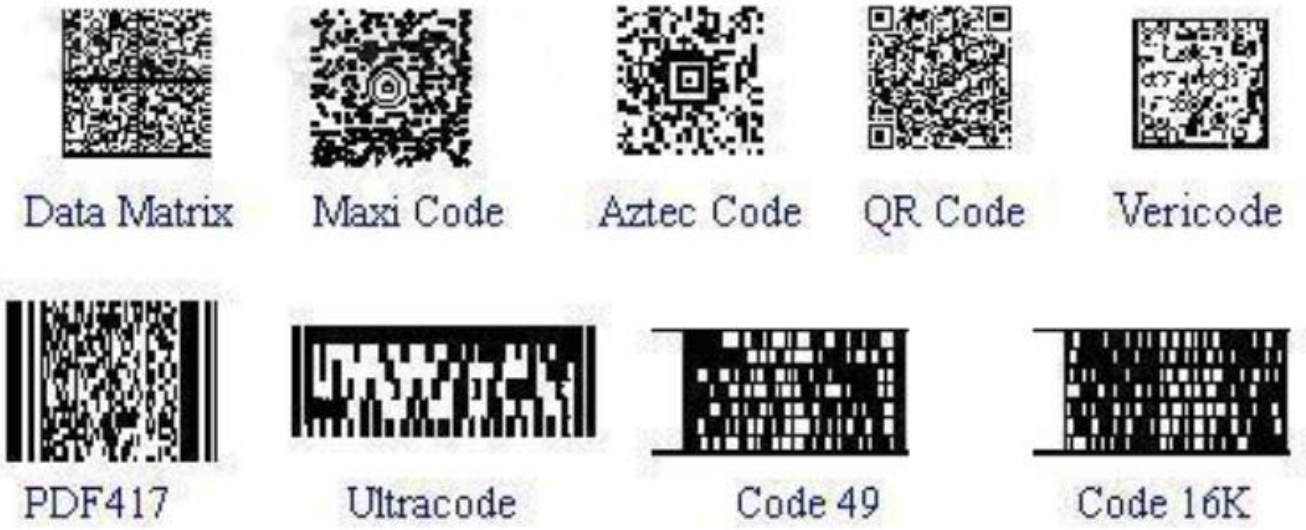
Ticketing machine—Two-dimensional code recognition function of HUSKYLENS

Bus card—Cards with two-dimensional code

Working Principle and Application Field of Object Tracking

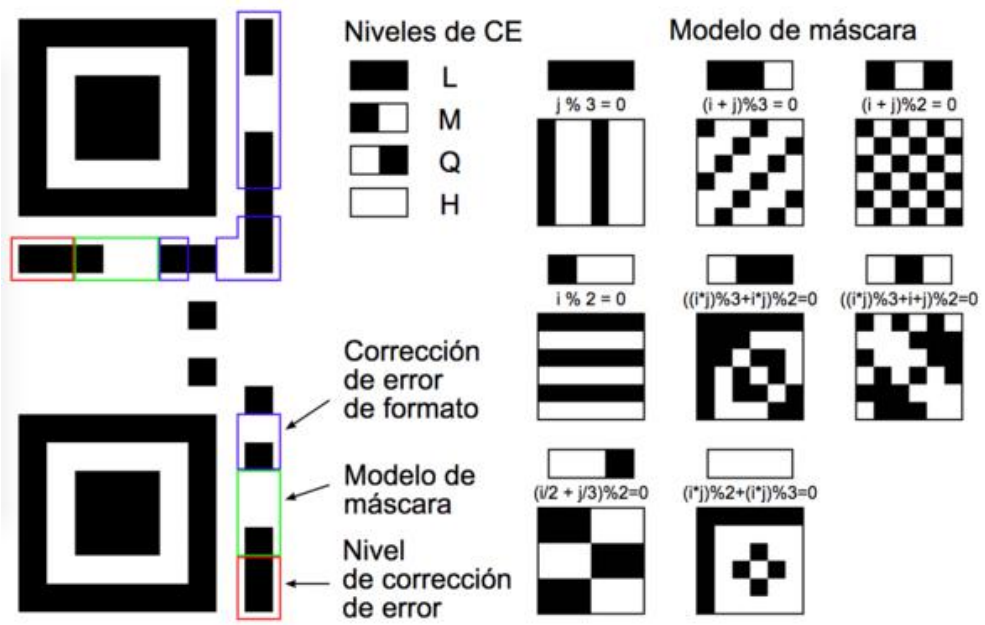
1. What is two-dimensional code?

Two-dimensional codes are also called **two-dimensional barcodes**. There are many types of two-dimensional codes. The common two-dimensional code, QR Code, which refers to Quick Response, is a popular encoding mode on mobile devices. QR code can store more information, and can also represent more data types than traditional two-dimensional codes.



2. Working principle of two-dimensional code

In fact, the two-dimensional code is not mysterious. It mainly converts information into small black and white squares, and then fills them into a large square. How information and squares correspond to each other? Here we have to mention an invention of age---binary.



Through the binary system, each word, number, and symbol are "translated" into a string of "0" and "1". Use white squares to represent "0" and black squares to represent "1". Then arrange these white and black squares according to a specific rule to obtain a two-dimensional code. The essence of a two-dimensional code is to convert information (data) into a binary code, and then fill the binary code into this large square.



From the picture above, we can see that there are black squares in the upper right corner, upper left corner, and lower left corner of the QR code. What are these three squares used for? They are used for positioning, so that the information recorded in the QR code can be accurately identified whether you scan the code horizontally or vertically. Compared with the horizontal direction of the barcode, the two-dimensional code records information in two dimensions, increasing the amount of information storage.

3. Application field of two-dimensional recognition

The two-dimensional code is widely used in daily life, you can scan it to pay, enter a website, and so on.

E-Menu

Many restaurants have adopted the form of QR code electronic menu. Consumers can see the introduction and price of related dishes as long as they enter the online menu by scanning a QR code.



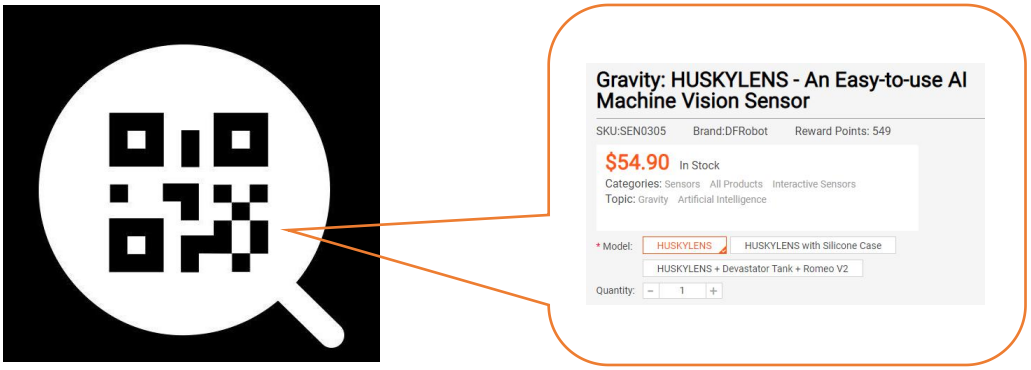
E-Coupon

The QR code can also be made into an electronic coupon, and the user can obtain the coupon or check which products are discounted by scanning it.



Anti-counterfeiting traceability

You can check the product production place by scanning the code.



Question: Do you also know that where are QR codes are used? Or do you think where can QR codes be used?

Demonstration of HUSKYLENS QR Code Recognition Function

1. Detect QR code

When HUSKYLENS detects QR codes, all the detected QR codes will be automatically selected by orange frames.



2. Learn QR codes

Next, make HuskyLens learn and recognize the standard card and student card.

Point the "+" symbol at the QR code, long or short press the "Learning button" to complete the first code learning. After releasing the "Learning button", the screen will display: "Press the button again to continue! Press other buttons to end. " To continue learning the next code, press the "Learning button" before the countdown ends. If you no longer need to learn other codes, press the "Function button" before the countdown ends, or do not operate and wait for the end of the countdown.





In this project, we are going to learn multiple QR codes. Therefore, we need to short press the "Learning Button" before the countdown ends. Then, point the "+" at the second QR code, pressing or holding the "Learning Button" to complete this learning. And so on.

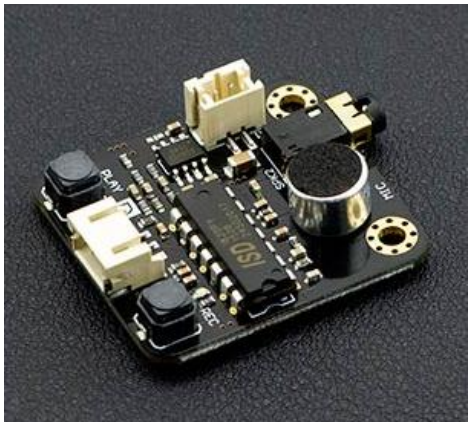
The code ID is in the same order as the code entered. That is, the learned codes will be tagged as "background: ID1", " background: ID2", " background: ID3", and so on. Moreover, different codes correspond to different frame colors.

3. Recognize QR codes

When the HUSKYLENS encounters the learned QR codes again, there will be a colored frame to select them and display their IDs on the screen. The size of the frame can change with the size of QR codes and track them automatically.



Introduction of Voice Recorder Module



1. What is voice recorder module?

The voice recorder module is an audio module that integrates recording and playback functions. There are buttons and microphones onboard, you can directly record your voice by pressing the record button, speaking and then releasing the button when finished.

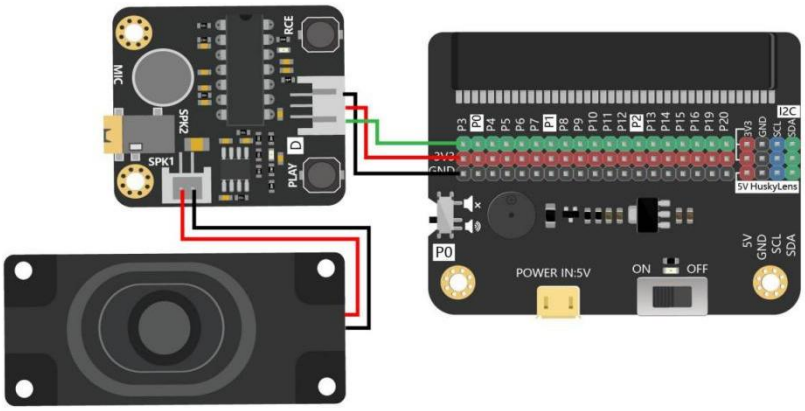
Note: The voice recorder module can realize audio recording within 10s. When you release the button or reach 10s, the red LED on the board will flash once.

2. Application of voice recorder module in project

In the project, the voice recorder module needs to be used with the stereo enclosed speaker. The stereo enclosed speaker is used to play the pre-recorded audio, which makes our work greater. How to use the voice recorder module? Let's try it together next!

First learn how to use the voice recorder module to record and play sounds. The use of the voice recorder module with stereo enclosed speaker is very simple. No programming is required. Power up the module, then you are ready to go!

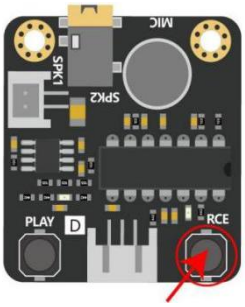
Hardware connection



Note: When recording, the IO expansion board needs to be powered.

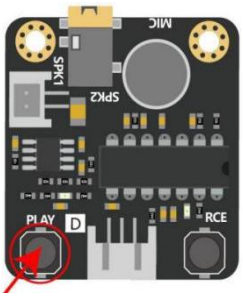
Recording

There is an "RCE" recording button on the voice recorder module. Press this button to start recording, and release the button to end the recording. Next, record an audio by yourself, the audio content is "Payment successful, get in please."



Playback

Press the "PLAY" button, and the stereo enclosed speaker will play the recorded audio. Each time you press the "PLAY" button, the audio will be played (press once to play the entire audio). After the playback is over, the red LED on the voice recorder flashes once.



Project Practice

We will finish the project in two steps. First, use multiple QR codes recognition function of HUSKYLENS to learn and recognize the bus cards with QR codes, and then display the fees that should be paid for each card on the HUSKYLENS screen.

Based on the previous step, the balance display and voice broadcast functions are added. Each time the card is successfully recognized, the corresponding fee will be subtracted from the balance, and then the voice broadcast " Payment successful, get in please.", which basically realizes the function of the ticketing machine.

Task1: Charge by cards

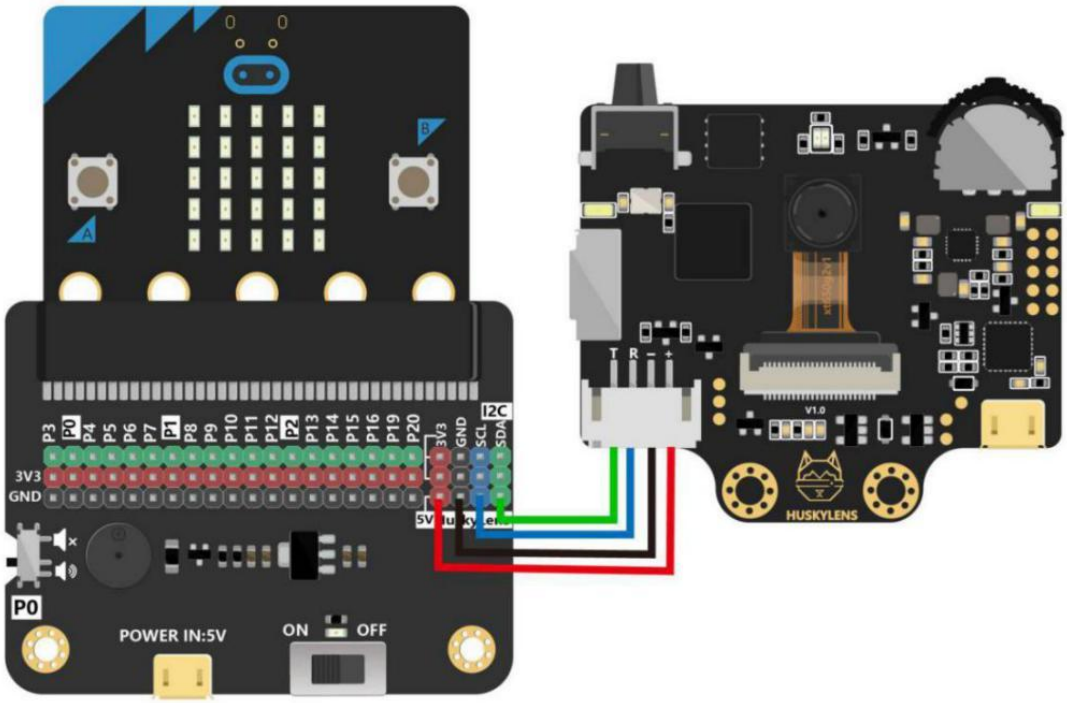
HUSKYLENS learns and recognizes the bus card with a QR code. Write a program to display the fee on the HUSKYLENS screen.

Task 2: Voice broadcast " Payment successful, get in please."

On the basis of task 1, the balance display and voice broadcast function are added. When the learned bus card is recognized, the voice broadcast is performed. It is assumed that each card is charged with 20 yuan, and the corresponding fee will be deducted on the basis of the remaining fee once the payment is successful.

Task1: Charge by card

Hardware connection



Program design

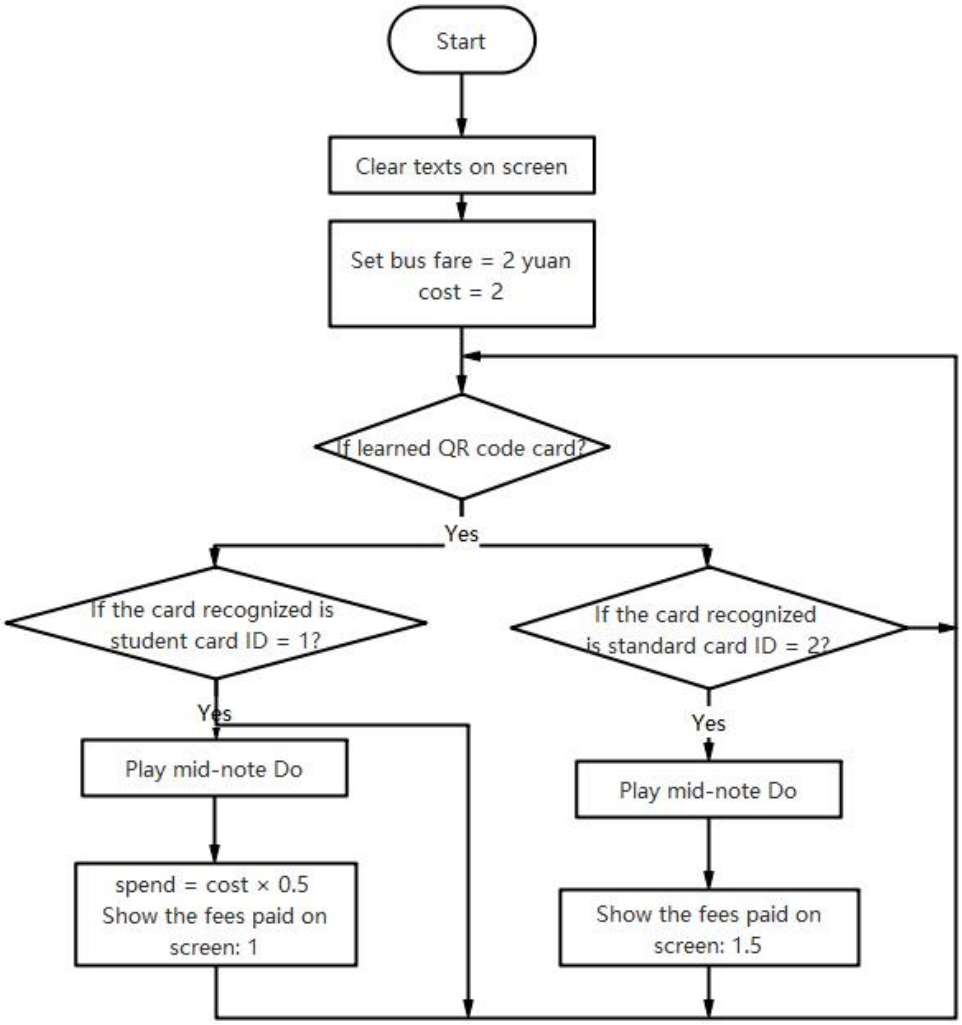
Function instruction:

Different types of bus cards, such as student cards, and standard cards are designed for different groups of people. Each type of bus card corresponds to a charging standard (take 2 yuan as an example, the charging standard and discount can be modified by yourself):

Bus Card	Discount	Fee
Student Card（ID1）	50%	1 yuan
Standard Card（ID2）	25%	1.5 yuan

Before designing the program, it has defaulted that HUSKYLENS has learned different bus cards (for specific operations, please refer to the demonstration of HUSKYLENS QR code recognition function).

Flowchart analysis



Sample program



Operating effect

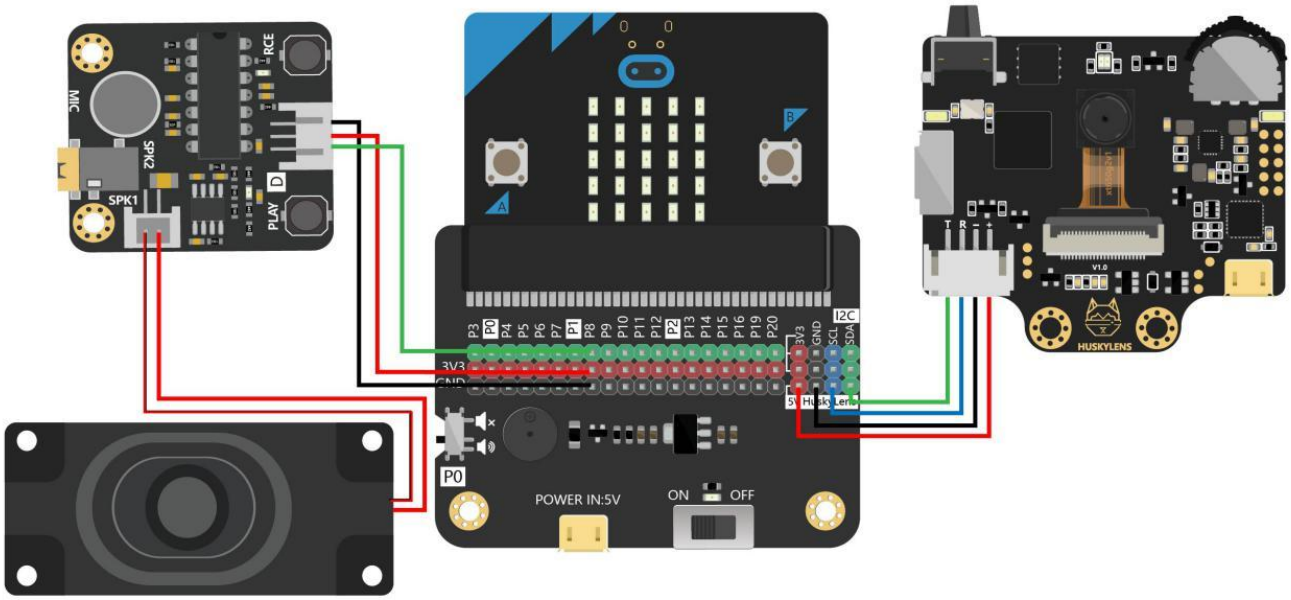
When a certain kind of card is recognized, the corresponding fee to be deducted will be displayed on the HUSKYLENS screen. The mid-note Do played in the program is mainly used as a reminder. As shown in the figure, the student card pays 1 yuan, and the standard card pays 1.5 yuan.





Task2: Voice broadcasting “payment successful”

Hardware connection



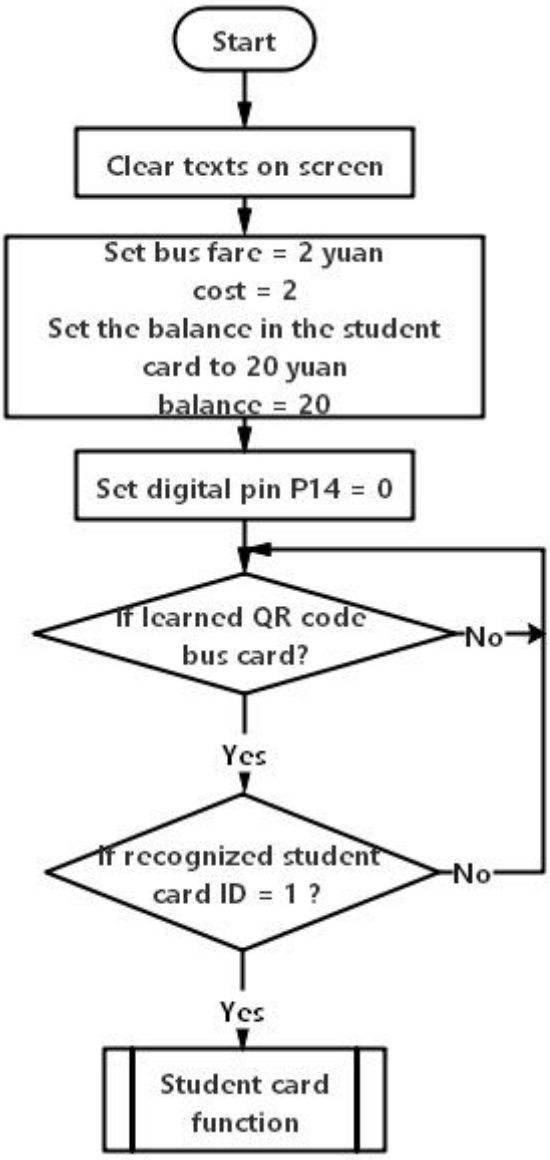
Program design

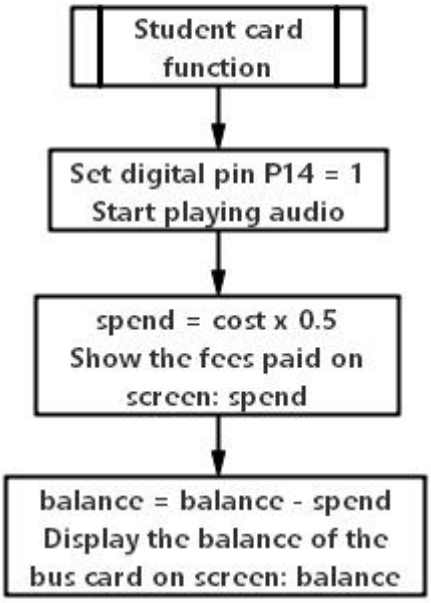
Function instruction:

In this task, I take the elderly card as an example, and add two new functions on the basis of task 1: When the HUSKYLENS camera recognizes the QR code bus card, the voice broadcast "Payment successes, get in please.", and the fee and the balance will be displayed on the screen.

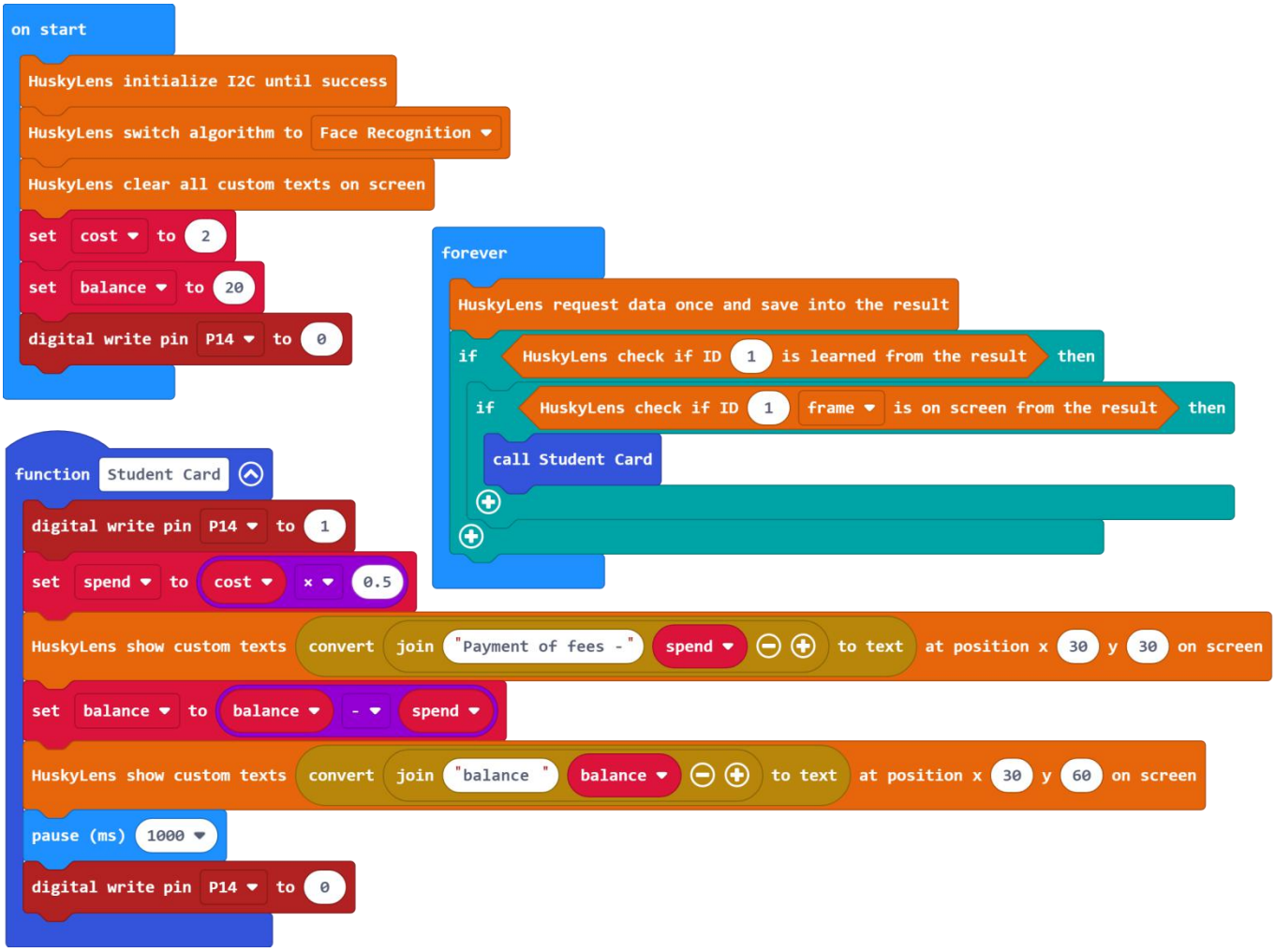
Homework: The following flowchart and program only realize the function of the student card, the rest of the standard cards need to be realized by yourself!

Flowchart analysis



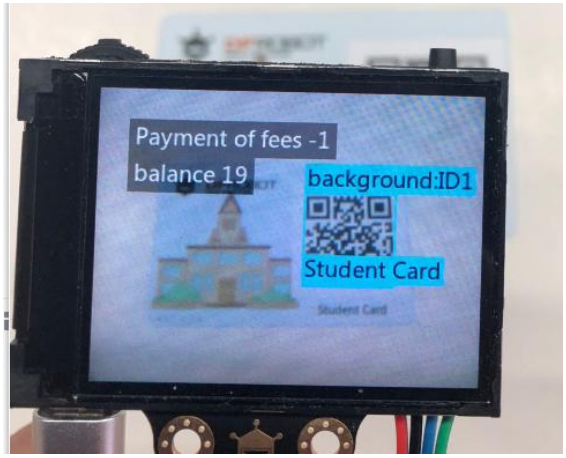


Sample program:



Operating effect

After HUSKYLENS successfully recognizes the QR code bus card, it will play " Payment successful, get in please." and then display the fee and the balance in the card on the screen.



Project Review

This project mainly uses QR code recognition technology to recognize the corresponding bus card type and output the corresponding ID. Then calculate the fees through different charging standards, so as to realize charging different types of cards according to different payment standards. In Task 2, the functions of voice broadcast and balance display are also added, which simulates the working principle of the smart bus ticketing machine.

Project Development

Although there is a balance display function of the smart bus card machine project, when the balance in the card is insufficient, there is no notification. How to remind users when the balance is insufficient?

Note: You need to use conditional judgment statements to judge the balance, use a buzzer to prompt according to the judgment result, or use an audio module+a passive speaker.

Lesson 9: Safety Helmet Reminder

As we all know, everyone must wear a safety helmet before entering the construction site. Don't look down upon the helmet, it can always save a life in the nick of time. Some people might question how much protection the helmet -- just a hard plastic shell -- can provide?

Exactly, the helmet may not protect you adequately when it comes to some large stones and mechanical components that fall from high. However, the probability is not that high. Usually are something like small stones, bricks, screws, etc. They are not big but have a large power when falling from high, which are easy to get workers injured.



If the worker wears a safety helmet, it can block and buffer most of the energy from the falling object, which can protect people.



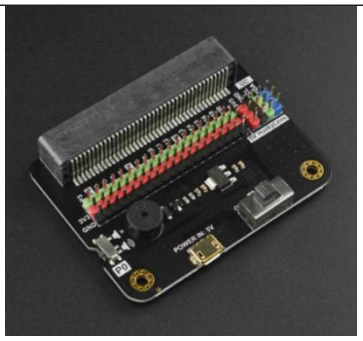
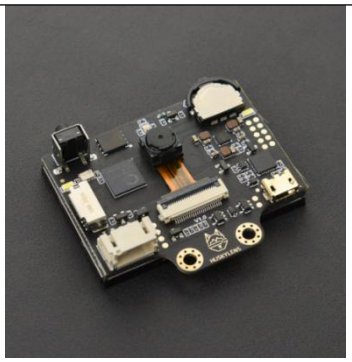
Although it' s so important for workers to wear the helmet, there is still someone forgetting to wear it. So, how can we avoid this?

Next, let' s use the **object classification** function of HuskyLens to make a safety helmet reminder! Only those workers wearing safety helmets can enter the construction site, while others without helmets cannot.

Learning Goals

1. Understand the principle of KNN algorithm for object classification
2. Learn the object classification function of HuskyLens
3. Use the object classification function of HuskyLens to make a safety helmet reminder

Preparation

	
Micro:bit x1	IO Extender for micro:bit V2.0 ×1
	

HuskyLens -- AI Camera ×1	DF9GMS 180° Micro Servo ×1
	
RGB Full Color Light With × 1	Object Classification Card × 2

Note: The object classification card is the same as the face card.

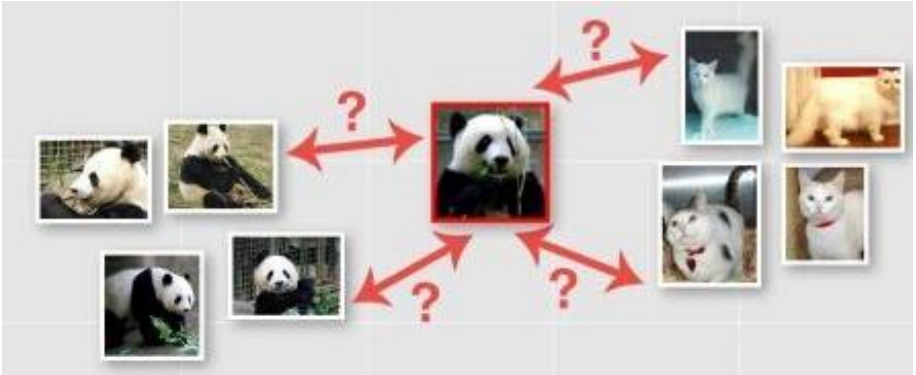
Learning Contents

Operating Principle of Object Classification

KNN can be said to be one of the simplest classification algorithms. At the same time, it is also one of the most commonly used classification algorithms.

1. What is KNN Algorithm

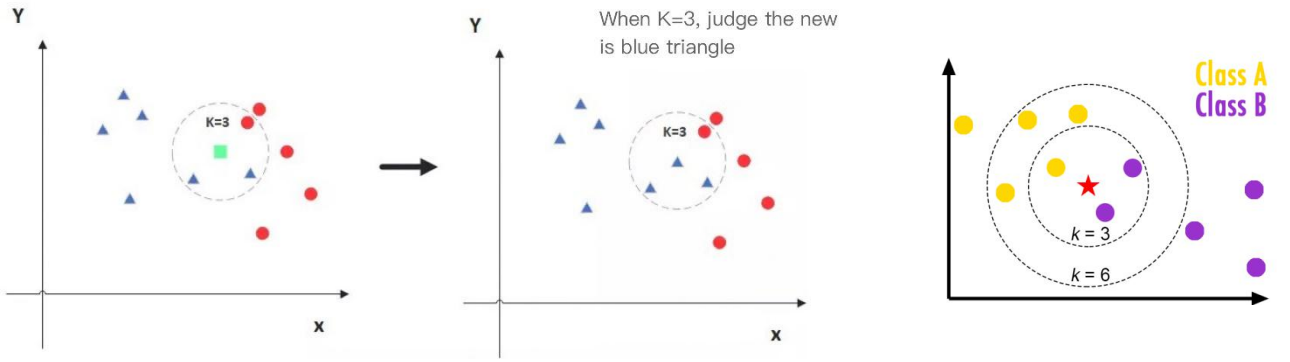
We can see KNN as a kind of rote-learning classifier. It will remember all the training data and will directly match the new data with them. If there are data with the same attributes, the classification of the training data will be used as the classification of the new.



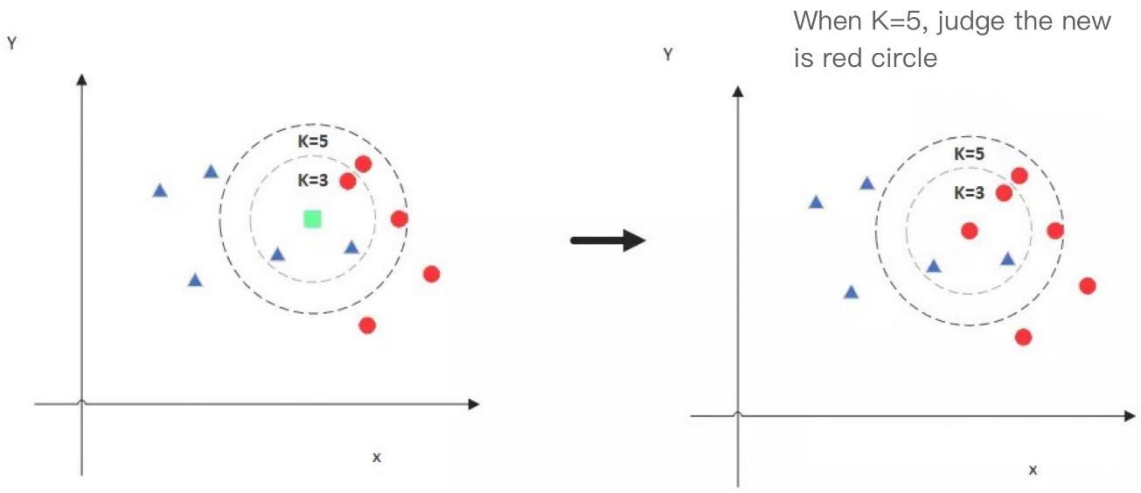
2. KNN Algorithm Principle

The full name of KNN is K Nearest Neighbors. There is no doubt that the value of K is definitely crucial. So what' s the nearest neighbor? Let's take a look at the picture.

The green point in the figure below is the data to be predicted. Assuming K=3, that is to say the KNN algorithm will find the three points closest to it (here we circle it) to see which kind is more. For example, as there are more blue triangles in the picture, and the new green points will be classified into it.



However, when K=5, the judgment will be different. In this case, there are more red circles, so the new green dots are classified as red circles. From this example, we can see that the value of K is very important.



The KNN algorithm will first train multiple data and extracts their characteristics. When predicting a new value, it judges which category x belongs to according to the category of the nearest K points. In the figure on the right, the yellow circle and the purple circle are the training data, and the red five-pointed star is the value to be predicted. When K=3 and K=6, what the predicted results respectively will be?

3. KNN Arithmetic of HuskyLens

In the object classification algorithm of HuskyLens, the K value is 1. Therefore, when HuskyLens predicts new data, it will compare the data with all the training data, and select the classification of the one with the highest similarity as its output result.

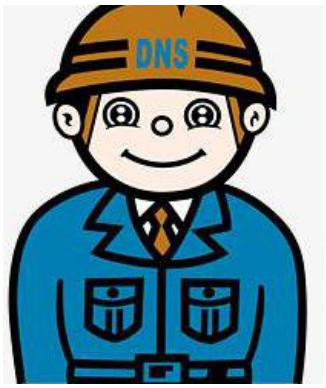
Demonstration of HuskyLens Object classification function

If you want to judge whether a worker is wearing a helmet, you need to classify the image according to the characteristics of the helmet. One type is those wearing the helmet, the other is those not wearing.

This project is to use the **object classification** function of HuskyLens to perform safety helmet recognition function.

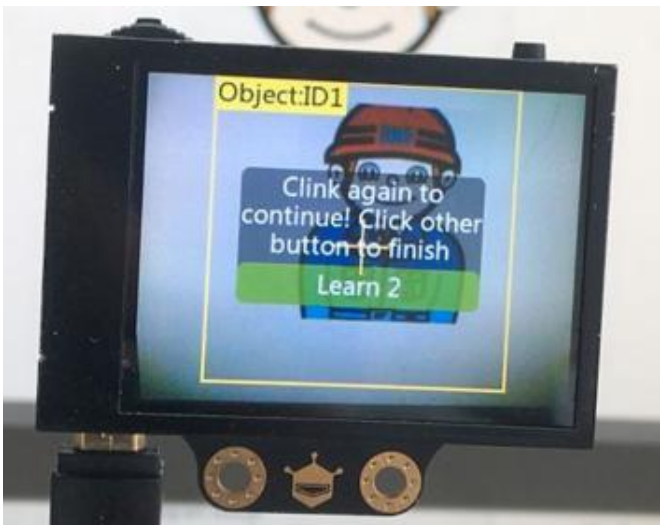
1. Learn Objects

It can be tested by the following cards.





Firstly, align the big orange frame in the center of the HuskyLens screen with the object to be learned (the worker with a helmet), press and hold the "learning button" to learn from different angles and distances. Then release the button after completing learning the first object. At this point, the screen will prompt: "Press the button again to continue! Press other buttons to end". If you want to continue to learn the next object, you can continue to learn the next object by pressing the "learning button" before the countdown ends. If you no longer need to learn other objects, just press the "function button" before the end of the countdown, or do nothing and wait for its end.

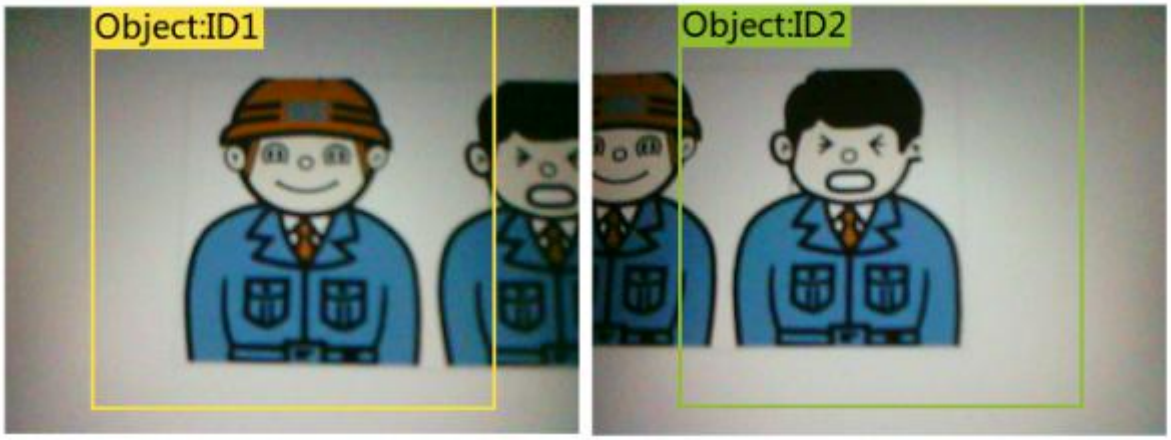


2. Recognize Objects

In the course, we need to continue to learn the next object (the worker without helmet), so press the "learning button" before the countdown ends, and then align the orange frame in the center of the screen with the object to be learned, press and hold "Learning button" to complete learning the second object. So on and so forth.

The object ID is consistent with its input order. That is, the learned objects will be marked as "object: ID1", "object: ID2", "object: ID3", and so on in order. Moreover, different objects have different border colors.

When HuskyLens detect the learned object again, its ID number will be displayed on the screen. As shown in the figure below, when HuskyLens recognizes that the worker is wearing a helmet, the screen displays ID1. If he does not wear it, it will display ID2.



3. FAQ

Q: Can the object classification algorithm get the relative position of the object on the HuskyLens screen?

A: No, it can’ t.

Under the object classification algorithm, the position of the output box is fixed, and its x and y center coordinates on the screen remain unchanged, so HuskyLens cannot get the relative position of the object on the screen. But it can learn different positions of the object as different IDs, and judge the position by them. For example, in an unmanned vehicle, learn ID 1, 2, 3 as the left, middle, and right sides of the road respectively. By judging the ID, the position of the unmanned vehicle relative to the road can be known.

Q: How to improve the accuracy of recognition under the object classification algorithm?

A: Press and hold the "learning button", you can record the target picture from multiple angles and distances to improve the accuracy.

Project Practice

This project will be divided into two tasks. First, we will learn to use the object classification function of HuskyLens to identify whether there is a helmet or not. Second, add LED strip lights and servos on this basis, and achieve the reminder effect according to their status.

Task 1: Show whether the one wears a safety helmet

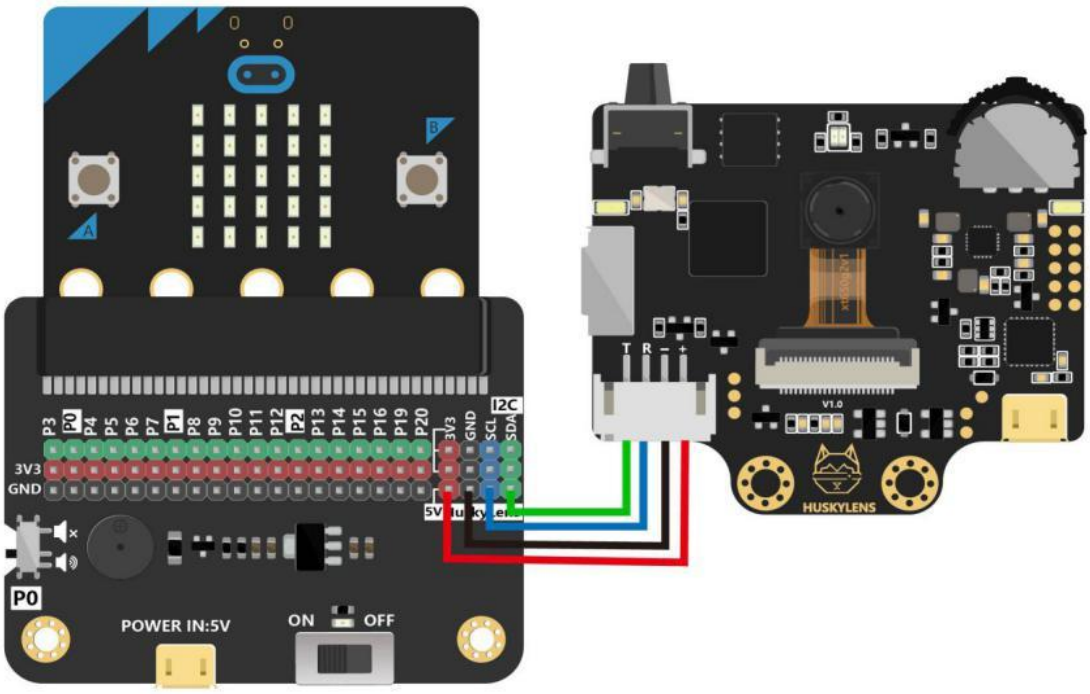
Use the HuskyLens sensor to learn three types of ID – the background, wearing a safety helmet, and not wearing a safety helmet. Write a program to rename the ID. In this way, we can identify whether the worker is wearing a safety helmet.

Task 2: Safety Helmet Reminder

On the basis of task 1, add LED strip lights and servos to judge whether workers entering the construction site with safety helmets, and perform corresponding operations and reminders.

Task 1: Show Whether One Wears a Safety Helmet

Hardware Connection



Program Design

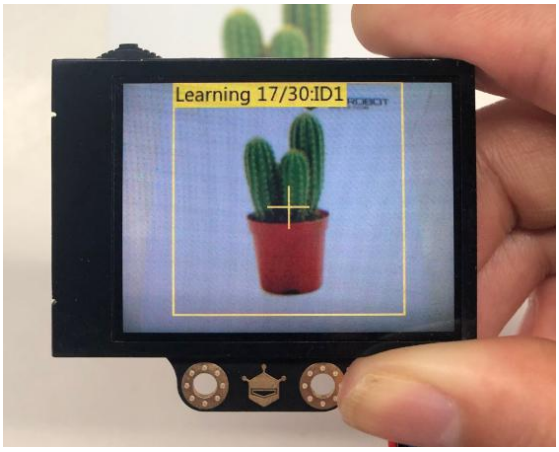
Function Description:

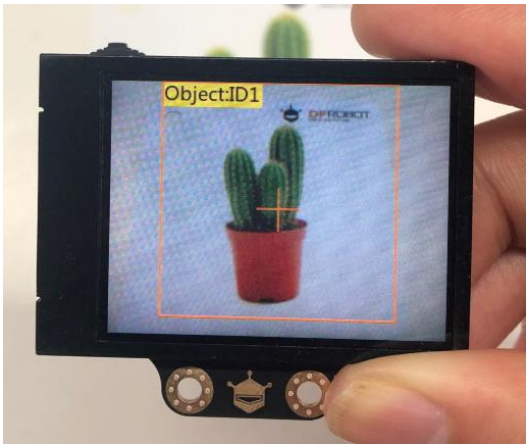
The object classification function can learn different photos of one object, and then use the built-in machine learning algorithm for training. After the training is completed, the algorithm will be named again. When the learned objects reappear on HuskyLens camera screen, HuskyLens can recognize them and display the algorithm name and its ID number.

Note: The more photos of a object HuskyLens learns, the more accurate the recognition will be. The default setting is to learn more than one photo.

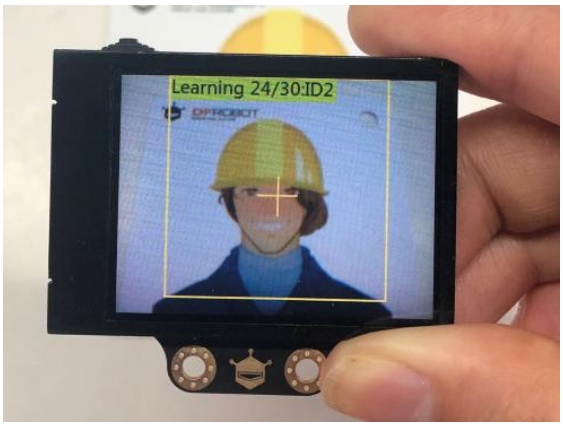
Learn and Recognize

Select the HuskyLens object classification function, press and hold the "learning button" to learn multiple background pictures to get ID1.





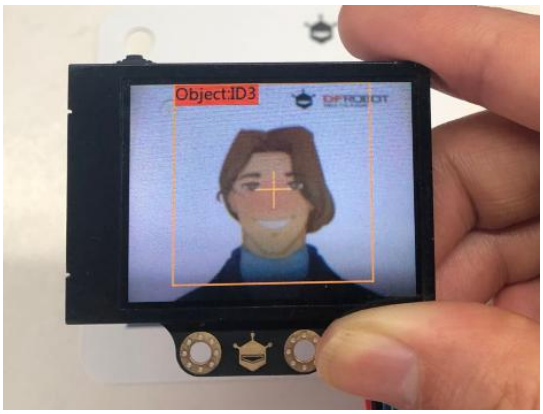
Then learn the character wearing a helmet, long press the "learning button" to learn multiple characters with helmets to get ID2.



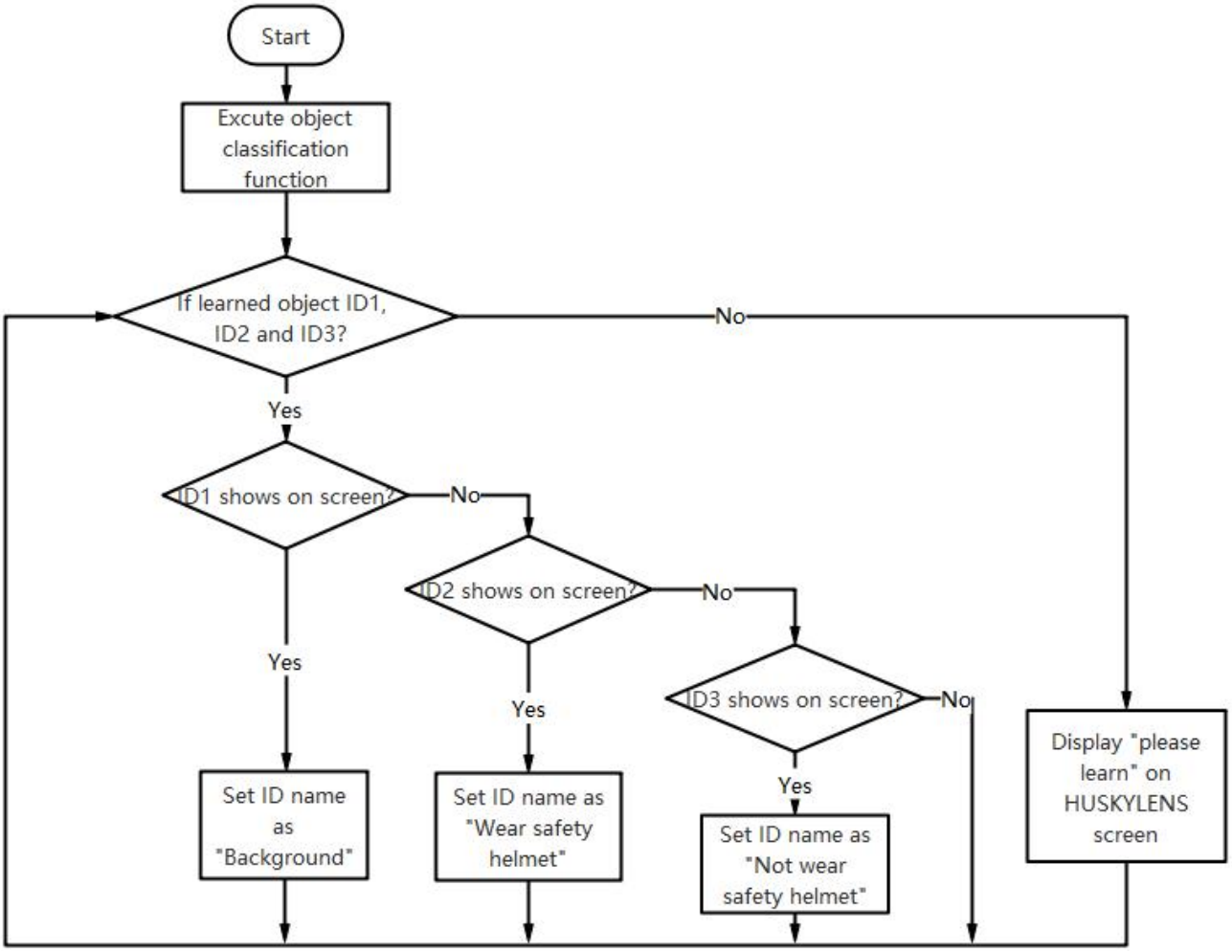


Finally, learn the character without a helmet to get ID3.

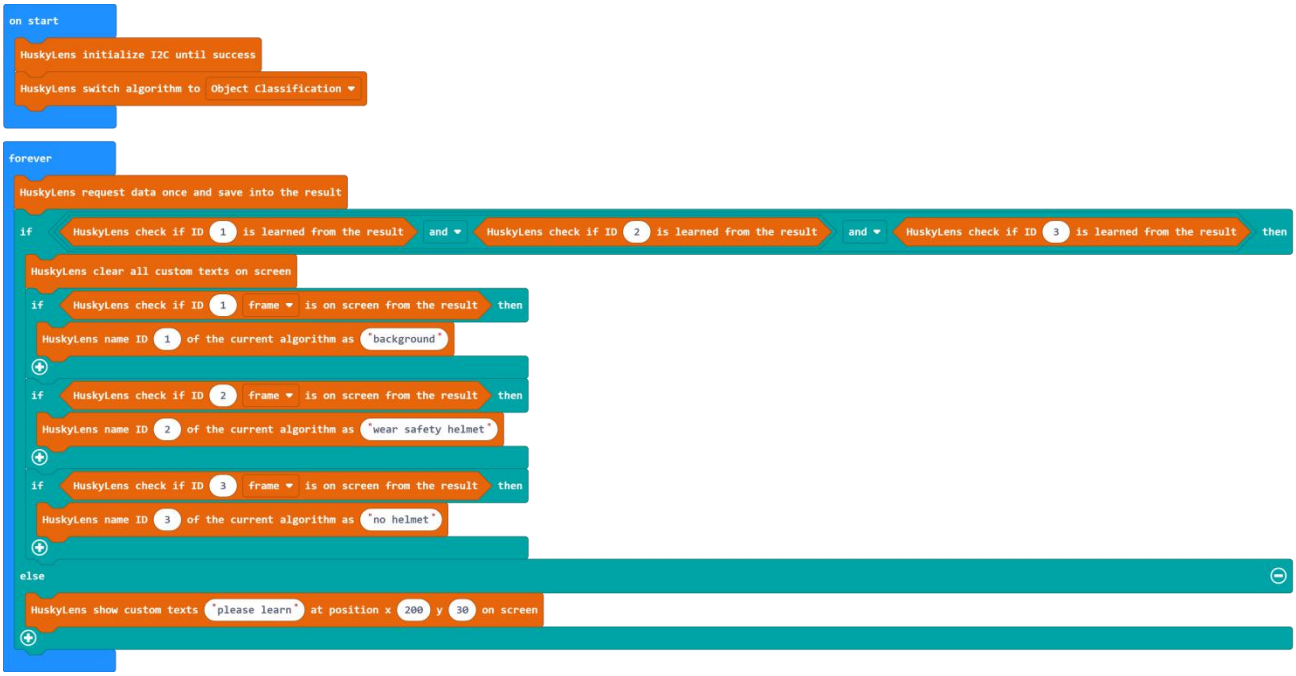




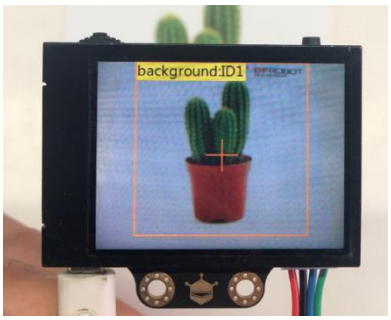
Program Flow Chart Analysis:



Sample Program:



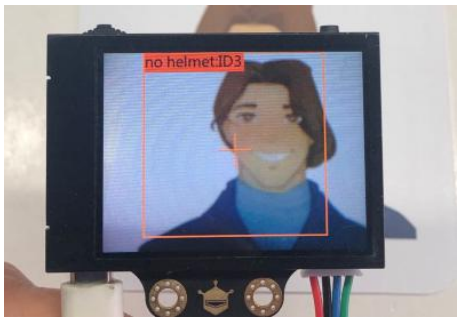
Operating Effect



When the learned background is recognized, display “background: ID1” on the screen.



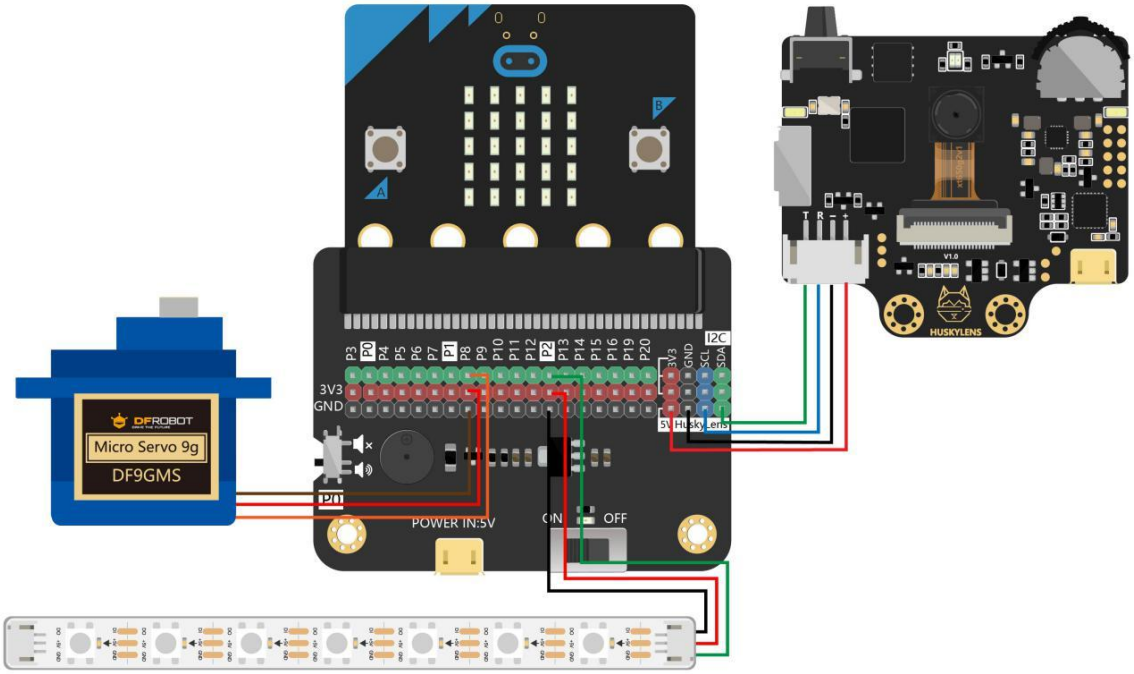
When a person wearing a safety helmet is recognized, display “wear safely helmet: ID2” on the screen.



When a person without a helmet is recognized, display “no Helmet: ID3” on the screen.

Task 2: Safety Helmet Reminder

Hardware Connection

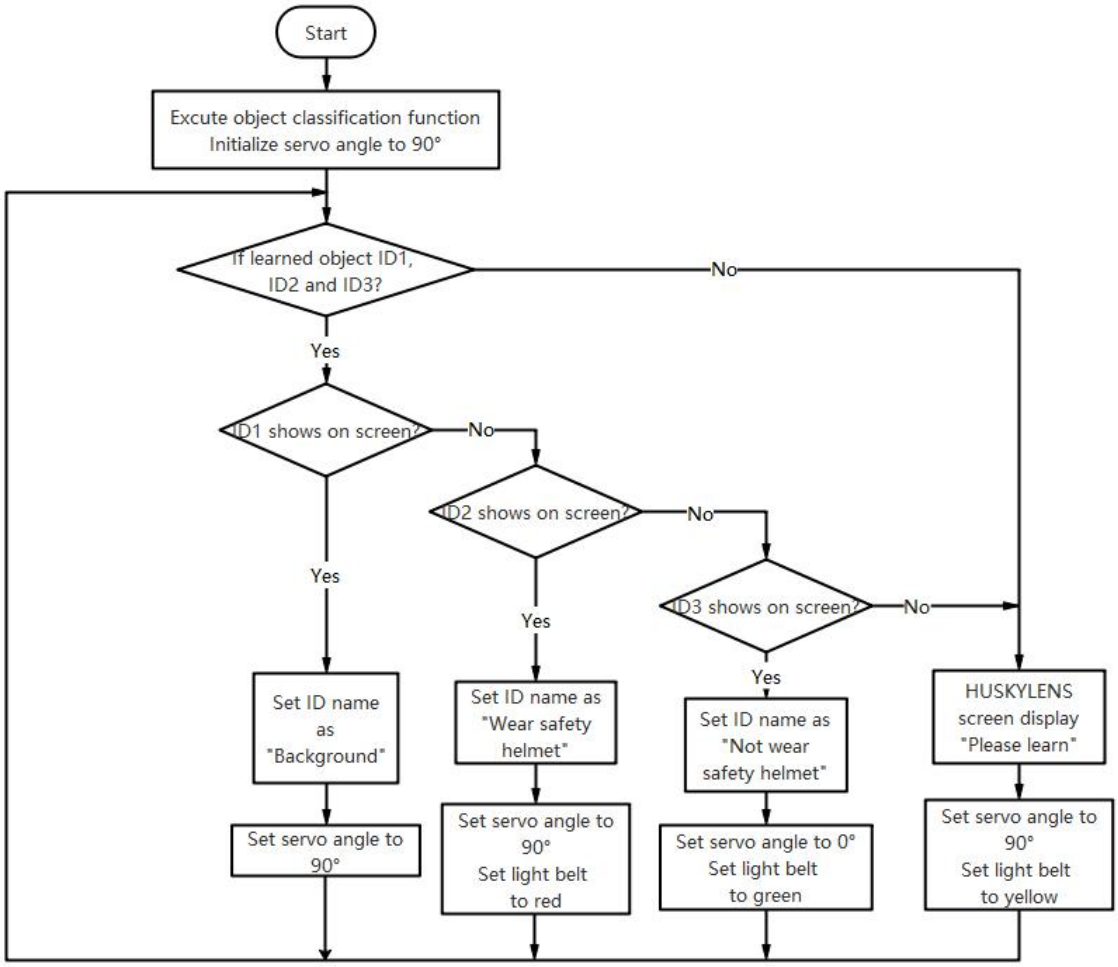


Program Design

Function Explain:

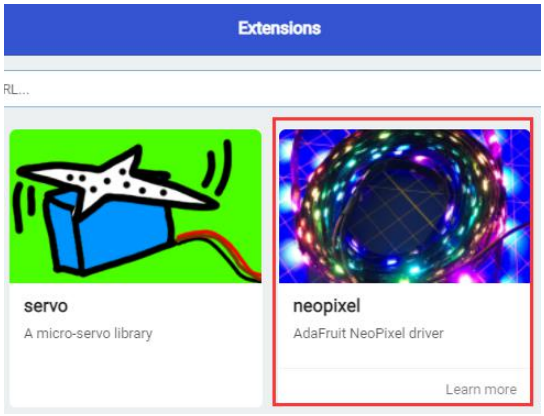
As the recognition of whether there is a helmet has been realized in task 1, we only need to add and set the state of the servos and the LED strip lights on its basis.

Program Flow Chart Analysis:



Sample Program:

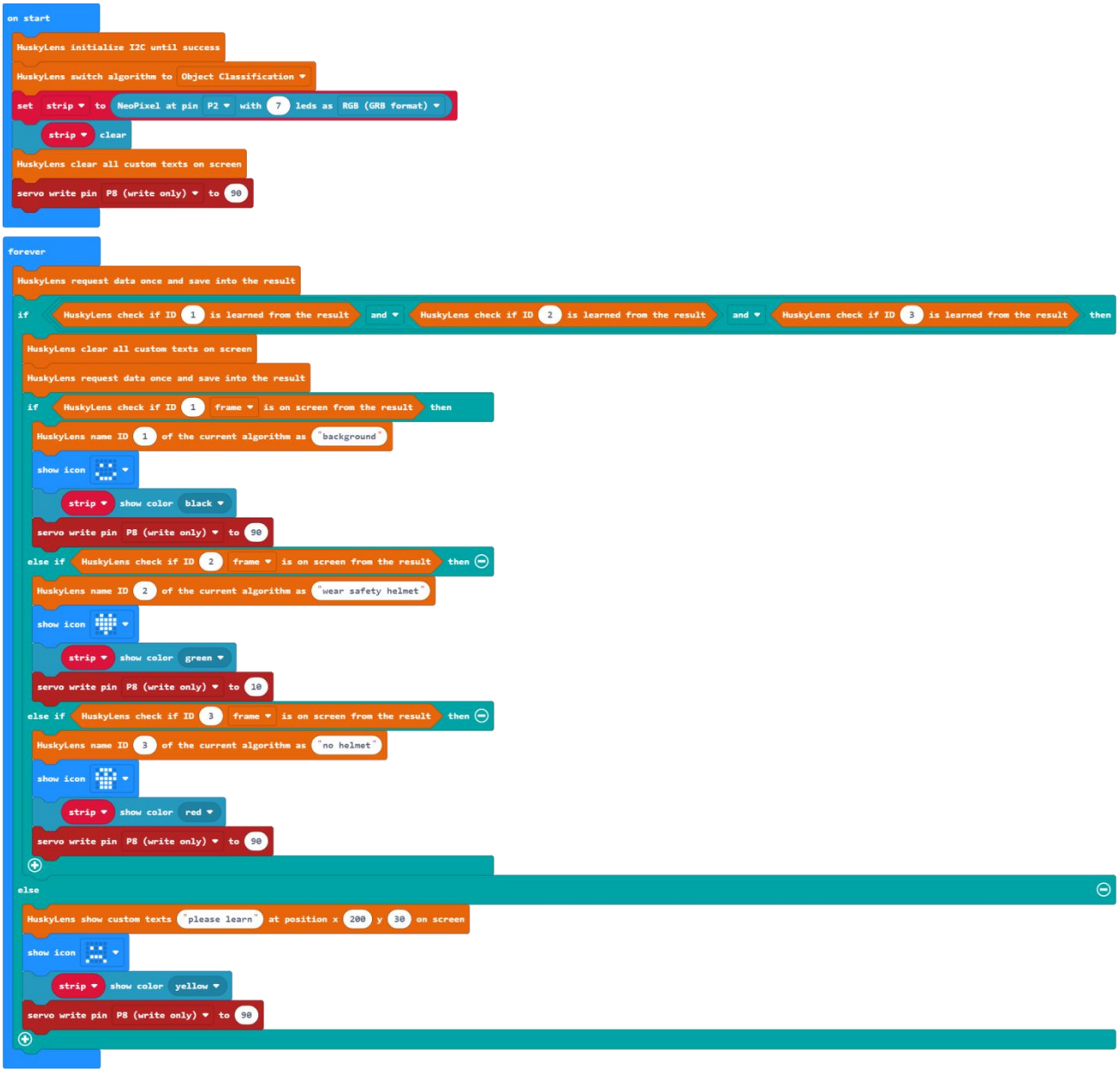
In the program, we need to use the function of the NeoPixel blocks. There are a special RGB light library. The method to add RGB LED strip library is as follows:



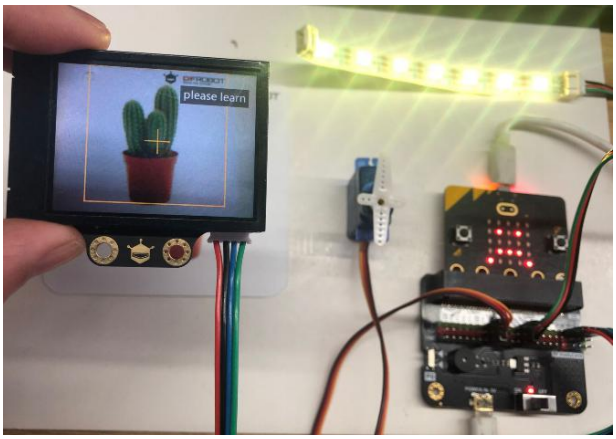
1. Select "NeoPixel" in Expansion.



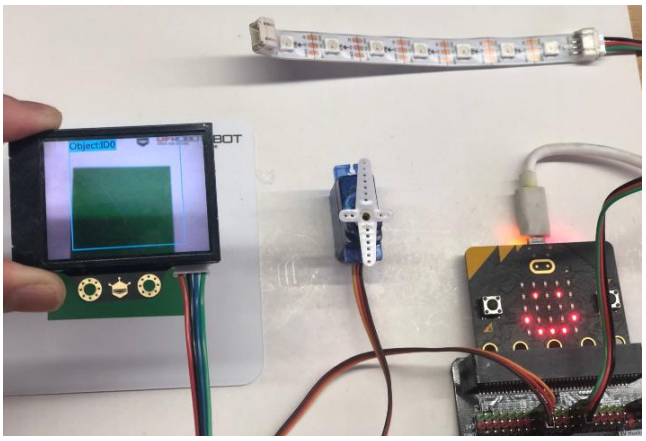
2. The "NeoPixel" blocks will be showed then.



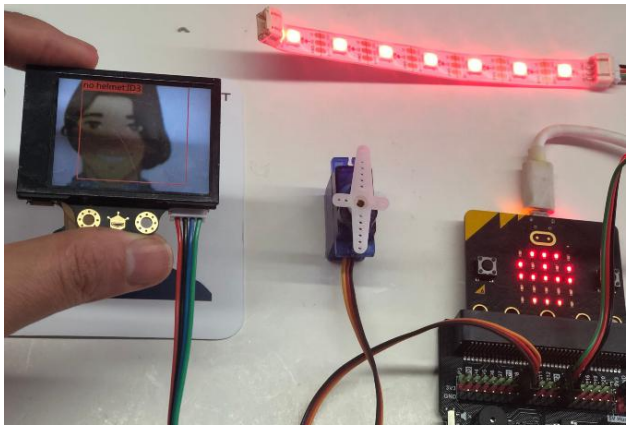
Operating Effect



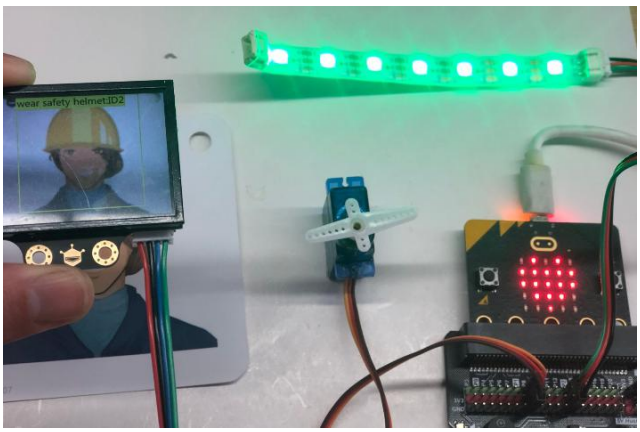
1. Unlearned or incomplete ID, closed, LED strip lights up in yellow.



2. No one at the door, closed, light off.



3. Detected a person without a helmet, closed, LED strip lights up in red.



4. Detected a person with a helmet, open, LED strip lights up in green.

Project Review

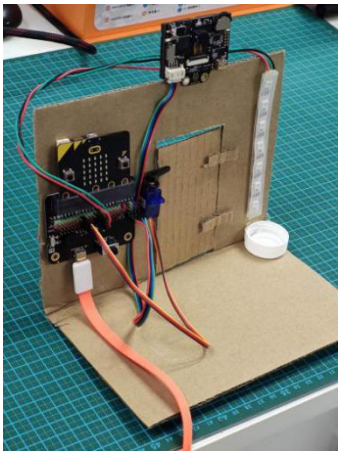
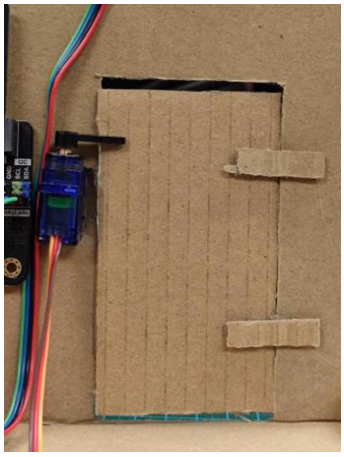
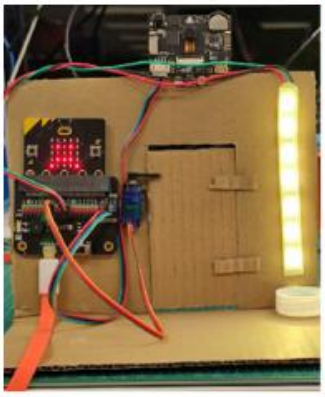
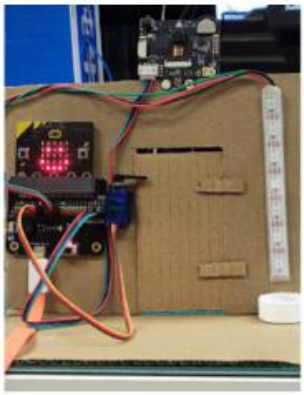
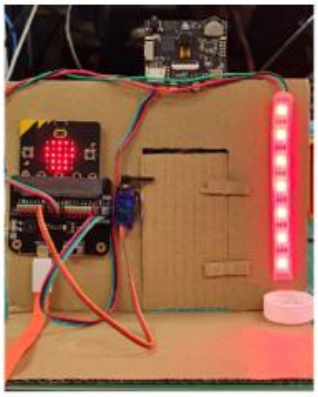
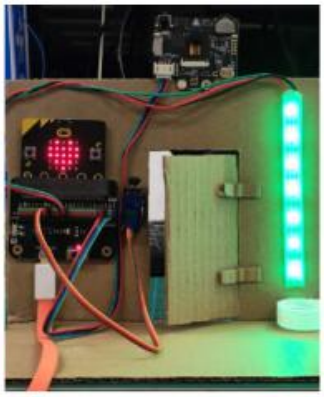
In this project, we learned how to use the object classification function of HuskyLens to made a safety helmet reminder with the servo and led strip. It not only allows workers to pay attention to wearing helmet, but also prevents workers who are not wearing helmets from entering the construction site.

Project Expansion

In this project, we mainly used the object classification function to classify people wearing safety helmets into one category, and those who do not wear into the other category. In fact, there is a wide

range of applications of object classification. Can you come up with other interesting projects by using the object classification function of HuskyLens?

Of course, you can continue to improve the helmet reminder and use the resources at hand to build a real model. The picture below is to use discarded corrugated paper and fix the hardware on it to make a safety helmet detection door on the construction site. Come and have a try!

Lesson 10: Decrypt Binary

There are only 10 types of people in the world-- those who understand binary, and those who don’ t. If you think there is a mistake in this joke, then you must be the latter type of person. The pun is that the binary translation for decimal digit 2 is “10”. Not decimal TEN, but a binary one-zero. People who understand binary codes would know that the actual meaning of this line is “There are only 2 types of people in the world....”

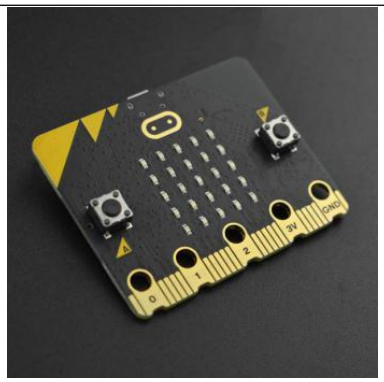
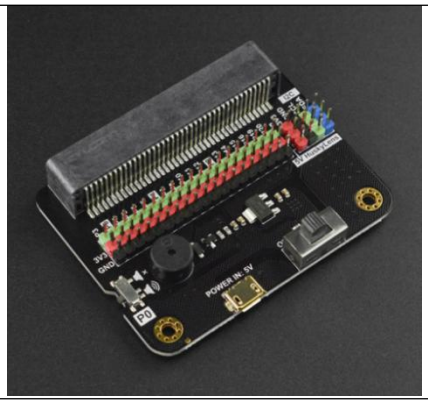


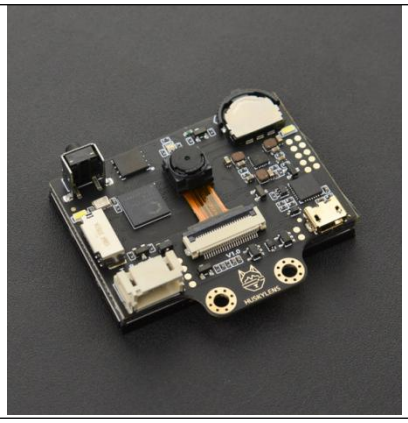
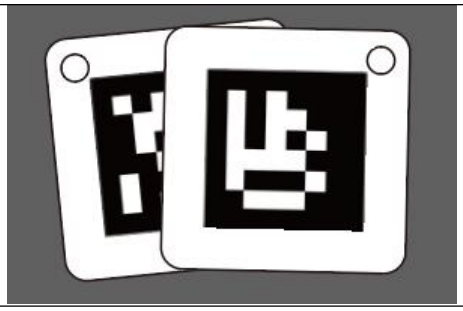
We often see scenes similar to the following picture in science fiction or hacker movies: countless numbers moving downward on the screen. Upon closer observation, we will find that there are only two types of codes: 1 and 0. This is the most basic number system that computers run: binary. So, why does the "10" in the binary system correspond to the "2" in the decimal system? Now, let's learn about the binary conversion method together.

Learning Objectives

1. What is binary?
2. How to count in binary
3. Binary and decimal conversion

Preparation

	
micro:bit v2 ×1	IO Extender for micro:bit V2.0 ×1

	
HUSKYLENS ×1	LED switch-Blue ×1
	
Tag card × 2	

Learning Content

Before learning binary, it is useful to briefly discuss the more familiar decimal counting system as a frame of reference.

Decimal counting uses ten symbols 0 through 9. Counting begins with the incremental substitution of the least significant digit (rightmost digit) which is often called the first digit. When the available symbols for this position are exhausted, the least significant digit is reset to 0, and the next digit of higher significance (one position to the left) is incremented (overflow), and incremental substitution of the low-order digit resumes. This method of reset and overflow is repeated for each digit of significance. Counting progresses as follows:

000, 001, 002, ... 007, 008, 009, (rightmost digit is reset to zero, and the digit to its left is incremented)

010, 011, 012, ...

...

090, 091, 092, ... 097, 098, 099, (rightmost two digits are reset to zeroes, and next digit is incremented)

100, 101, 102, ...

Learn Binary

1. What is a binary number?

The binary system works the same way as decimal, except that only the two symbols 0 and 1 are available. And instead of multiplying the digit by a power of 10, we multiply it by a power of 2. Thus, after a digit reaches 1 in binary, an increment resets it to 0 but also causes an increment of the next digit to the left:

0000,

0001, (rightmost digit starts over, and next digit is incremented)

0010, 0011, (rightmost two digits start over, and next digit is incremented)

0100, 0101, 0110, 0111, (rightmost three digits start over, and the next digit is incremented)

1000, 1001, 1010, 1011, 1100, 1101, 1110, 1111 ...

You may not believe it! Everything you see from mobile phones and computers, whether it is text, pictures, movies, games, or the document you are looking at now, is composed of s 0 and 1.

2. Why do computers ultimately use binary?



① Binary is convenient for the realization of hardware technology. We know that for general circuits, there are only two states: on and off, and binary also only has two symbols: 0 and 1.

② Binary is more suitable for complex computer logic operations. The two values, 0 and 1, correspond exactly to the two states of computer logic algebra "true" and "false".

③ Binary can simplify mathematical operations. Take addition as an example. For binary, there are only two values, 0 and 1, and there are only three combinations [0+0], [0+1], and [1+1], while the decimal system has 55 arithmetic combinations.

How does binary count?

Suppose we were traversing to an ancient time. Now, we were on two beacon towers.



We have agreed that when the enemy comes, we will notify each other and ask for support. When one enemy comes, light the torch once. In this way, we saw the light once, which proved that there was only one enemy and no support was needed. What if 5 enemies come at the same time? There are two ways to convey information:

① Light and extinguish the torch 5 times.

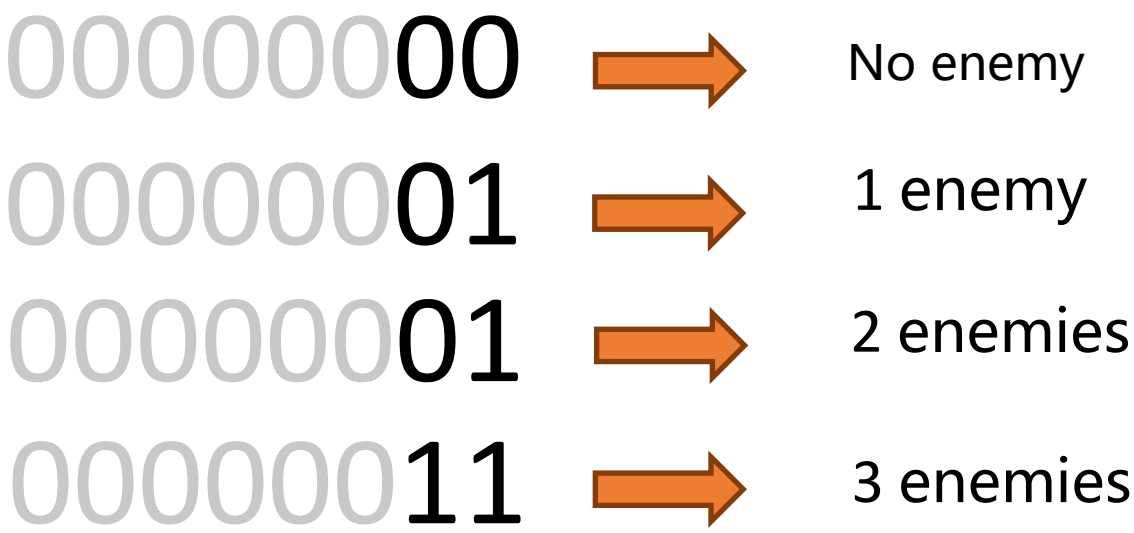
② Light 5 torches at the same time.

When I received the signal, I knew that you were facing 5 enemies, and it was enough to send 3 soldiers to support you. But what if two hundred enemies come? According to the above two methods, the enemy has already approached the city until the torches have been lighted 200 times; if 200 torches are lit at the same time, it may cause a fire in the city and the enemy could win before arrival. Now it is certainly not realistic for us to use torches to transmit signals, but what about replacing torches with signal lights? Although the signal light can transmit the signal by turning on/off quickly, when a large number of enemies come, it is still not convenient. Do you have a better way to transmit the signal?

The "on" and "off" of the bulb are actually two states. If a few bulbs can be used to indicate all possible situations, isn’t it convenient and simple?



The following situations may occur when two indicators are used for signal transmission:



By analogy, using such 8 light bulbs, by setting the "on" or "off" of the light bulbs in different positions, the situation of more than 8 enemies coming can be expressed.

How to convert between binary and decimal

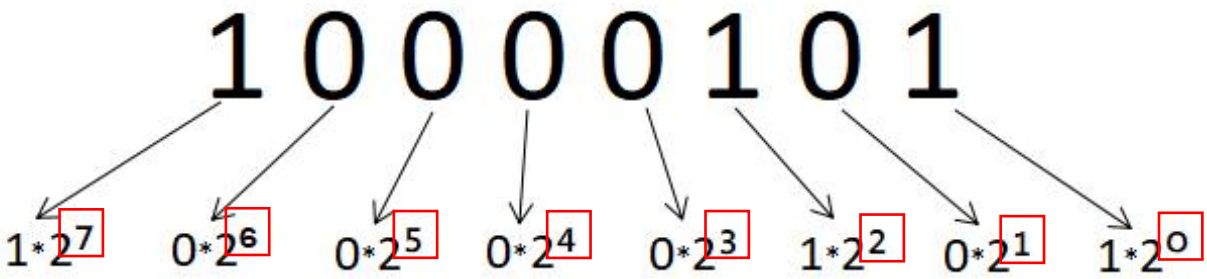
There is a certain degree of difficulty in the method of converting between binary and decimal. Here we mainly introduce how to convert binary to decimal.

1. Binary to decimal

The conversion of binary to decimal is to sum according to the addition rules of decimal. That is, multiply the digits starting from the least significant bit by powers of 2 (2^0, 2^1, 2^2 and so on) till the most significant bit and then sum up the results.

Take the binary 10000101 as an example:

The number circled in red is the power, which you need to count from right to left and start from 0.



$$10000101 = 1*2^7+0*2^6+0*2^5+0*2^4+0*2^3+1*2^2+0*2^1+1*2^0$$
$$= 128+0+0+0+0+4+0+1$$
$$= 133$$

Therefore, the decimal number of the binary 10000101 is 133.

Note: The binary number system employs 2 as the base and requires only two different symbols for its digits, 0 and 1. In the binary system, each digit represents an increasing power of 2, with the rightmost digit representing 2^0, the next representing 2^1, then 2^2, and so on. The table below can be used as a reference:

2 to the power of n								
Power of 2	2^0	2^1	2^2	2^3	2^4	2^5	2^6	2^7
Result	1	2	4	8	16	32	64	128

Through binary numbers, a specific number of signals can be transmitted. If all 8 light bulbs are lit, how many enemies are there? When the 8-digit binary numbers are all 1, that is, 11111111, what number does it represent in decimal?

Binary	1	1	1	1	1	1	1	1
Digit	7th	6th	5th	4th	3th	2th	1th	0th
Power of 2	2^7	2^6	2^5	2^4	2^3	2^2	2^1	2^0
Result	128	64	32	16	8	4	2	1
Decimal	11111111=（ ？ ）							

2. Decimal to binary

The conversion of a decimal integer to a binary integer is as follows:

Step - 1 Divide the decimal number which is to be converted by two which is the base of the binary number.

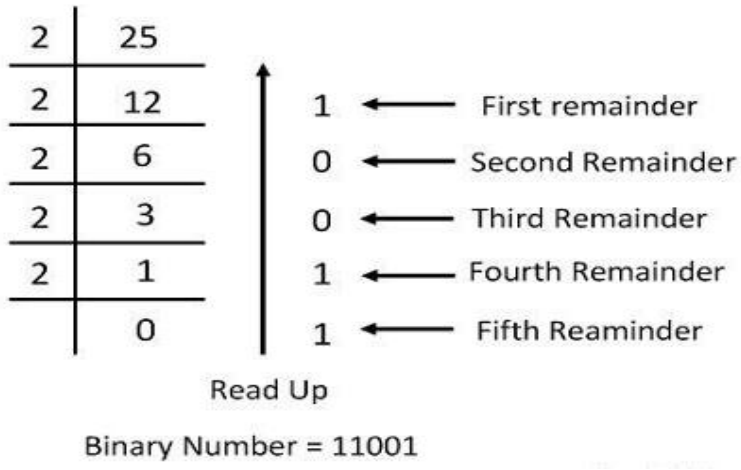
Step - 2 The remainder which is obtained from step 1 is the rightmost bit of the new binary number.

Step - 3 Divide the quotient which is obtained from step 2 and the remainder obtained from this is the second rightmost bit of the binary number.

Step - 4 Repeat the process until the quotient remains zero.

Step - 5 The last remainder obtained from the division is the leftmost bit of the binary number. Hence arrange the number from leftmost bit to the rightmost bit (i.e., from bottom to top).

This method can easily be understood by considering an example which is explained below. Take 25 as an example:



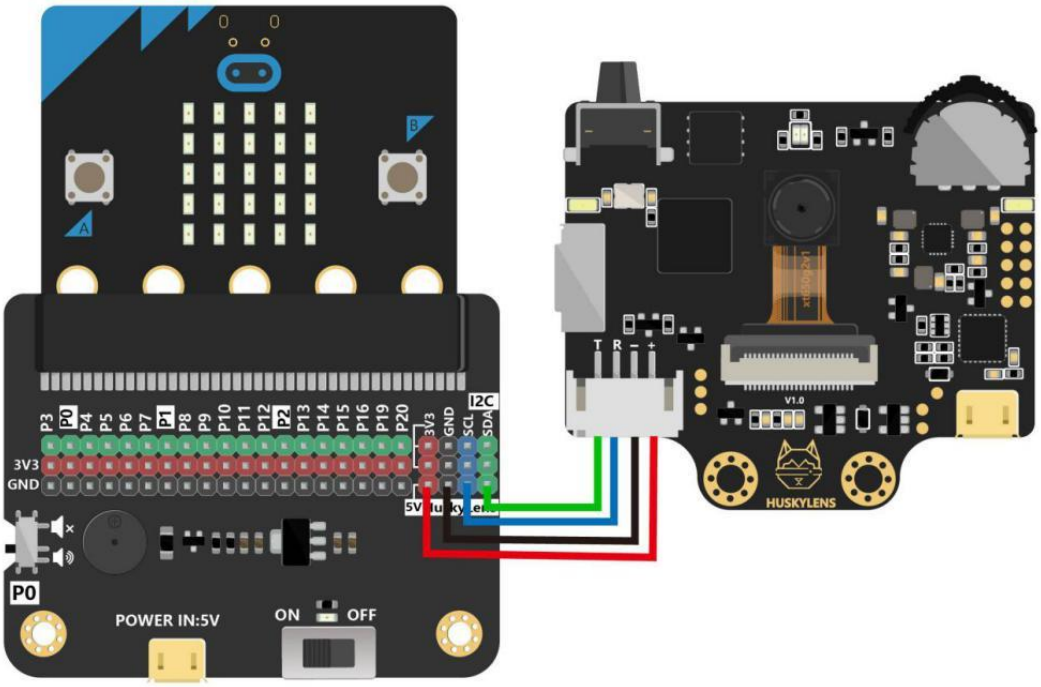
Note: The order of binary numbers is from bottom to top, from high digit to low digit, so the binary number of 25 is 11001.

Project practice

This project only needs two tag paper cards. For the demonstration of the tag recognition function, please refer to Lesson 7. First of all, we default that those two paper cards are learned and can be recognized by HuskyLens. Press the Bistable Button, when HuskyLens recognizes ID1, the screen will display binary 1; when HuskyLens recognizes ID2, it will display binary 0 on the screen. Press the button again to convert the binary data into the familiar decimal number.

Task1: Display binary

Hardware connection

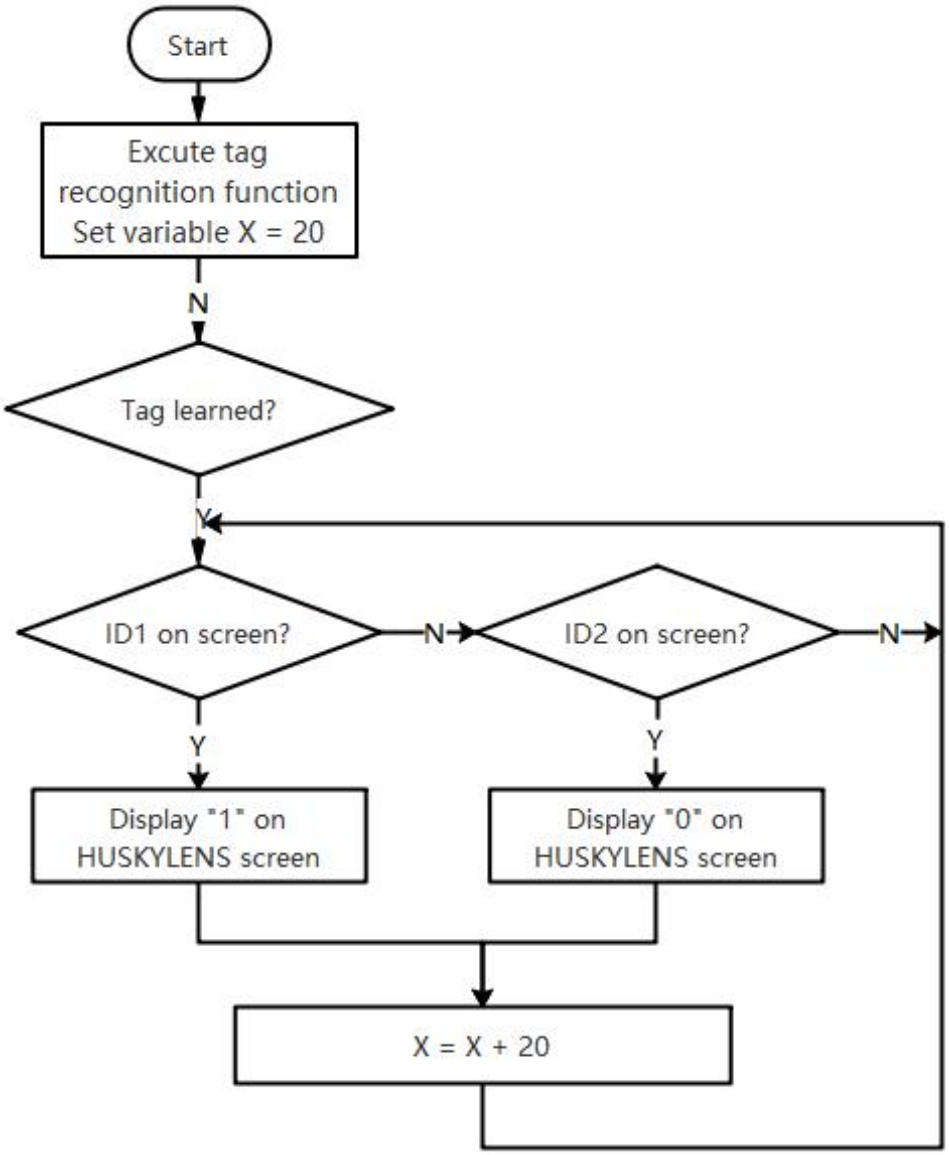


Program design

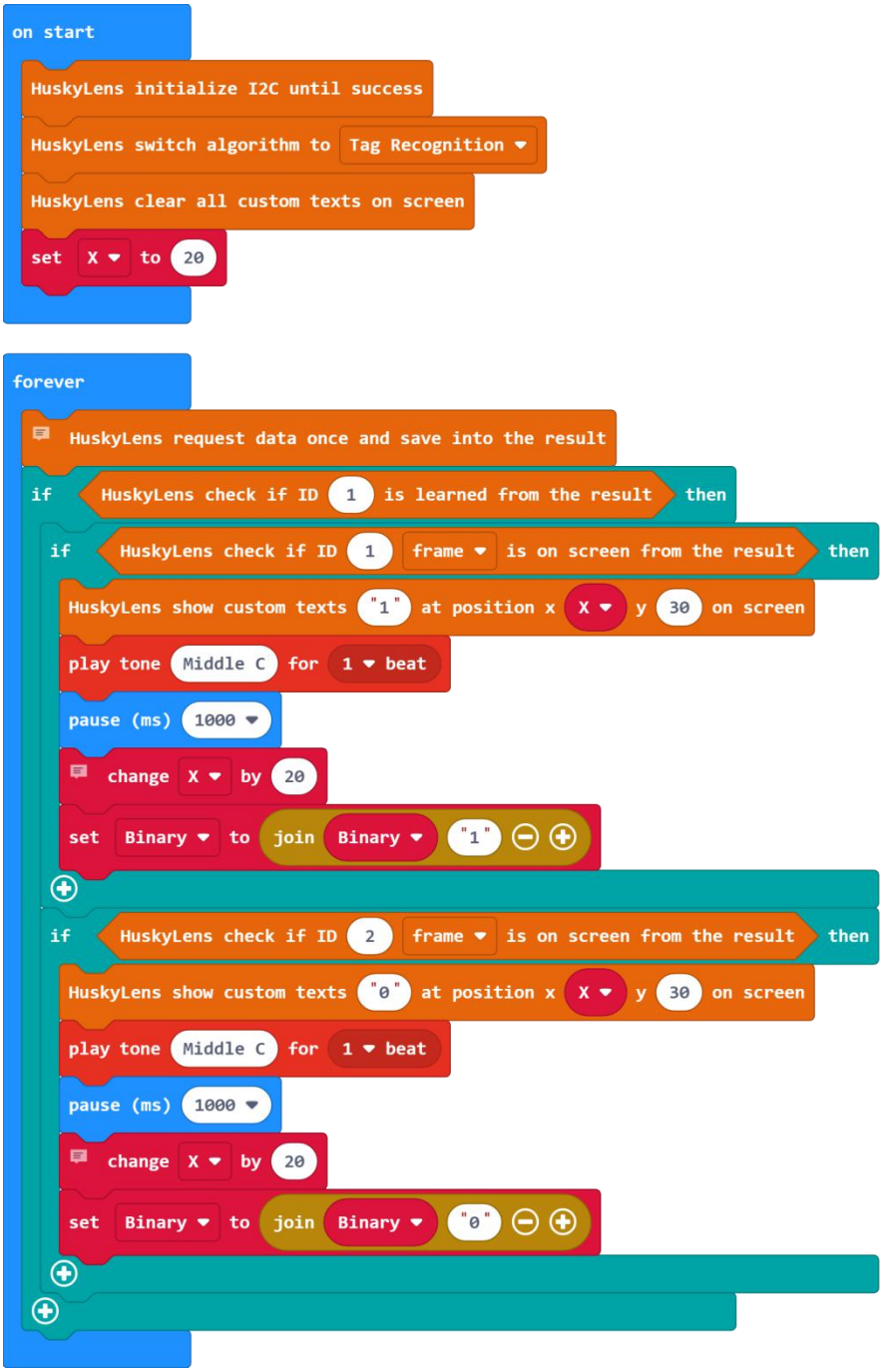
Function instruction:

By recognizing the learned tags, the binary numbers 0 and 1 are displayed on the HuskyLens screen. How can a complete set of binary numbers be displayed? Each time a tag is recognized, a number is displayed, and the following number should not cover the previous number. It needs to be controlled by changing the X coordinate value of the displaying position.

Flowchart Analysis



The Sample Program



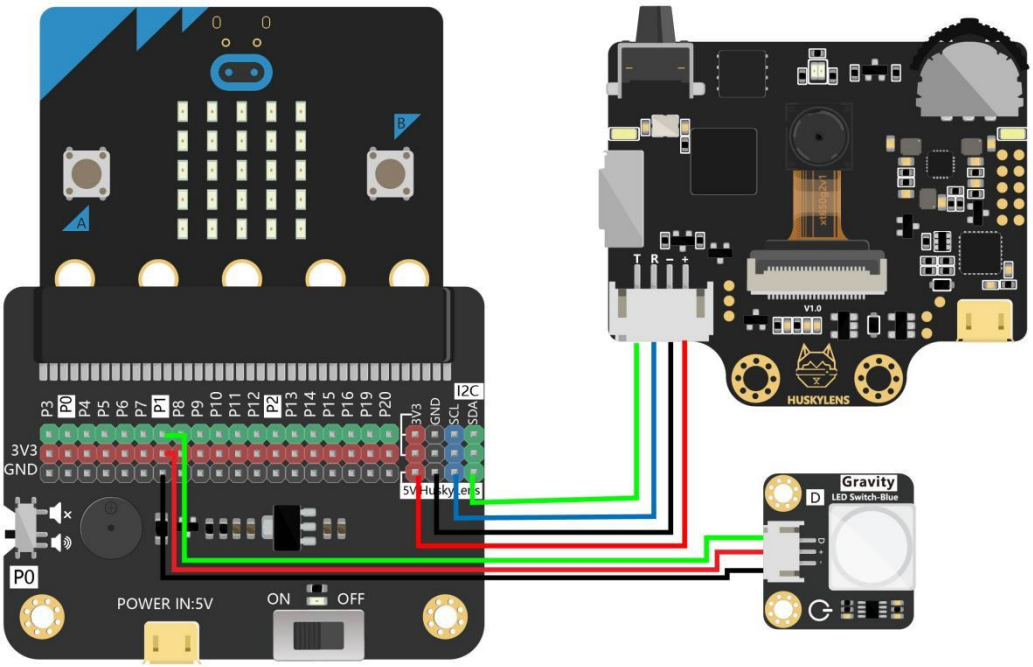
Operating Effect

Run the program, when HuskyLens recognizes the tag ID1, the binary number 1 is displayed on the screen; when HuskyLens recognizes the tag ID2, the binary number 0 is displayed on the screen, and the latter number is positioned behind the former one.



Task2: Decimal to binary

Hardware connection



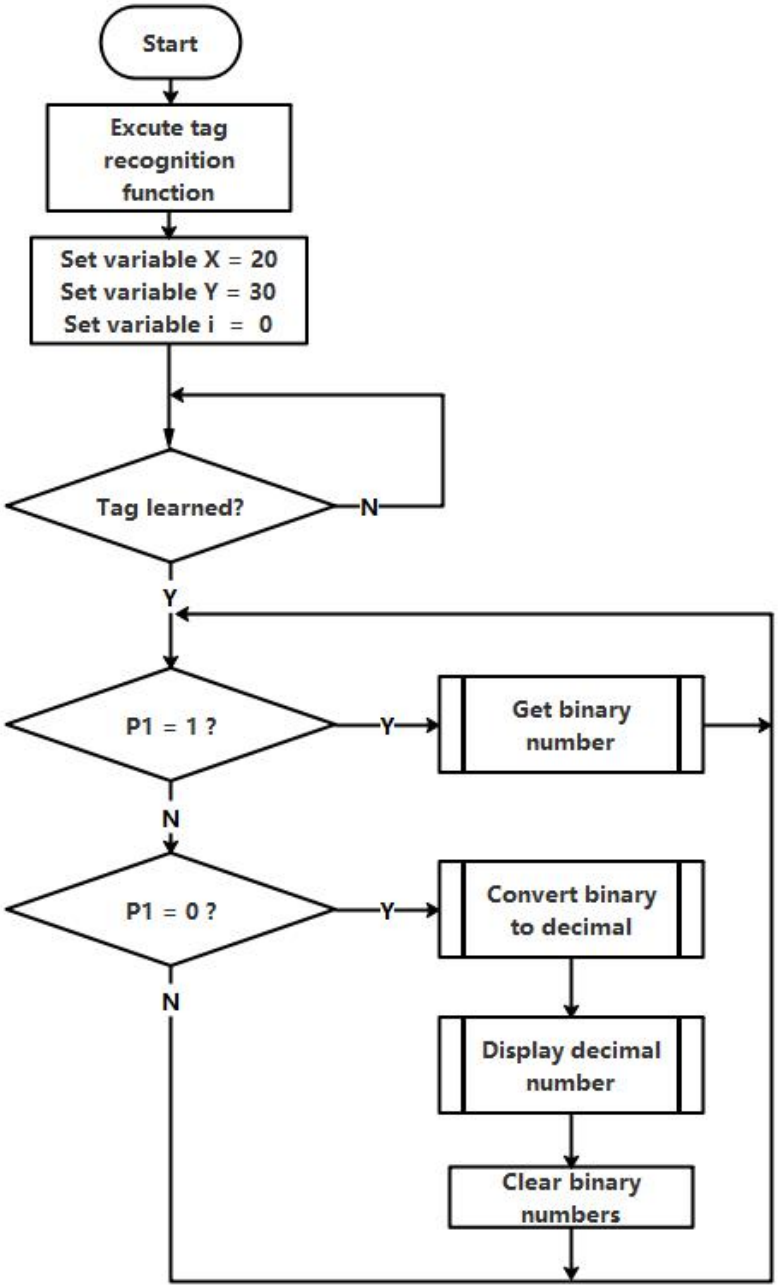
Program design

Function instruction:

This task will perform function encapsulation according to the three different functions of obtaining binary, converting binary to decimal, and displaying characters. Obtaining binary: After pressing the switch, you can perform tag recognition to obtain the binary number; Converting binary to decimal:

press the switch again to convert the binary number to decimal; Displaying: display the decimal numbers on the screen when the button is pressed.

Flowchart Analysis



Note: The flowchart about the three sub-functions is omitted here. Students who are interested can draw the flowchart of the sub-function according to the complete program example.

The Sample Program



Operating Effect

Run the program and press the button. When the indicator is on, the binary number is obtained through the tag. Press the button again, the indicator is off, and display the corresponding decimal number on the next line. Take the binary 100101 as an example, verify whether the decimal number of it is 37.

Get binary



Convert binary to decimal



Project Review

In this project, we learned how to convert between binary and decimal, and use the tag recognition function of HuskyLens to obtain binary numbers, and then convert the obtained binary numbers into decimal numbers. After mastering the binary conversion method, you can play some games with your friends and give them a string of binary numbers to see if he or she understands what you mean.

Project Development

In the binary part, we mentioned that all the text that we see on a computer or mobile phone is composed of binary. What does this mean? In fact, a set of binary numbers can not only be converted into decimal numbers, but also correspond to the corresponding characters (text). Let's use a table to understand the relationship between binary numbers and characters.

Binary	Decimal	Character
0100 0001	65	A
0100 0010	66	B
0100 0011	67	C
0100 0100	68	D
0100 0101	69	E
0100 0110	70	F
0100 0111	71	G
0100 1000	72	H
0100 1001	73	I
0100 1010	74	J
0100 1011	75	K

0100 1100	76	L
0100 1101	77	M
0100 1110	78	N
0100 1111	79	O
0101 0000	80	P
0101 0001	81	Q
0101 0010	82	R
0101 0011	83	S
0101 0100	84	T
0101 0101	85	U
0101 0110	86	V
0101 0111	87	W
0101 1000	88	X
0101 1001	89	Y
0101 1010	90	Z

Simply change the display function in the task 2 program. Just add this block "



" , and then you can also convert binary numbers into A~Z characters and show them on HuskyLens screen, as shown in the figure:



Lesson11: HuskyLens Camera

Photographs are the universal language of our era. With the rapid development of smartphones, phone photography has become more and more popular. Convenient operations and numerous creative apps are changing people's habits of recording images. There are no strict rules for phone photography, no professional equipment is needed, only a pair of eyes that can discover beauty would suffice.



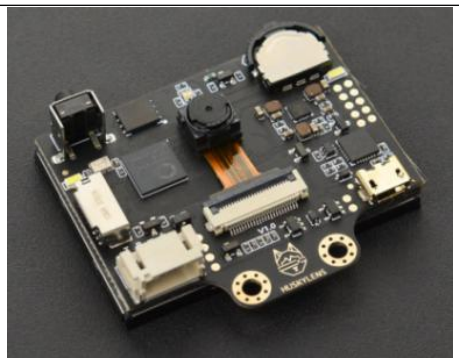
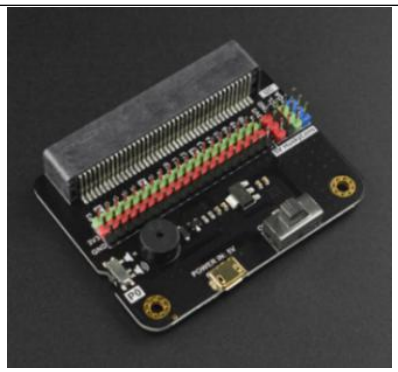
It is not difficult to find in life that more and more people are holding mobile phones to record life, scenery, food, flowers and plants. Different phone photography themes all express their perception of beauty and love for life!

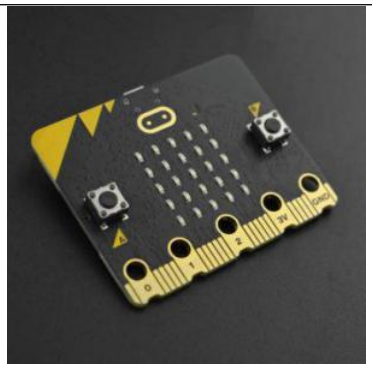
Is HuskyLens related to the camera? Of course, HuskyLens also has a camera function, let' s try making a HuskyLens camera!

Learning Objectives

1. Learn the principle of pinhole image and the origin of the first camera
2. Learn the principle of HuskyLens photography
3. Learn to use HuskyLens to make a camera

Preparation

	
HuskyLens ×1	IO Extender for micro:bit V2.0 ×1

	
micro:bit v2 ×1	SD Card ×1
	
Card Reader ×1	

Note: SD card and card reader are not included in the kit.

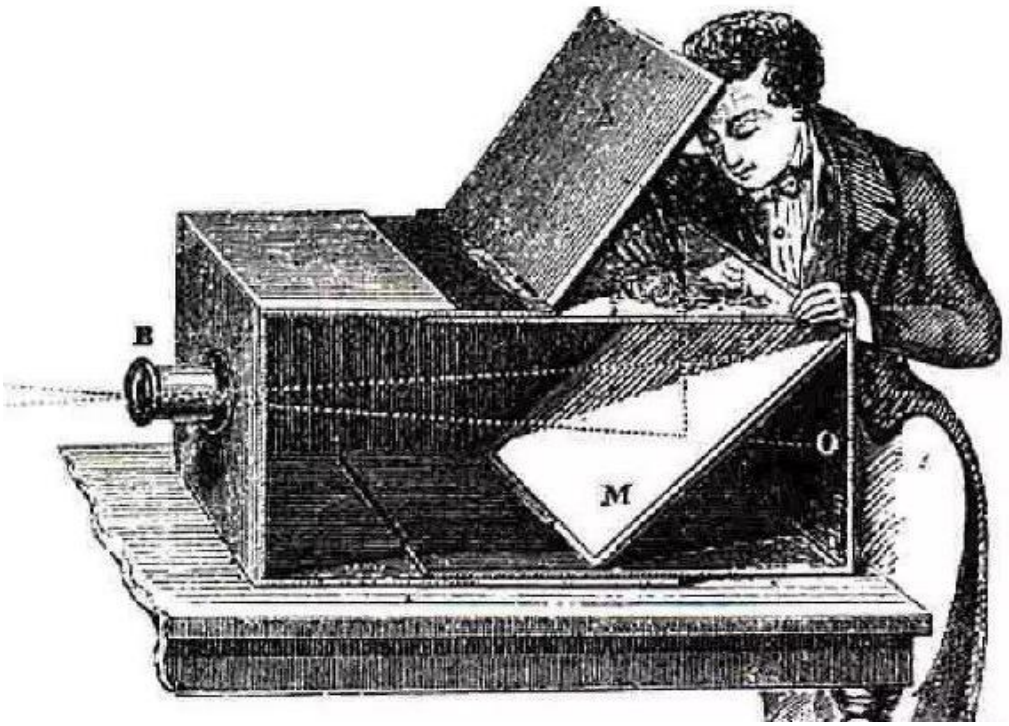
Learning Content

What is a camera? Why does the objective scene freeze on the film as soon as the shutter rings? Especially nowadays, digital cameras and mobile phones are popular with people, and the fixed images that are shot can be immediately seen in the viewfinder of the LCD screen. How are these almost magical operations realized?

The Origin of the First Camera and the Principle of Pinhole Imaging

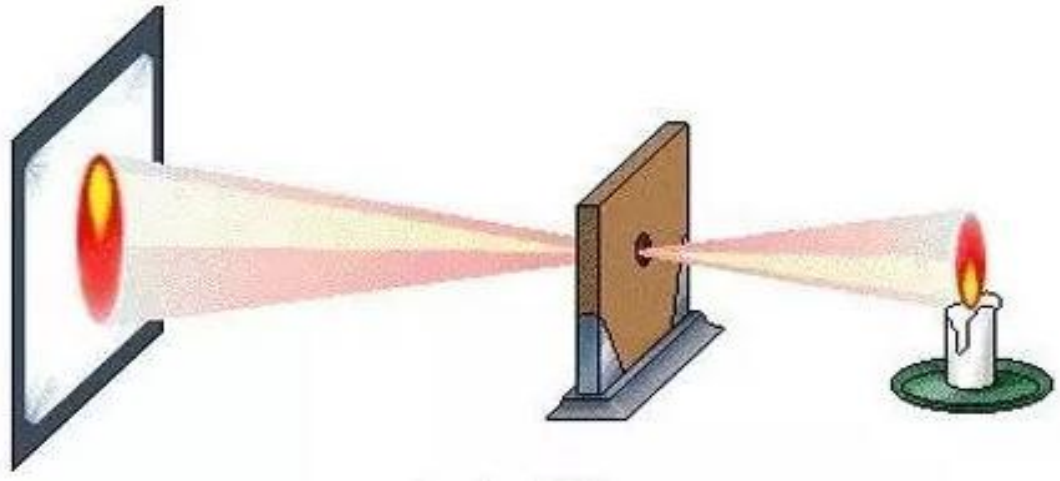
1. The first camera-Camera obscura

Before the camera was invented, people had begun to use the principle of "pinhole imaging" to make optical imaging devices, which are called "Camera obscura". In the first half of the 19th century, people finally found the method and medium to preserve the optical image on the projection surface in the camera obscura. Therefore, the camera obscura is considered the ancestor of the camera.



2. What is pinhole imaging?

Use a plate with a pinhole between the screen and the object, and an inverted image of the object will be formed on the screen. We call this phenomenon pinhole imaging. Moving the plate back and forth, the size of the image will also change accordingly. This functional phenomenon reflects the nature of light propagating along a straight line. The imaging principle of the camera is to use "pinhole imaging". The light enters through the pinhole and maps on the photosensitive film to form an image.

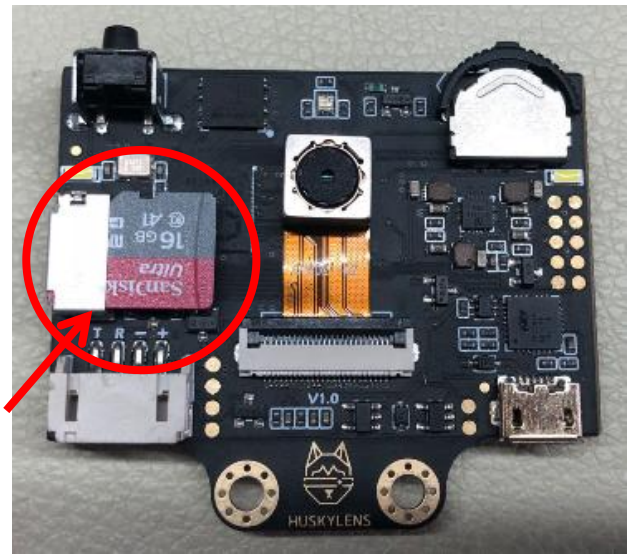


HuskyLens Photograph Principle

1. Can HuskyLens photograph?

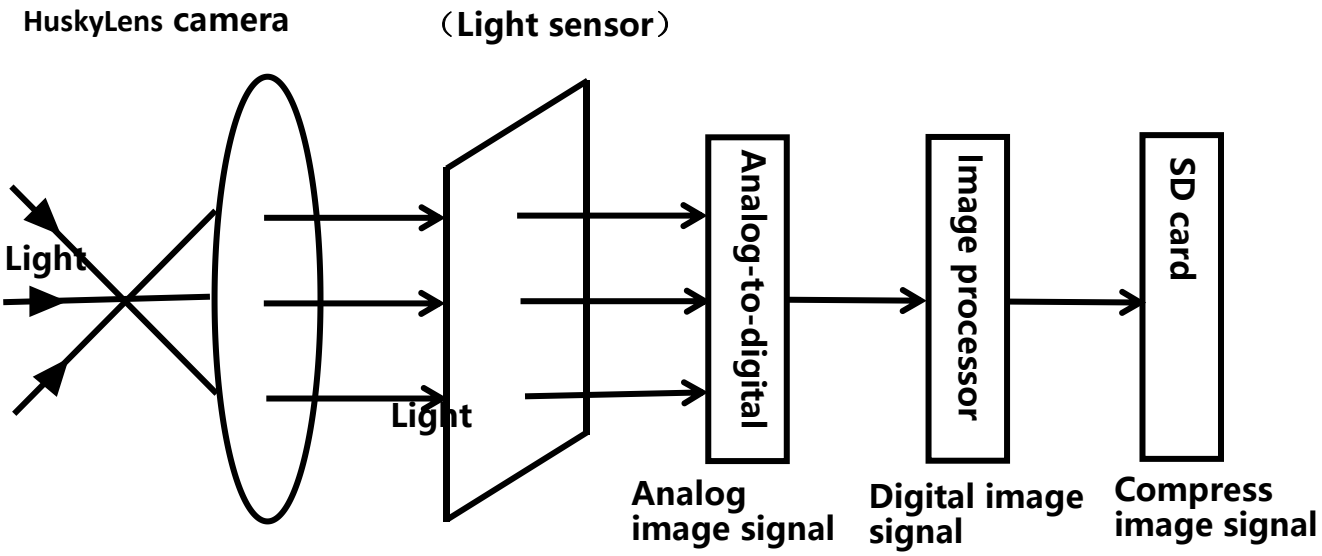
You must be thinking that you are already familiar with the usage and functions of HuskyLens, but you have not found the camera function at all, and there is no shoot button. Is it impossible for HuskyLens to take pictures?

HuskyLens can be used for photograph. As a vision sensor, the HuskyLens has a rear camera with 5 million pixels. Taking pictures is the most basic function. Whether you are using a mobile phone or a camera to take a photo, the photo you take needs to be stored on the SD card or the memory space. Therefore, HuskyLens also needs an SD card to store photos. The SD card is inserted as follows:



2. HuskyLens photography principle

Traditional cameras use "film" as the carrier for recording information, and the "film" of HuskyLens cameras is its photosensitive sensor. The light sensor is the core of the HuskyLens camera and the key technology for taking pictures. It can convert light into electric charge, which is converted into digital signal through an analog-to-digital converter, and the digital signal is saved by the SD card after being compressed.



Project practice

When using a mobile phone to take a photo, you need to trigger the camera button to take a photo, and then three different ways to trigger the camera function.

Task 1: Touch

Use the touch LOGO of micro:bit V2 to simulate the camera button of a mobile phone. Touch the LOGO once, and HuskyLens will take a picture and store it on the SD card.

Task 2: Voice-activated

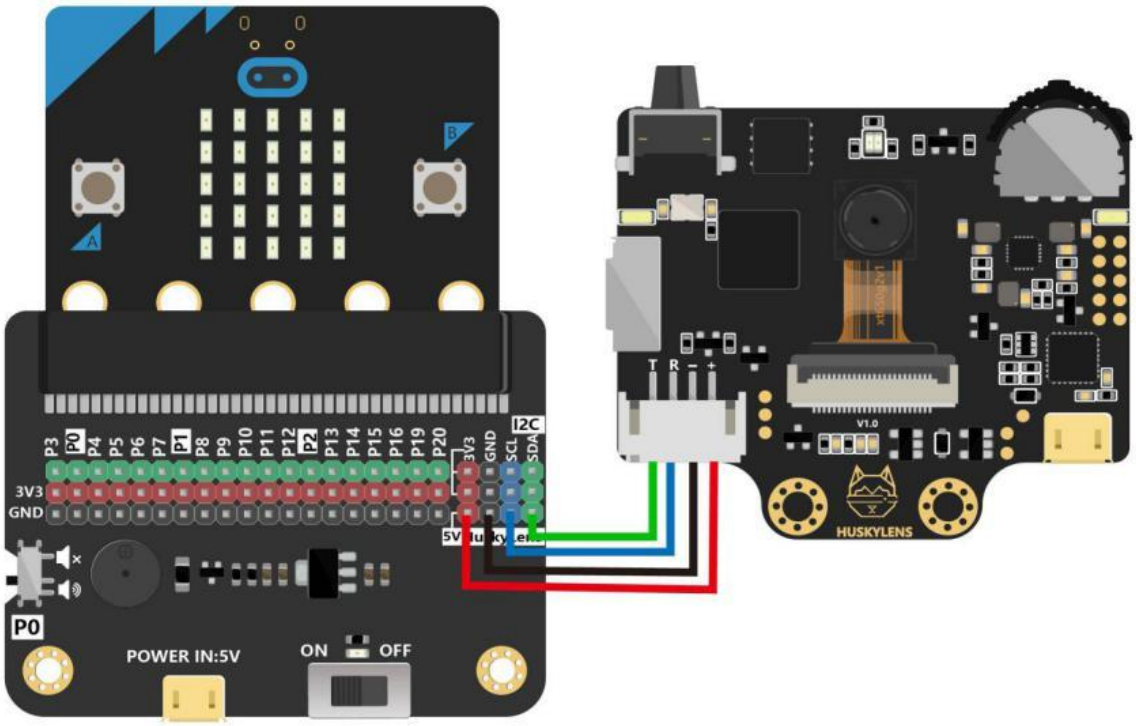
A Voice-activated camera is mainly used to take photos by voice. A microphone which can detect sound is added to the micro:bit V2 main control board. When the detected sound value is greater than 100, HuskyLens will perform the camera function.

Task 3: Time-lapse

Time-lapse photography is mainly used to take group photos. Usually, the camera is fixed in a position and set at an appropriate time-lapse, and then the photo is taken.

Task1: Touch

Hardware connection

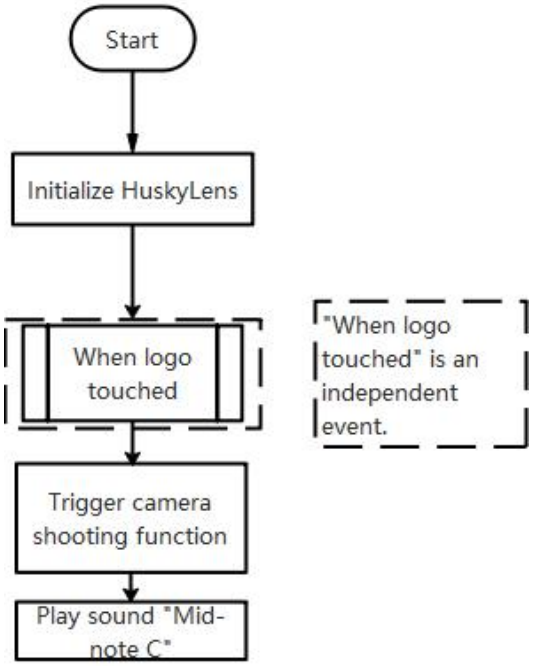


Program design

Function instruction:

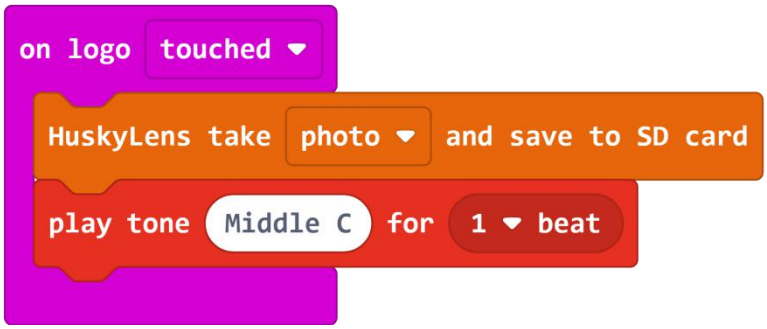
Touching to take pictures is the easiest way to take pictures in HuskyLens cameras. The HuskyLens camera program is called by triggering the touch LOGO of the micro:bit.

Flowchart Analysis



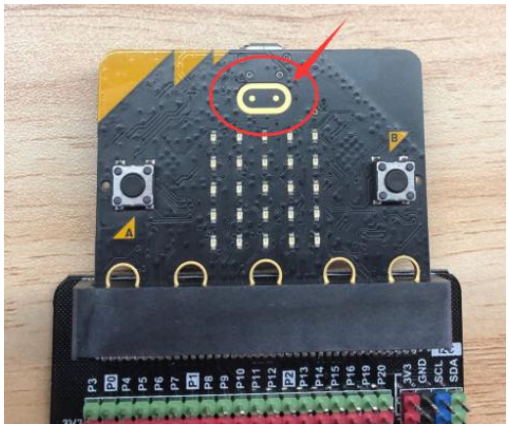
The Sample Program



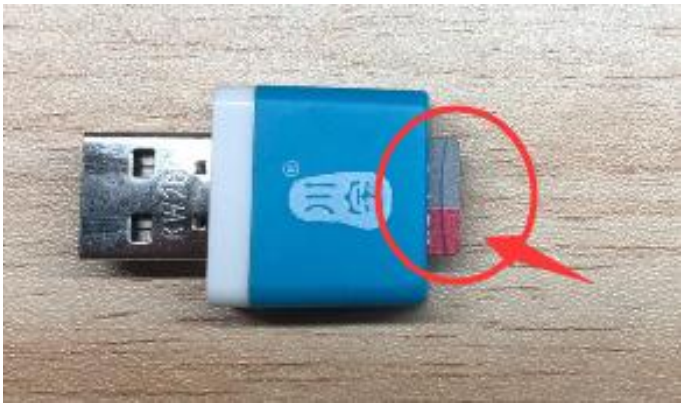


Operating Effect

Run the program and touch the micro:bit LOGO. If a "middle C" tone is played, it means that the photo was taken successfully. Then you can view the photos using the card reader.



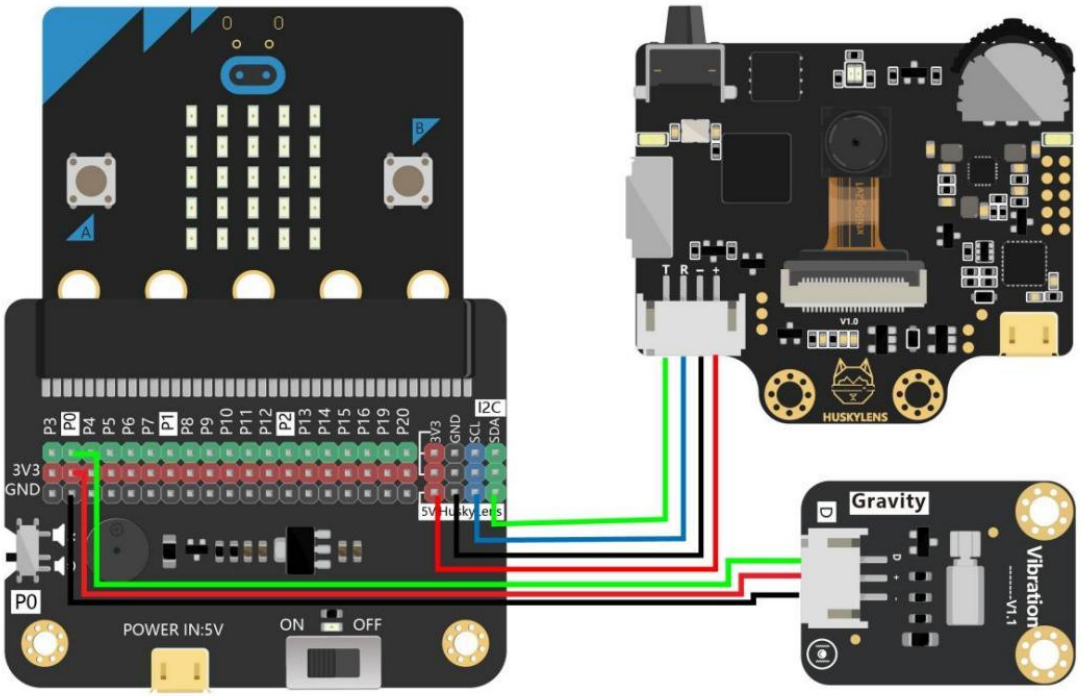
You can take a photo with one touch of the LOGO.



Insert the memory card into the card reader. After inserting the card reader into the USB port of the computer, you can view the photos you have taken.

Task2: Voice-activated

Hardware connection

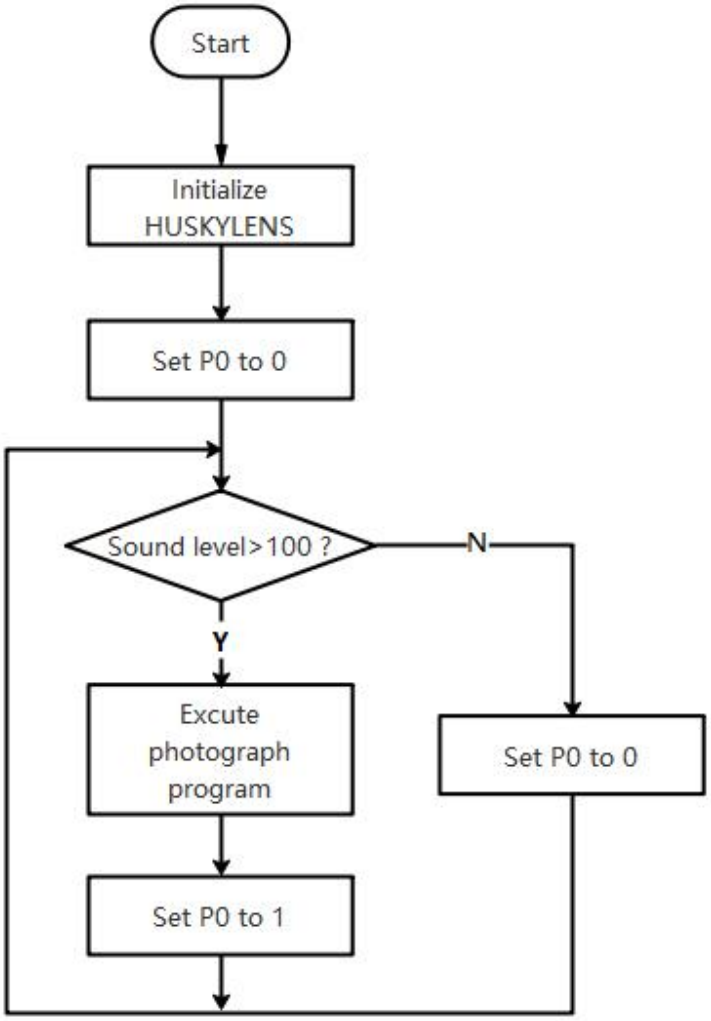


Program design

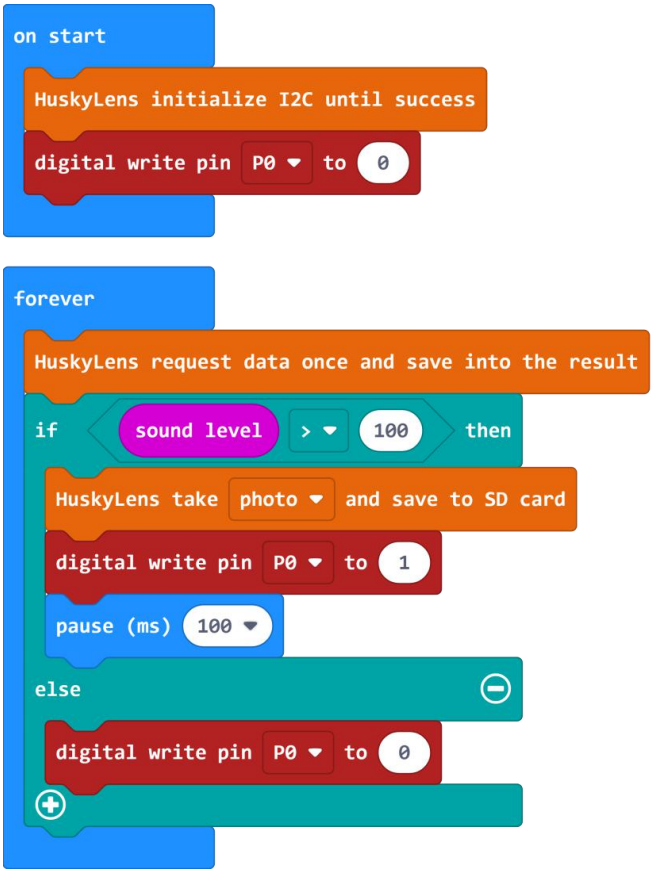
Function instruction:

Voice-activated photography is mainly achieved by judging the value of the voice. A microphone is integrated on the micro:bit V2 main control board, and the sound level can be detected through the microphone. When the detected "sound loudness"> 100, the camera program is executed and the micro vibration module vibrates.

Flowchart Analysis

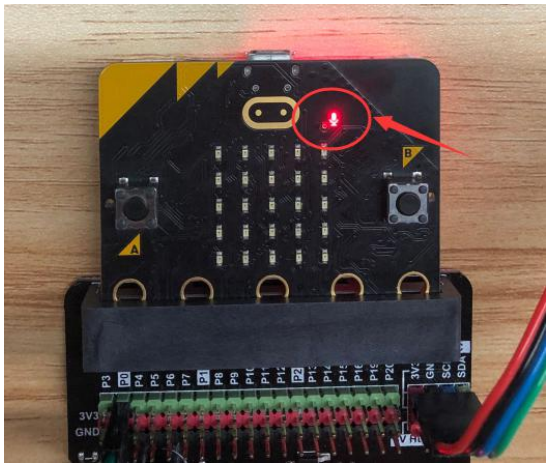


The Sample Program



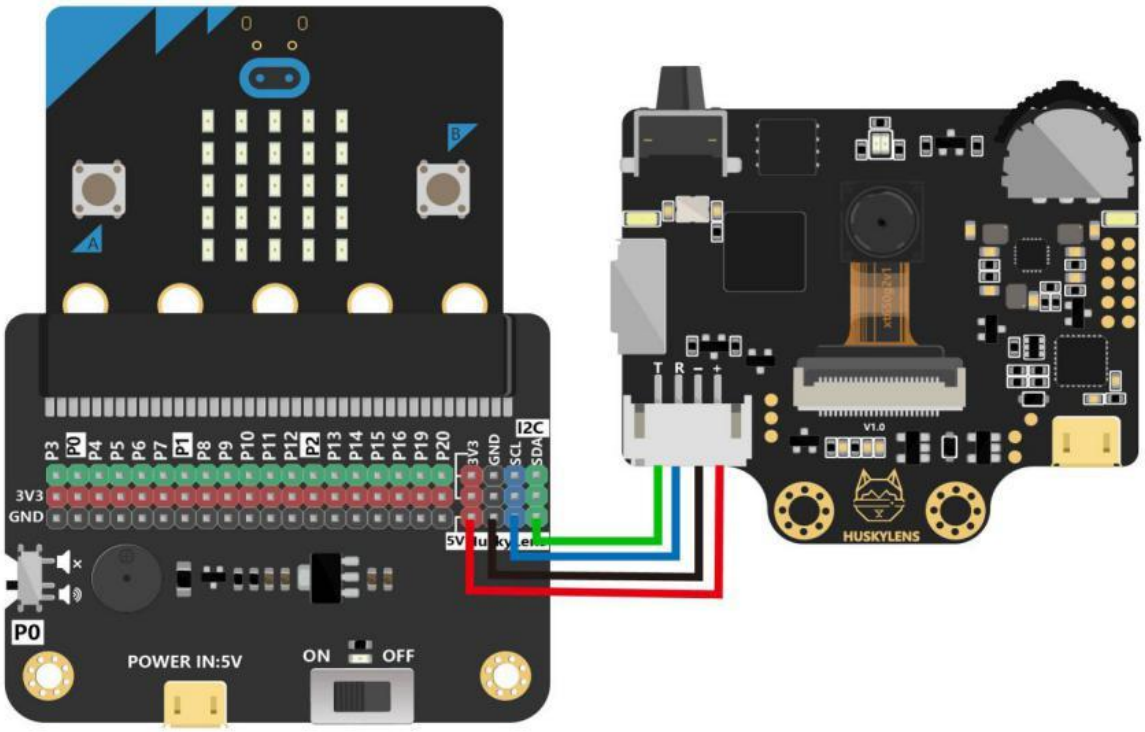
Operating Effect

Run the program, the small red microphone on the micro:bit will be lit up. When we speak loudly, HuskyLens will execute the camera command and the vibration module will vibrate. The vibration module vibrates every shoot.



Task3: Time-lapse

Hardware connection

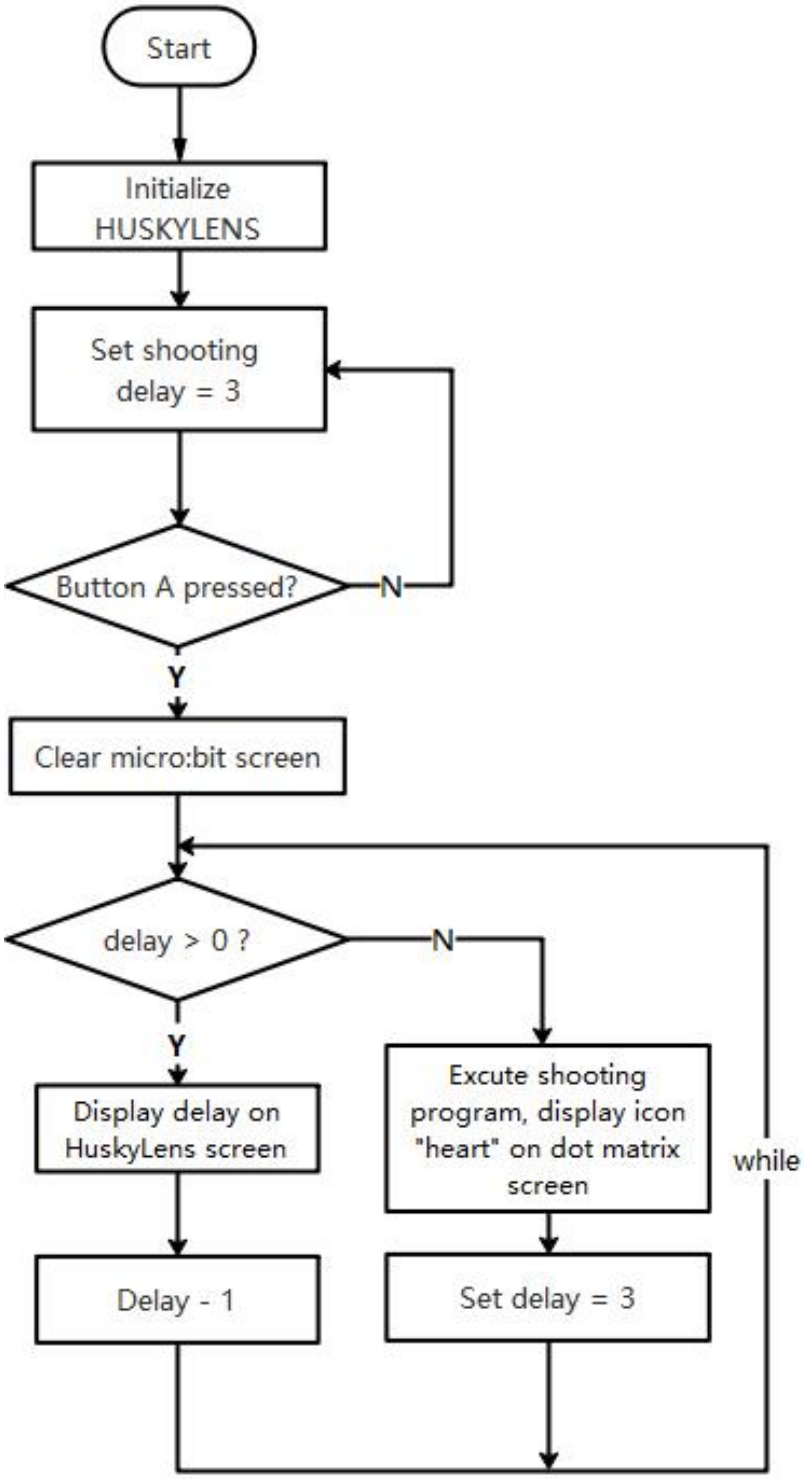


Program design

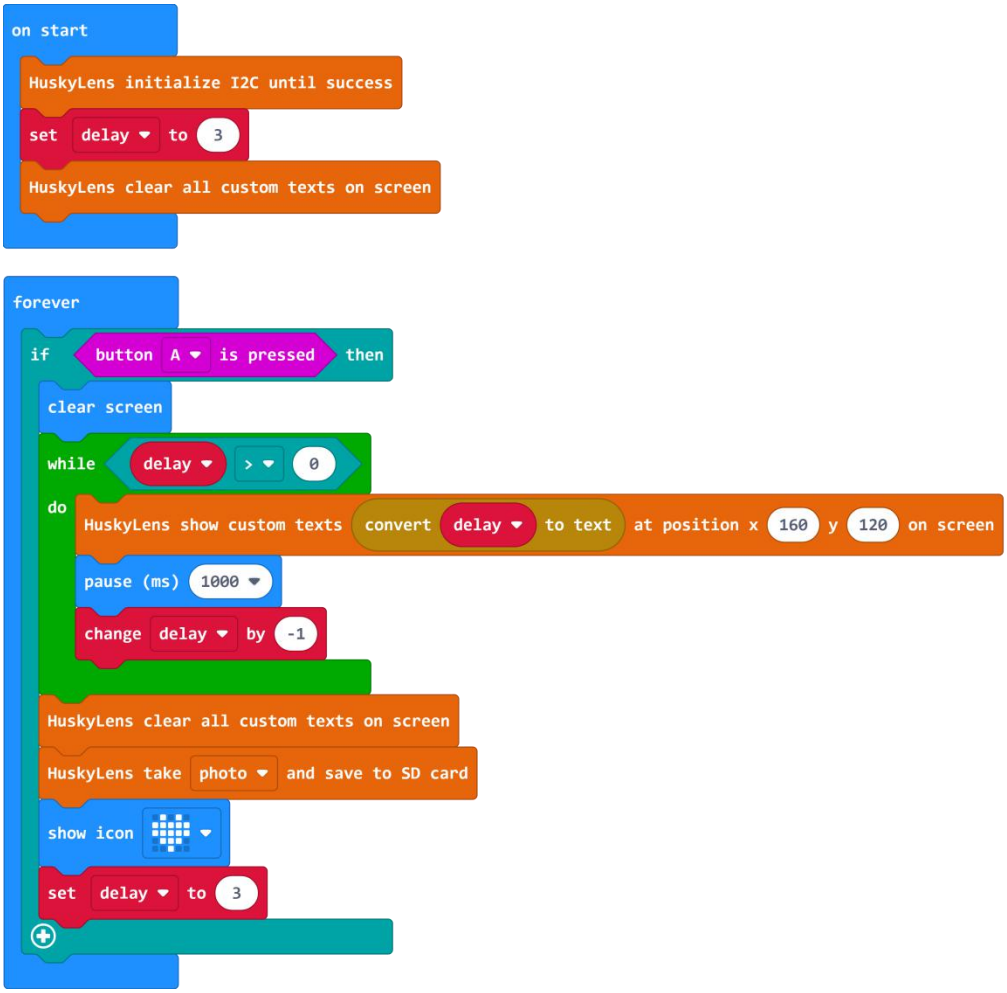
Function instruction:

In this project, set a 3S delay and use button A on the micro:bit to trigger the camera program. Click button A, the words 3, 2, and 1 will be displayed on the HuskyLens screen, and the "heart" pattern will be displayed on the micro:bit matrix screen to indicate the success of the photography.

Flowchart Analysis



The Sample Program



Operating Effect

Run the program and press button A, the HuskyLens screen displays the numbers 3, 2, and 1 in a decreasing manner. When the micro:bit dot matrix screen displays the "heart" pattern, it means the photo is taken.

Project Review

In this project, we mainly learned the principle of pinhole imaging and the principle of HuskyLens photography. Using HuskyLens to take pictures, we simulate the most commonly used phone photography in our lives.

The three tasks are mainly to trigger the camera function through three different hardware methods, which make up for the disadvantage of HuskyLens without a camera button. What other hardware methods can we use to trigger the camera function?

Project Development

Have you ever thought of using the eight built-in functions of HuskyLens to trigger the camera function? For example, what should you do if you only want to take pictures with puppies?

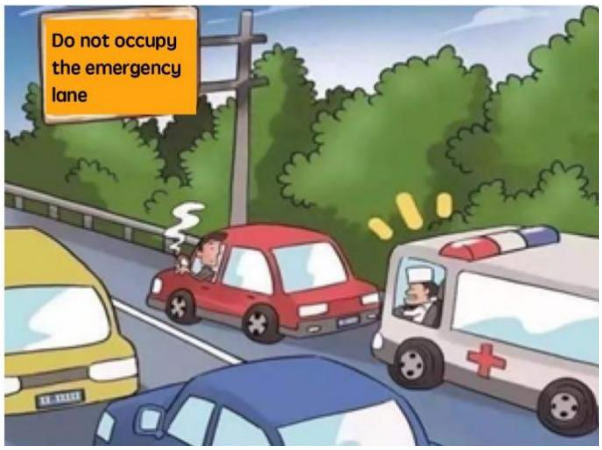
Tip: You can use the object classification function to learn pictures of puppies, and trigger the camera function by judging whether there are puppies in the picture.

Lesson 12: Emergency Lane Watchdog

The emergency lane is the only "passage of life" for rescue vehicles in the event of an emergency. According to the Road Traffic Safety Law, except for police cars, engineering rescue vehicles, and ambulances that perform emergency tasks, other motor vehicles are not allowed to enter the "emergency lane".



With the increasing number of vehicles, road traffic is seeing mounting pressure. At the same time, the violation of highway emergency lanes takes up a great proportion of all kinds of traffic violations, which need to be tackled urgently. Currently, people mainly deal with it through patrol cars and manual photography. However, these two methods are extremely inefficient, costing manpower and material resources greatly and bringing inconvenience to management.



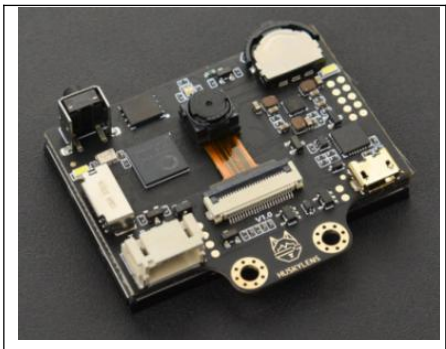
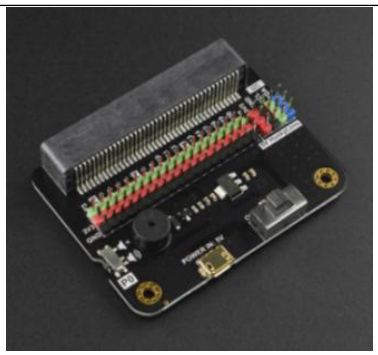
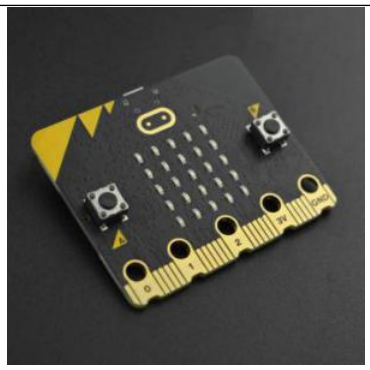


So, for intensifying investigation and punishment of occupation of emergency lanes, how can we use HuskyLens to make an emergency lane watchdog make up for the shortage of the two traditional methods?

Learning Goals

1. Use the program to trigger the camera function
2. Use the HuskyLens to make an emergency lane watchdog

Preparation

		
HuskyLens ×1	IO Extender for micro:bit V2.0 ×1	micro:bit v2 ×1
		

Memory Card ×1	RGB full color light strip ×1	Object Recognition Card

Note: The memory card is not included in this kit.

Learning Contents

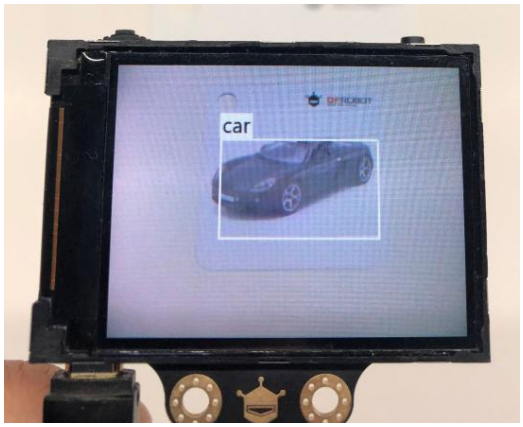
How to judge whether the vehicle occupies the emergency lane? We can make it by installing a camera above the emergency lane. In this case, when HuskyLens detects that a vehicle is in the lane, it will take a picture of it, and then use the flashing RGB lights to warn the vehicle to leave quickly.

Later, the traffic police only need to check the photos in the memory card to know which vehicles have occupied the emergency lane in violation of regulations, which can save them a lot of time and energy.

HuskyLens Object Recognition Function Demonstration

1. Detect Objects

Align HuskyLens to the target object – car, there will be a white box to select it automatically and its name will be displayed as well on the screen.



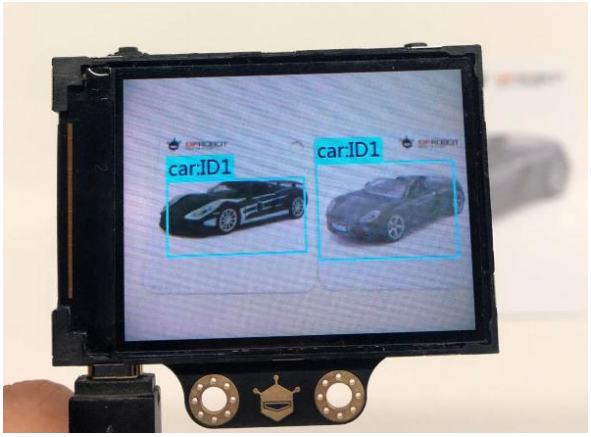
2. Mark Objects

Firstly, align HuskyLens to the target object. When it is detected and its name is displayed on screen, align the “+” on screen to the center of the white box of the object, short press the "learning button" to mark it. At this point, the color of the box will change from white to blue, and the name and ID1 of the object will be displayed as well.



3. Recognize Objects

When HuskyLens comes across objects it marked again, there will be a blue box to select them on the screen, as well as their names and ID.



Note: For this project, the functions of learning multiple objects and learning a single object are both suitable.

Project Practice

Next, we will complete this project by dividing it into two tasks. First, we must use object recognition and camera functions of HuskyLens to take photos of the vehicles peccancy in the emergency lane. Then add the alarm warning function on the basis of the previous function.

Task 1: Vehicle Photo

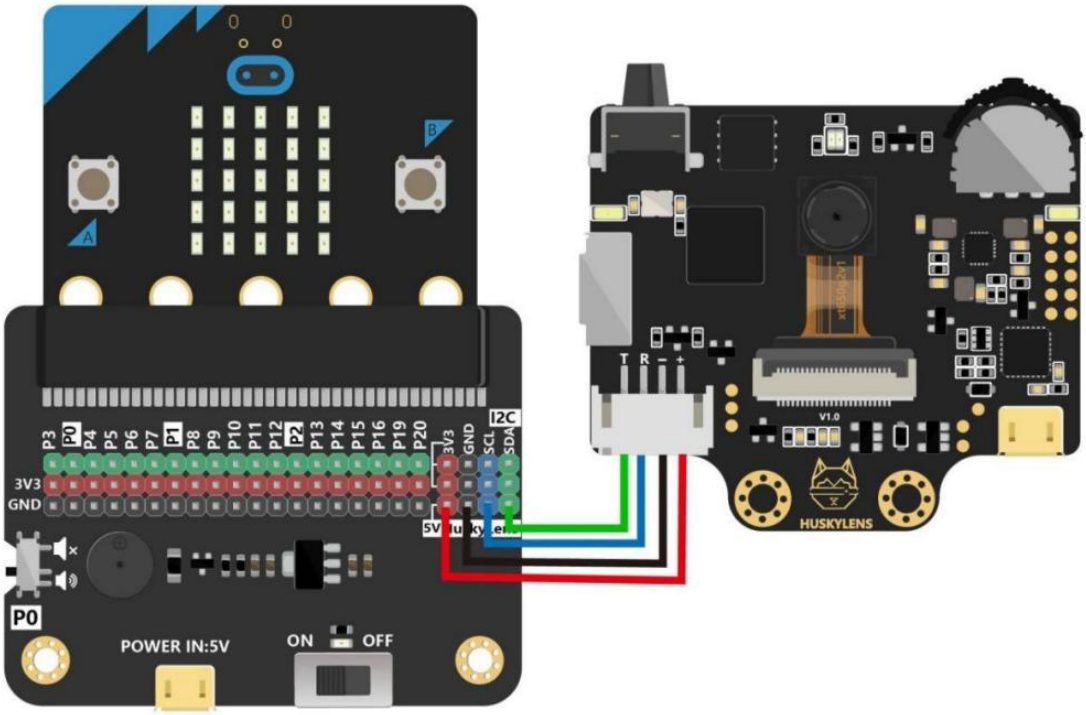
In the program, all the vehicles appearing on the HuskyLens screen will be regarded as vehicles that occupy emergency lanes illegally and HuskyLens will take pictures of them.

Task 2: Alarm Warning

Based on task 1, add RGB lights and alarm to warn the vehicle to leave the emergency lane.

Task 1: Vehicle Photo

Hardware Connection

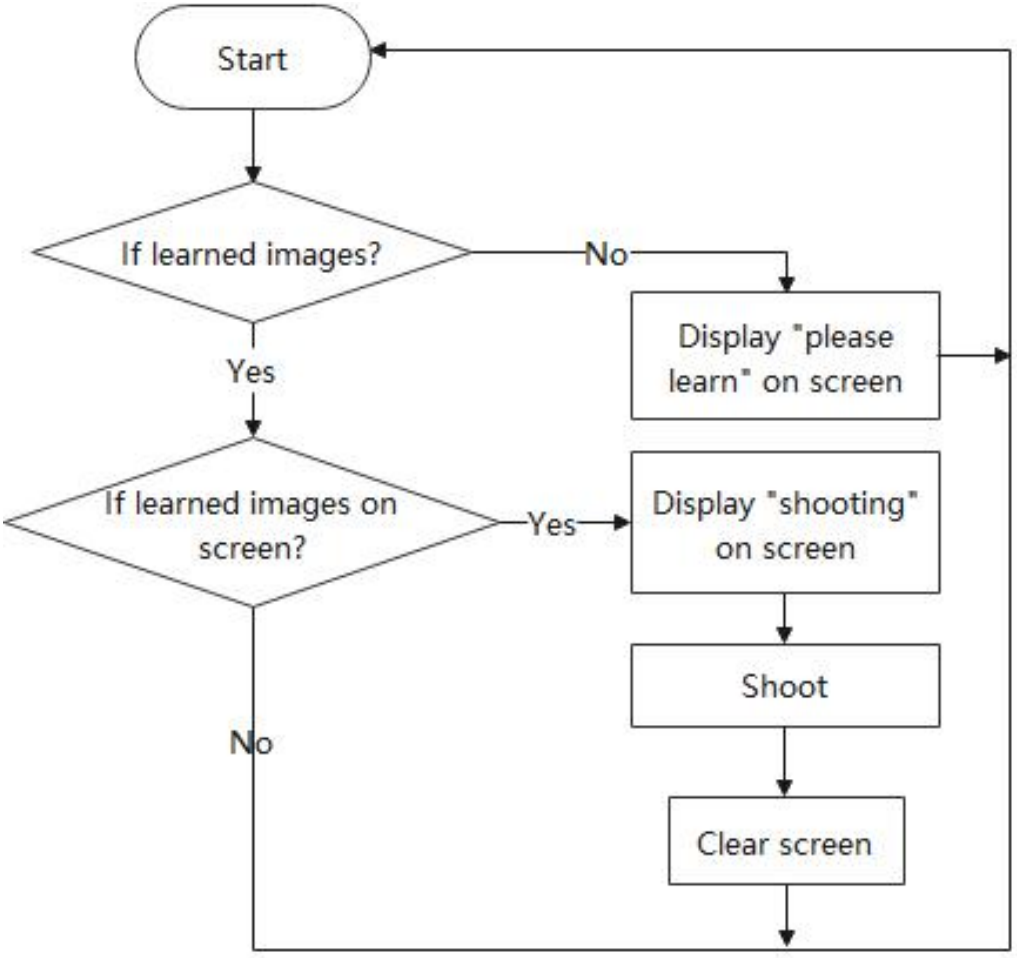


Program Design

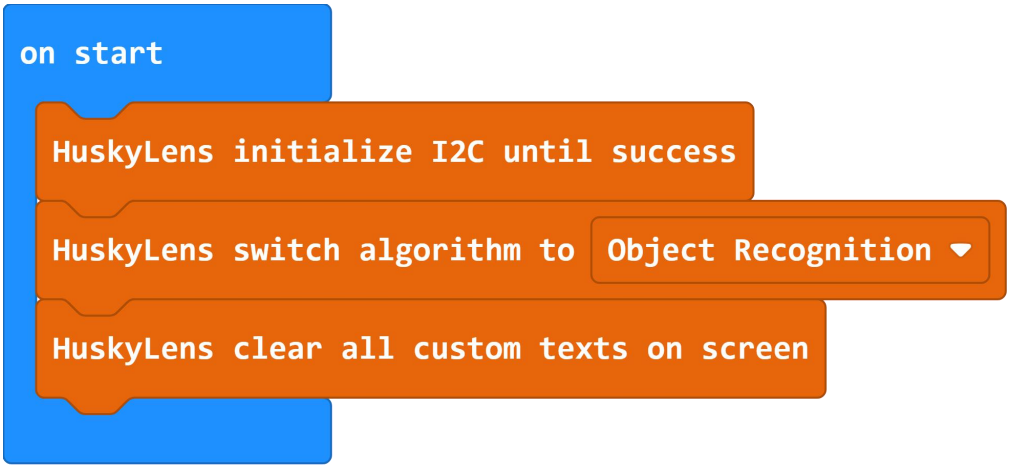
Function Description:

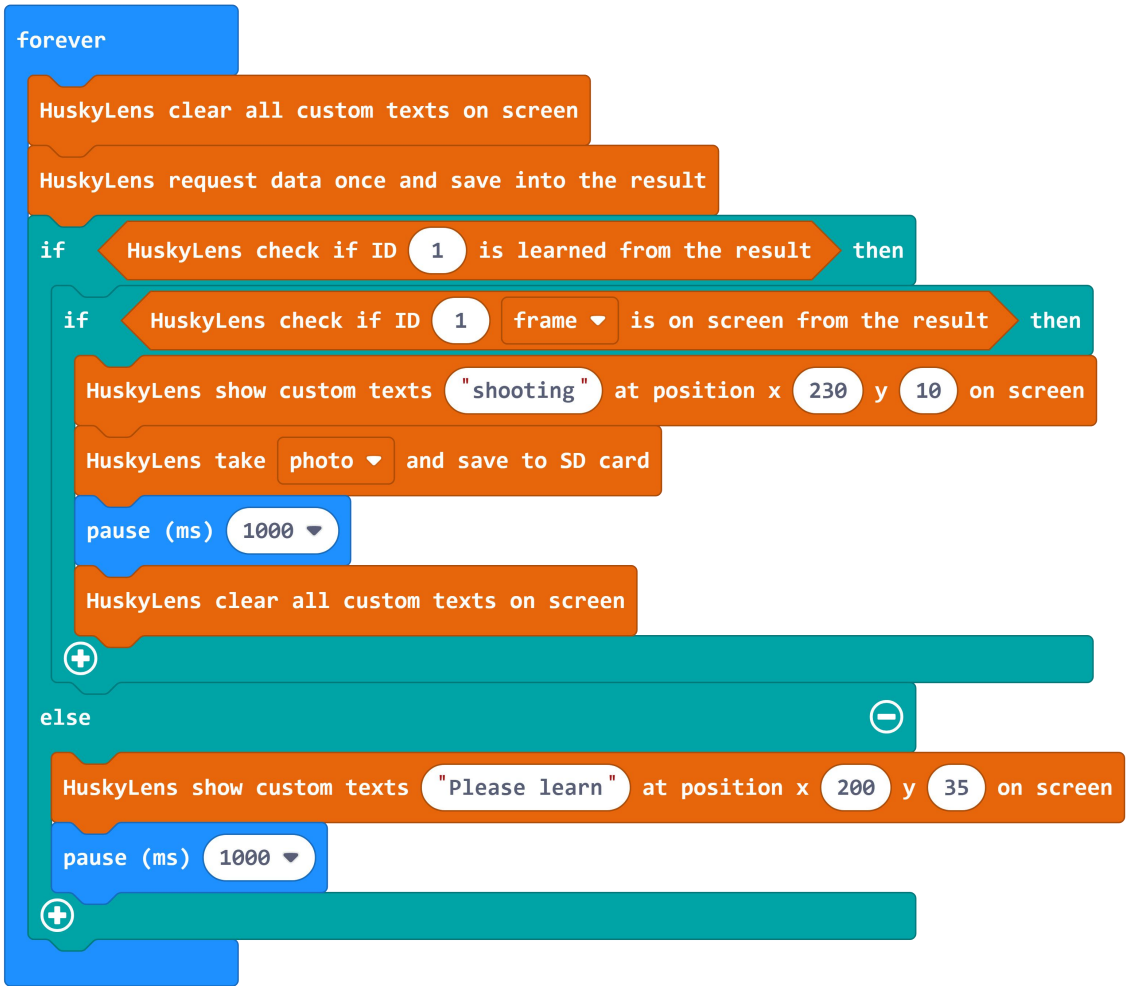
When HuskyLens detects that there is a vehicle in the emergency lane, it will display “shooting” on screen (x=230, y=10) and take a picture, which will be stored on the SD card.

Program Flow Chart Analysis:



Sample Program:





Operating Effect

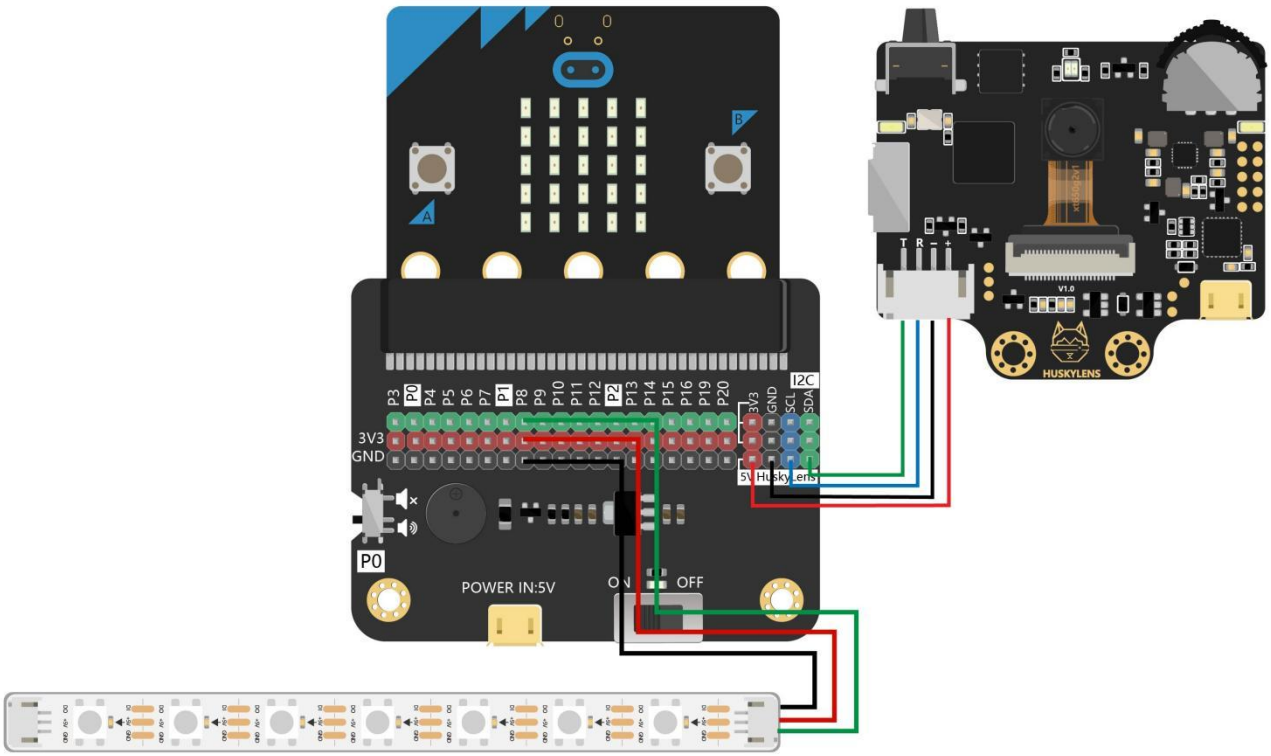
Run the program. The word "Please learn" will be displayed on the screen until HuskyLens learns something. After learning the car, when HuskyLens detects it, the word "shooting" will be displayed on the screen.





Task 2: Alarm Warning

Hardware Connection

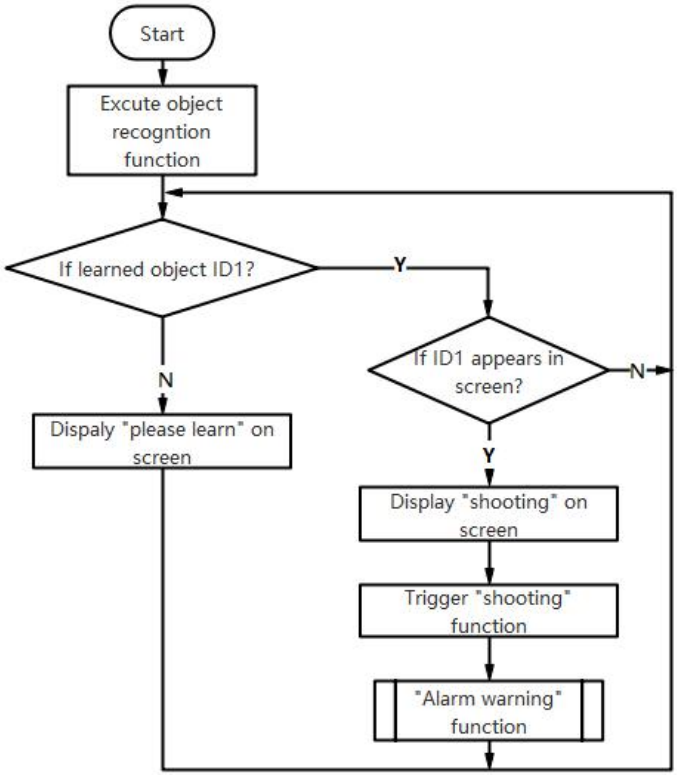


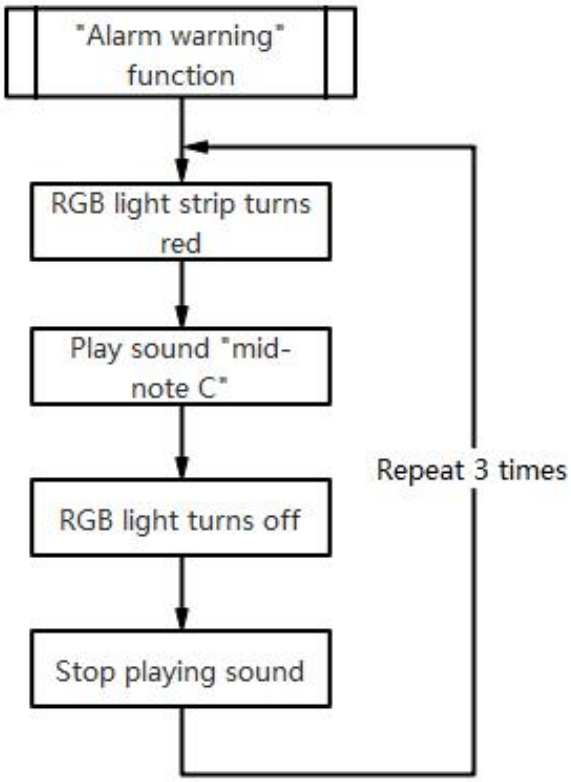
Program Design

Function Explain

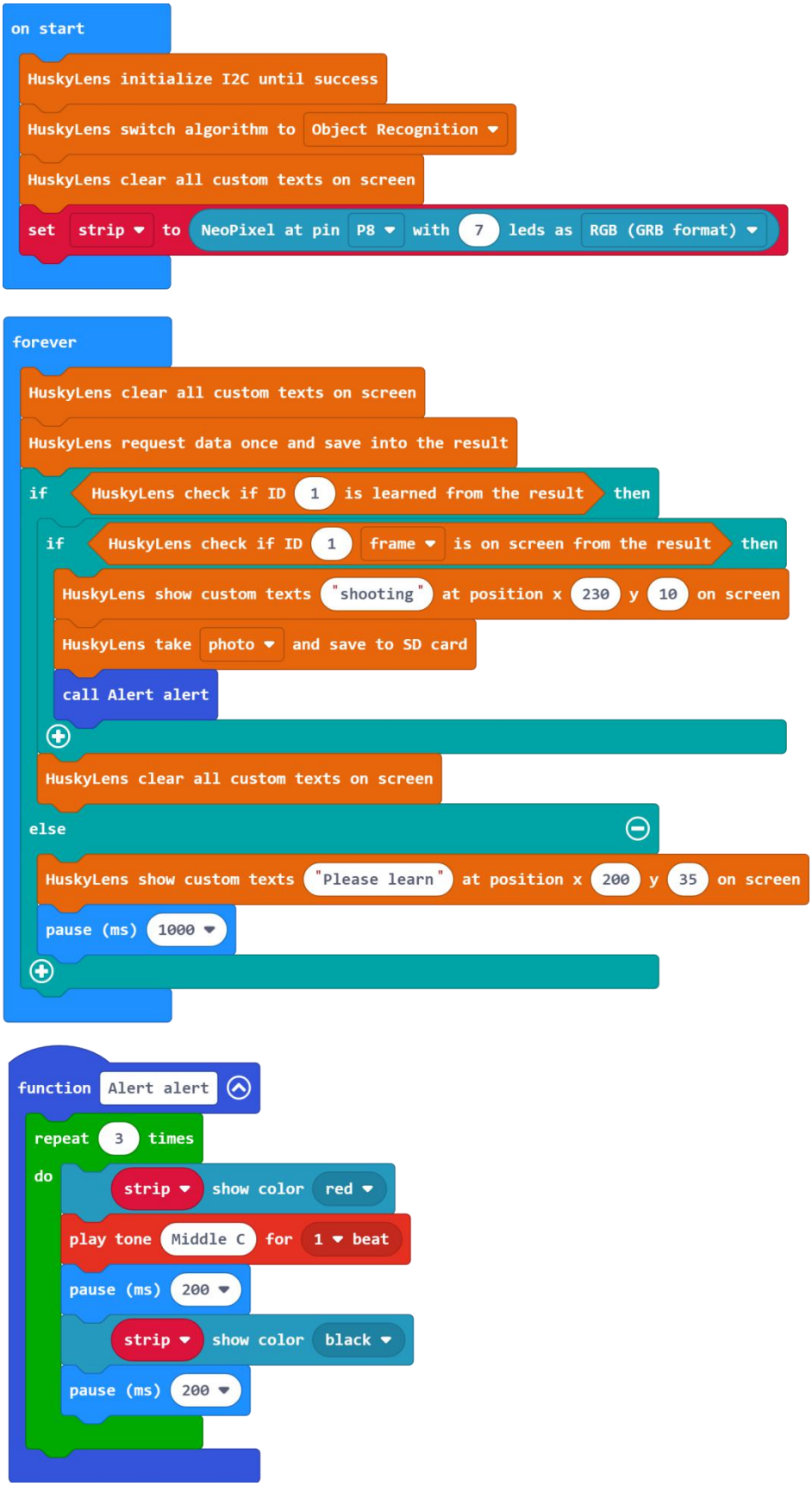
When the HuskyLens detects that vehicles are occupying the emergency lane illegally, how to signal them to leave quickly? In this case, we can warn the vehicles by the alarm sound and flashing RGB lights.

Program Flow Chart Analysis:



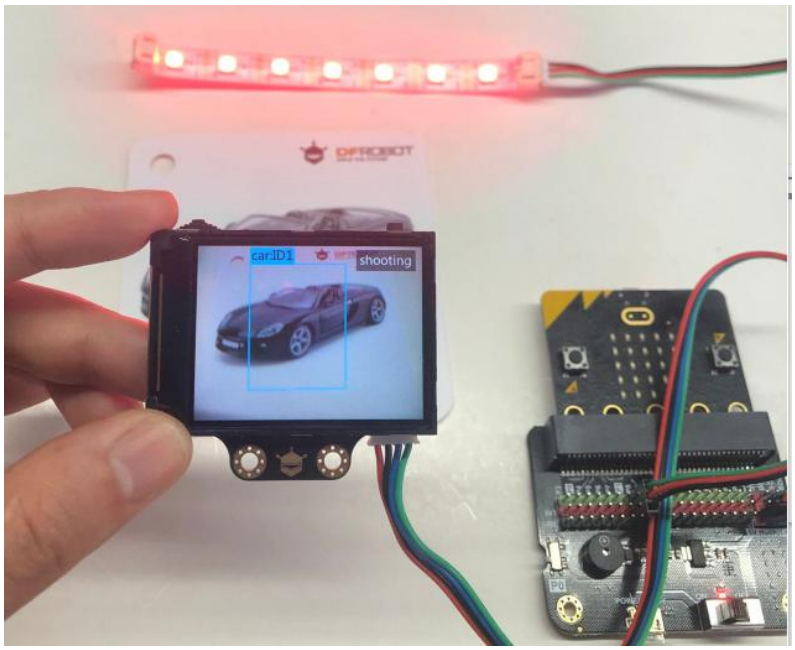

"Alarm warning" function
RGB light strip turns red
Play sound "mid-note C"
RGB light turns off
Stop playing sound
Repeat 3 times

Sample Program:



Operating Effect

Run the program. When HuskyLens detects a car occupying the emergency lane, it will take a picture of it, and then make some lights and alarm sound to warn the car to leave the lane quickly. Besides, you can also view the photos of vehicles on the memory card through the card reader.

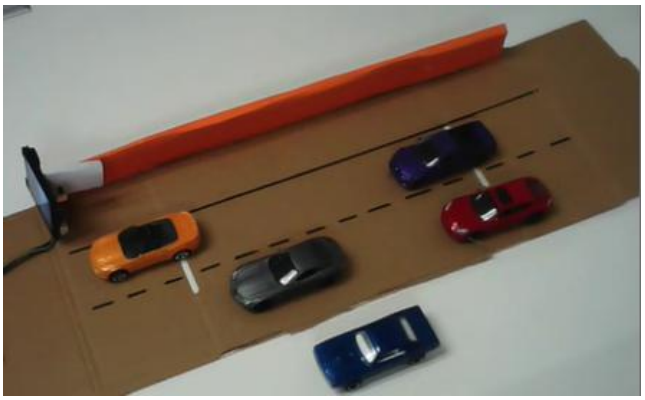


Project Review

In this project, we have used the function of object recognition and triggered the camera function by program judgment. That is, it will take a picture once the car is recognized. Moreover, the project can also remind the vehicles occupying the emergency lane to leave by lighting up RGB lights and making alarm sounds.

The emergency lane watchdog, to a large extent, saves traffic police a lot of time and energy. Their work efficiency can be improved owing to the fact that manual photography is no longer required. Because they can know which vehicles have illegally occupied the emergency lane by checking the picture on the memory card.

Project Expansion



You can use waste corrugated paper to draw a complete lane, and then fix HuskyLens above the emergency lane to ensure that HuskyLens can detect vehicles in it.